

6 Transmitter characteristics

6.1 General

Unless otherwise stated the transmitter characteristics are specified at the antenna connector(s) of the UE for the bands operating on frequency range 1 and over the air of the UE for the bands operating on frequency range 2. The requirements for frequency range 1 and frequency range 2 can be verified separately. For the carrier in frequency range 1, requirements can be verified with NR FR2 link disabled. For the carrier in frequency range 2, requirements can be verified in OTA mode with E-UTRA or NR FR1 connecting to the network by OTA without calibration.

Unless otherwise stated, requirements for NR transmitter written in TS 38.101-1 [2] and TS 38.101-2 [3] apply and are assumed anchor agnostic. Unless otherwise stated, if UE indicates IE maxNumberSRS-Ports-PerResource = n2 in NR standalone operation mode, the said UE shall meet the NR requirements for either power class 2 or power class 3 in EN-DC within FR1 if UE indicates IE maxNumberSRS-Ports-PerResource = n1 for EN-DC on this NR band. Requirements are verified under conditions where anchor resources do not interfere NR operation.

6.2 Void

6.2A Transmitter power for CA

6.2A.1 UE maximum output power for CA

6.2A.1.1 Inter-band CA between FR1 and FR2

Table 6.2A.1.1-1: Void

For inter-band NR CA in FR1 and FR2 combined, the UE shall meet each transmitter power requirement specified in TS 38.101-1 [2] and TS 38.101-2 [3] for each frequency range independently.

6.2A.2 UE maximum output power reduction for CA

6.2A.2.1 Inter-band CA between FR1 and FR2

For inter-band NR CA between FR1 and FR2, UE maximum output power reduction specified in TS 38.101-1 [2] and TS 38.101-2 [3] apply for each frequency range respectively.

6.2A.3 UE additional maximum output power reduction for CA

For inter-band NR CA between FR1 and FR2, UE additional maximum output power reduction specified in TS 38.101-1 [2] and TS 38.101-2 [3] apply for each frequency range respectively.

6.2A.4 Configured output power for CA

6.2A.4.1 Configured output power level

For inter-band NR CA between FR1 and FR2, UE configured output power specified in TS 38.101-1 [2] and TS 38.101-2 [3] apply for each frequency range respectively.

6.2A.4.2 $\Delta T_{IB,c}$ for CA

6.2A.4.2.1 $\Delta T_{IB,c}$ for Inter-band CA between FR1 and FR2

Unless otherwise stated, $\Delta T_{IB,c}$ for NR FR1 band and FR2 band of inter-band CA defined in table 5.5A.1-1 is set to zero.

Table 6.2A.4.2.1-1: Void

6.2B Transmitter power for DC

6.2B.1 UE maximum output power for DC

6.2B.1.1 Intra-band contiguous EN-DC

The following UE Power Classes define the total maximum output power for any transmission bandwidth(s) of the CG(s) configured.

The maximum output power is measured as the total maximum output power across the UE antenna connector(s). The period of measurement shall be at least one sub frame.

Table 6.2B.1.1-1: Maximum output power for EN-DC (continuous sub-blocks)

EN-DC configuration	Power class 2 (dBm)	Tolerance (dB)	Power class 3 (dBm)	Tolerance (dB)
DC_(n)71AA			23	+2/-3
DC_(n)41AA	26	+2/-3 ¹	23	+2/-3 ¹
NOTE 1: If all transmitted resource blocks over all component carriers are confined within F_{UL_low} and $F_{UL_low} + 4$ MHz or/and $F_{UL_high} - 4$ MHz and F_{UL_high} , the maximum output power requirement is relaxed by reducing the lower tolerance limit by 1.5 dB				
NOTE 2: Power Class 3 is the default power class unless otherwise stated.				

If UE supports a different power class than the default UE power class for EN-DC band combination, and the supported power class enables higher maximum output power than that of the default power class:

- if the E-UTRA UL/DL configuration is 0 or 6; or
- if the E-UTRA UL/DL configuration is 1 and special subframe configuration is 0 or 5; or
- if the IE *p-maxUE-FR1-r15* as defined in TS 36.331 [8] is provided and set to the maximum output power of the default power class or lower;
 - apply all requirements for the default power class, and set the configured transmitted power as specified in clause 6.2B.4;
- else
 - apply all requirements for the supported power class, and set the configured transmitted power class as specified in clause 6.2B.4;

6.2B.1.2 Intra-band non-contiguous EN-DC

Table 6.2B.1.2-1: Maximum output power for EN-DC (non-continuous sub-blocks)

EN-DC configuration	Power class 2 (dBm)	Tolerance (dB)	Power class 3 (dBm)	Tolerance (dB)
DC_3A_n3A ²			23	+2/-3
DC_41A_n41A	26	+2/-3 ¹	23	+2/-3 ¹
NOTE 1: If all transmitted resource blocks over all component carriers are confined within F_{UL_low} and $F_{UL_low} + 4$ MHz or/and $F_{UL_high} - 4$ MHz and F_{UL_high} , the maximum output power requirement is relaxed by reducing the lower tolerance limit by 1.5 dB				
NOTE 2: Only single switched UL is supported in Rel.15				
NOTE 3: Power Class 3 is the default power class unless otherwise stated.				

If UE supports a different power class than the default UE power class for EN-DC band combination, and the supported power class enables higher maximum output power than that of the default power class:

- if the E-UTRA UL/DL configuration is 0 or 6; or
- if the E-UTRA UL/DL configuration is 1 and special subframe configuration is 0 or 5; or
- if the IE *p-maxUE-FR1-r15* as defined in TS 36.331 [8] is provided and set to the maximum output power of the default power class or lower;
 - apply all requirements for the default power class, and set the configured transmitted power as specified in clause 6.2B.4;
- else
- apply all requirements for the supported power class, and set the configured transmitted power class as specified in clause 6.2B.4;

6.2B.1.3 Inter-band EN-DC within FR1

For inter-band EN-DC of E-UTRA and NR in FR1, the following UE Power Classes define the maximum output power for any transmission bandwidth within the aggregated channel bandwidth. The maximum output power is measured as the sum of the maximum output power at each UE antenna connector. The period of measurement shall be at least one sub frame (1ms). UE maximum output power shall be measured over all component carriers from different bands. If each band has separate antenna connectors, maximum output power is measured as the sum of maximum output power at each UE antenna connector.

Table 6.2B.1.3-1: Maximum output power for inter-band EN-DC (two bands)

EN-DC configuration	Power class 3 (dBm)	Tolerance (dB)
DC_1A_n28A	23	+2/-3
DC_1A_n40A	23	+2/-3
DC_1A_n51A	23	+2/-3
DC_1A_n77A	23	+2/-3
DC_1A_n78A DC_1A_n84A_ULSUP-TDM_n78A	23	+2/-3
DC_1A_n79A	23	+2/-3
DC_2A_n5A	23	+2/-3 ¹

DC_2A_n66A	23	+2/-3 ¹
DC_2A_n71A	23	+2/-3
DC_2A_n78A	23	+2/-3
DC_3A_n7A	23	+2/-3 ¹
DC_3A_n28A	23	+2/-3 ¹
DC_3A_n40A	23	+2/-3 ¹
DC_3A_n51A	23	+2/-3 ¹
DC_3A_n77A	23	+2/-3 ¹
DC_3A_n78A DC_3A_n80A_ULSUP-TDM_n78A,	23	+2/-3 ¹
DC_3A_n79A DC_3A_n80A_ULSUP-TDM_n79A,	23	+2/-3 ¹
DC_3A_n82A	23	+2/-3 ¹
DC_5A_n40A	23	+2/-3 ¹
DC_5A_n66A	23	+2/-3 ¹
DC_5A_n78A	23	+2/-3
DC_7A_n28A	23	+2/-3 ¹
DC_7A_n51A	23	+2/-3 ¹
DC_7A_n78A DC_7C_n78A	23	+2/-3
DC_8A_n40A	23	+2/-3 ¹
DC_8A_n77A	23	+2/-3
DC_8A_n78A DC_8A_n81A_ULSUP-TDM_n78A,	23	+2/-3
DC_8A_n79A DC_8A_n81A_ULSUP-TDM_n79A,	23	+2/-3
DC_11A_n77A	23	+2/-3
DC_11A_n78A	23	+2/-3
DC_11A_n79A	23	+2/-3
DC_12A_n5A	23	+2/-3
DC_12A_n66A	23	+2/-3
DC_18A_n77A	23	+2/-3
DC_18A_n78A	23	+2/-3
DC_18A_n79A	23	+2/-3
DC_19A_n77A	23	+2/-3
DC_19A_n78A	23	+2/-3
DC_19A_n79A	23	+2/-3
DC_20A_n8A	23	+2/-3
DC_20A_n28A DC_20A_n83A	23	+2/-3
DC_20A_n51A	23	+2/-3
DC_20A_n77A	23	+2/-3
DC_20A_n78A DC_20A_n82A_ULSUP-TDM_n78A,	23	+2/-3
DC_21A_n77A	23	+2/-3
DC_21A_n78A	23	+2/-3

DC_21A_n79A	23	+2/-3
DC_25A_n41A	23	+2/-3
DC_26A_n41A	23	+2/-3
DC_26A_n77A	23	+2/-3
DC_26A_n78A	23	+2/-3
DC_26A_n79A	23	+2/-3
DC_28A_n51A	23	+2/-3
DC_28A_n77A	23	+2/-3
DC_28A_n78A DC_28A_n83A_ULSUP-TDM_n78A,	23	+2/-3
DC_28A_n79A	23	+2/-3
DC_30A_n5A	23	+2/-3
DC_30A_n66A	23	+2/-3
DC_38A_n78A	N/A	N/A
DC_39A_n78A	23	+2/-3 ¹
DC_39A_n79A	23	+2/-3 ¹
DC_40A_n77A	N/A	N/A
DC_41A_n77A DC_41C_n77A	23	+2/-3 ¹
DC_41A_n78A DC_41C_n78A	23	+2/-3 ¹
DC_41A_n79A DC_41C_n79A	23	+2/-3 ¹
DC_42A_n51A	23	+2/-3
DC_42A_n77A	N/A	N/A
DC_42A_n78A	N/A	N/A
DC_42A_n79A	N/A	N/A
DC_66A_n5A	23	+2/-3 ¹
DC_66A_n71A	23	+2/-3
DC_66A_n78A, DC_66A_n86A_ULSUP-TDM_n78A,	23	+2/-3
NOTE 1: For the transmission bandwidths confined within F_{UL_low} and $F_{UL_low} + 4$ MHz or $F_{UL_high} - 4$ MHz and F_{UL_high}, the maximum output power requirement is relaxed by reducing the lower tolerance limit by 1.5 dB NOTE 2: $P_{PowerClass, EN-DC}$ is the maximum UE power specified without taking into account the tolerance NOTE 3: For inter-band EN-DC the maximum power requirement should apply to the total transmitted power over all component carriers (per UE). NOTE 4: Power Class 3 is the default power class unless otherwise stated.		

6.2B.1.3a Inter-band NE-DC within FR1

For inter-band NE-DC of E-UTRA and NR in FR1, the following UE power classes define the maximum output power for any transmission bandwidth within the aggregated channel bandwidth. The maximum output power is measured as the sum of the maximum output power at each UE antenna connector. The period of measurement shall be at least one sub frame (1 ms). UE maximum output power shall be measured over all component carriers from different bands. If each band has separate antenna connectors, maximum output power is measured as the sum of maximum output power at each UE antenna connector.

Table 6.2B.1.3a-1: Maximum output power for inter-band NE-DC (two bands)

NE-DC configuration	Power class 3 (dBm)	Tolerance (dB)
DC_n1A_28A	23	+2/-3

6.2B.1.4 Inter-band EN-DC including FR2

UE maximum output power requirement for E-UTRA single carrier and CA operation specified in clauses 6.2.2 and 6.2.2A of TS 36.101 [4] and for NR single carrier, CA operation and UL-MIMO specified in clause 6.2.1, 6.2A.1, and 6.2D.1 of TS 38.101-2 [3] apply.

6.2B.1.5 Inter-band EN-DC including both FR1 and FR2

UE maximum output power requirement for E-UTRA single carrier and CA operation specified in clauses 6.2.2 and 6.2.2A of TS 36.101 [4] and for NR single carrier specified in clause 6.2.1 of TS 38.101-1 [2] and for NR single carrier, CA operation and UL-MIMO specified in clause 6.2.1, 6.2A.1, and 6.2D.1 of TS 38.101-2 [3] apply. When uplink is EN-DC mode within FR1 only then UE maximum output power requirement is specified in clause 6.2B.1.3 of this specification.

6.2B.2 UE maximum output power reduction for DC

6.2B.2.0 General

The UE maximum output power reduction (MPR) specified in this clause is applicable for UEs configured with EN-DC when NS_01 is indicated in the MCG and the SCG. The MPR applies subject to indication in the field *modifiedMPRbehavior* for the SCG [2].

6.2B.2.1 Intra-band contiguous EN-DC

6.2B.2.1.1 General

When the UE is configured for intra-band contiguous EN-DC, the UE determines the total allowed maximum output power reduction as specified in this clause.

For UE supporting dynamic power sharing the following:

- for the MCG, MPR_c in accordance with TS 36.101 [4]
- for the SCG,

$$MPR'_c = MPR_{NR} = \text{MAX}(MPR_{\text{single,NR}}, MPR_{\text{ENDC}})$$

- for the total configured transmission power,

$$MPR_{\text{tot}} = P_{\text{PowerClass,EN-DC}} - \min(P_{\text{PowerClass,EN-DC}}, 10 \cdot \log_{10}(10^{((P_{\text{PowerClass,E-UTRA}} - MPR_{\text{E-UTRA}})/10)} + 10^{((P_{\text{PowerClass,NR}} - MPR_{\text{NR}})/10)}))$$

where

$$MPR_{\text{E-UTRA}} = \text{MAX}(MPR_{\text{single,E-UTRA}}, MPR_{\text{ENDC}})$$

with

- $MPR_{\text{single,E-UTRA}}$ is the MPR defined for the E-UTRA transmission in TS 36.101 [4]

- $MPR_{\text{single,NR}}$ is the MPR defined for the NR transmission in TS 38.101-1 [2]

For UEs not supporting dynamic power sharing the following

- for the MCG,

$$MPR_c = \text{MAX}(MPR_{\text{single,E-UTRA}}, MPR_{\text{ENDC}})$$

- for the SCG,

$$MPR'_c = \text{MAX}(MPR_{\text{single,NR}}, MPR_{\text{ENDC}})$$

where

- $MPR_{\text{single,NR}}$ is the MPR defined for the NR transmission in TS 38.101-1 [2]
- $MPR_{\text{single,E-UTRA}}$ is the MPR defined for the E-UTRA transmission in TS 36.101 [4]

MPR_{ENDC} is defined in Clause 6.2B.2.1.2

6.2B.2.1.2 MPR for power class 3 and power class 2

MPR in this clause is applicable for power class 3 and power class 2 UEs indicating IE *dualPA-Architecture* supported with ENDC power class being the same as the E-UTRA and NR power class, otherwise the UE can use as much MPR as needed to fulfil emissions requirements when scheduled with dual uplink transmission. For UEs scheduled with single uplink transmission, MPR in clause 6.2.4 of TS 36.101 [4] and 6.2.2 of TS 38.101-1 [2] apply. For a UE supporting dynamic power sharing for DC_(n)71AA for which dual simultaneous uplink transmissions are mandatory and A-MPR defined in clause 6.2B.3.1.1 is applied as MPR. The allowed maximum output power reduction applied to transmission on the MCG and the SCG is defined as follows:

$$MPR_{\text{ENDC}} = M_A$$

Where M_A is defined as follows

$$M_A = \begin{array}{ll} 15 & ; \quad 0 \leq B < 0.5 \\ 10 & ; \quad 0.5 \leq B < 1.0 \\ 8 & ; \quad 1.0 \leq B < 2.0 \\ 6 & ; \quad 2.0 \leq B \end{array}$$

Where:

For UEs supporting dynamic power sharing,

$$B = (L_{\text{CRB_alloc, E-UTRA}} * 12 * \text{SCS}_{\text{E-UTRA}} + L_{\text{CRB_alloc,NR}} * 12 * \text{SCS}_{\text{NR}}) / 1,000,000$$

For UEs not supporting dynamic power sharing,

For E-UTRA

$$B = (L_{\text{CRB_alloc, E-UTRA}} * 12 * \text{SCS}_{\text{E-UTRA}} + 12 * \text{SCS}_{\text{NR}}) / 1,000,000$$

Where $\text{SCS}_{\text{NR}} = 15,000$ Hz is assumed in calculation of B.

For NR

$$B = (12 * \text{SCS}_{\text{E-UTRA}} + L_{\text{CRB_alloc,NR}} * 12 * \text{SCS}_{\text{NR}}) / 1,000,000$$

Where $\text{SCS}_{\text{E-UTRA}} = 15,000$ Hz is assumed in calculation of B.

and M_A is reduced by 1 dB for $B < 2$.

6.2B.2.2 Intra-band non-contiguous EN-DC

6.2B.2.2.1 General

When the UE is configured for intra-band non-contiguous EN-DC, the UE determines the total allowed maximum output power reduction as specified in this clause.

For UE supporting dynamic power sharing the following:

- for the MCG, MPR_c in accordance with TS 36.101 [4]
- for the SCG,

$$MPR'_c = MPR_{NR} = \text{MAX}(MPR_{\text{single,NR}}, MPR_{\text{ENDC}})$$

- for the total configured transmission power,

$$MPR_{\text{tot}} = P_{\text{PowerClass,EN-DC}} - \min(P_{\text{PowerClass,EN-DC}}, 10 \cdot \log_{10}(10^{(P_{\text{PowerClass,E-UTRA}} - MPR_{\text{E-UTRA}})/10}) + 10^{(P_{\text{PowerClass,NR}} - MPR_{\text{NR}})/10}))$$

where

$$MPR_{\text{E-UTRA}} = \text{MAX}(MPR_{\text{single,E-UTRA}}, MPR_{\text{ENDC}})$$

with

- $MPR_{\text{single,E-UTRA}}$ is the MPR defined for the E-UTRA transmission in TS 36.101 [4]
- $MPR_{\text{single,NR}}$ is the MPR defined for the NR transmission in TS 38.101-1 [2]

For UEs not supporting dynamic power sharing the following

- for the MCG,

$$MPR_c = \text{MAX}(MPR_{\text{single,E-UTRA}}, MPR_{\text{ENDC}})$$

- for the SCG,

$$MPR'_c = \text{MAX}(MPR_{\text{single,NR}}, MPR_{\text{ENDC}})$$

where

- $MPR_{\text{single,NR}}$ is the MPR defined for the NR transmission in TS 38.101-1 [2]
- $MPR_{\text{single,E-UTRA}}$ is the MPR defined for the E-UTRA transmission in TS 36.101 [4]

MPR_{ENDC} is defined in Clause 6.2B.2.2.2

6.2B.2.2.2 MPR for power class 3 and power class 2

MPR in this clause is applicable for power class 3 and power class 2 UEs indicating *IE dualPA-Architecture* supported with ENDC power class being the same as the E-UTRA and NR power class, otherwise the UE can use as much MPR as needed to fulfil emissions requirements when scheduled with dual uplink transmission. For UEs scheduled with single uplink transmission, MPR in clause 6.2.4 of TS 36.101 [4] and 6.2.2 of TS 38.101-1 [2] apply. The allowed maximum output power reduction for IM3

related emissions applied to transmission on the MCG and the SCG is defined as follows:

$$\text{MPR}_{\text{ENDC}} = M_A$$

Where M_A is defined as follows

$$M_A = \begin{array}{ll} 18 & ; \quad 0 \leq B < 1.0 \\ 17 & ; \quad 1.0 \leq B < 2.0 \\ 16 & ; \quad 2.0 \leq B < 5.0 \\ 15 & ; \quad 5.0 \leq B \end{array}$$

Where:

For UEs supporting dynamic power sharing,

$$B = (\text{L}_{\text{CRB_alloc, E-UTRA}} * 12 * \text{SCS}_{\text{E-UTRA}} + \text{L}_{\text{CRB_alloc, NR}} * 12 * \text{SCS}_{\text{NR}}) / 1,000,000$$

For UEs not supporting dynamic power sharing,

For E-UTRA

$$B = (\text{L}_{\text{CRB_alloc, E-UTRA}} * 12 * \text{SCS}_{\text{E-UTRA}} + 12 * \text{SCS}_{\text{NR}}) / 1,000,000$$

Where $\text{SCS}_{\text{NR}} = 15,000$ Hz is assumed in calculation of B.

For NR

$$B = (12 * \text{SCS}_{\text{E-UTRA}} + \text{L}_{\text{CRB_alloc, NR}} * 12 * \text{SCS}_{\text{NR}}) / 1,000,000$$

Where $\text{SCS}_{\text{E-UTRA}} = 15,000$ Hz is assumed in calculation of B.

and M_A is reduced by 1 dB for $B < 2$.

6.2B.2.3 Inter-band EN-DC within FR1

For inter-band EN-DC between E-UTRA and FR1 NR, UE maximum output power reduction specified in TS 36.101 [4] and TS 38.101-1 [2] apply for E-UTRA and NR respectively.

6.2B.2.3a Inter-band NE-DC within FR1

For inter-band NE-DC between E-UTRA and FR1 NR, UE maximum output power reduction specified in TS 36.101 [4] and TS 38.101-1 [2] apply for E-UTRA and NR respectively.

6.2B.2.4 Inter-band EN-DC including FR2

UE maximum output power reduction requirement for E-UTRA single carrier and CA operation specified in clauses 6.2.3 and 6.2.3A of TS 36.101 [4] and for NR single carrier, CA operation and UL-MIMO specified in clause 6.2.2, 6.2A.2, and 6.2D.2 of TS 38.101-2 [3] apply.

6.2B.2.5 Inter-band EN-DC including both FR1 and FR2

UE maximum output power reduction requirement for E-UTRA single carrier and CA operation specified in clauses 6.2.3 and 6.2.3A of TS 36.101 [4] and for NR single carrier specified in clause 6.2.2 of TS 38.101-1 [2] and for NR single carrier, CA operation and UL-MIMO specified in clause 6.2.2, 6.2A.2, and 6.2D.2 of TS 38.101-2 [3] apply.

6.2B.3 UE additional maximum output power reduction for EN-DC

6.2B.3.1 Intra-band contiguous EN-DC

6.2B.3.1.0 General

For intra-band contiguous EN-DC band combinations with additional requirements the allowed A-MPR is specified in Table 6.2B.3.1.0-1 for UEs configured with EN-DC and combinations of network signalling values indicated in the E-UTRA and NR cell groups.

Unless otherwise stated the A-MPR specified in clause 6.2B.3.1 for intra-band contiguous EN-DC configurations is the total power reduction allowed including MPR.

Table 6.2B.3.1.0-1: Additional maximum power reduction for Intra-band contiguous EN-DC

DC configuration	Requirement (clause)	E-UTRA network signalling value	NR network signalling value	A-MPR (clause)
DC_(n)71AA	6.5B.2.1.2.1	NS_35	NS_35	6.2B.3.1.1 ³
DC_(n)41AA ¹	6.5B.2.1.2.2	NS_01 or NS_04	NS_04	6.2B.3.1.2 ⁴
NOTE 1: Only applies to UEs that support dual UL transmission for this EN-DC combination. NOTE 2: The additional emission requirement is indicated when the combination of network signalling values in the two CGs is set (only for UEs configured with EN-DC). NOTE 3: The A-MPR is applied as MPR if NS_35 is not signalled. NOTE 4: Void				

6.2B.3.1.1 A-MPR for DC_(n)71AA

For UE supporting dynamic power sharing the following:

- for the MCG, $A\text{-MPR}_c$ in accordance with TS 36.101 [4]
- for the SCG, $A\text{-MPR}'_c = A\text{-MPR}_{DC}$
- for the total configured transmission power, $A\text{-MPR}_{tot} = A\text{-MPR}_{DC}$

with $A\text{-MPR}_{DC}$ as defined in this clause.

For UEs not supporting dynamic power sharing the following

- for the MCG,

$$A\text{-MPR}_c = A\text{-MPR}_{E\text{-UTRA}}$$

- for the SCG,

$$A\text{-MPR}'_c = A\text{-MPR}_{NR}$$

with $A\text{-MPR}_{E\text{-UTRA}}$ and $A\text{-MPR}_{NR}$ as defined in this clause.

For DC_(n)71AA with configured with network signaling values as per Table 6.2B.3.1.0-1 the allowed A-MPR is defined by

- for UE indicating support of dynamicPowerSharing in the *UE-MRDC-Capability* IE

$$A\text{-MPR}_{DC} = \text{CEIL}\{M_{A,DC}(A), 0.5\}$$

where $A\text{-MPR}_{DC}$ is the total power reduction allowed (dB),

- for OFDM:

$$M_{A,DC} = 11.00 - 11.67 \cdot A; \quad 0.00 < A \leq 0.30$$

$$8.10 - 2.00 \cdot A; \quad 0.30 < A \leq 0.80$$

$$6.50; \quad 0.80 < A \leq 1.00$$

- for DFT-S-OFDM:

$$M_{A,DC} = \begin{cases} 11.00 - 13.33 \cdot A; & 0.00 < A \leq 0.30 \\ 8.00 - 3.33 \cdot A; & 0.30 < A \leq 0.60 \\ 6.00; & 0.60 < A \leq 1.00 \end{cases}$$

where

with $L_{CRB, E-UTRA}$ and $N_{RB, E-UTRA}$ the number of allocated PRB and transmission bandwidth for MCG, $L_{CRB, NR}$ and $N_{RB, NR}$ the number of allocated PRB and transmission bandwidth for SCG with SCS = 15 kHz.

- for UE not indicating support of dynamicPowerSharing

$$A-MPR_{E-UTRA} = \text{CEIL}\{ M_{A,E-UTRA}, 0.5\}$$

$$A-MPR_{NR} = \text{CEIL}\{ M_{A,NR}, 0.5\}$$

where A-MPR is the total power reduction allowed per CG with

Where $L_{CRB, NR}$ and $N_{RB, NR}$ the number of allocated PRB and transmission bandwidth for SCG with SCS = 15 kHz.

6.2B.3.1.2 A-MPR for NS_04

6.2B.3.1.2.0 General

When the UE is configured for B41/n41 intra-band contiguous EN-DC and it receives IE NS_04, the UE determines the total allowed maximum output power reduction as specified in this clause. The A-MPR for EN-DC defined in this clause is used instead of MPR defined in 6.2B.2.1, not additively, so EN-DC MPR = 0 when NS_04 is signaled. For UEs scheduled with single uplink transmission, AMPR in clause 6.2.4 of [4] and 6.2.3 of [2] apply.

For UE supporting dynamic power sharing the following:

- for the MCG, $A-MPR_c$ in accordance with TS 36.101 [4]
- for the SCG,

$$A-MPR'_c = A-MPR_{NR} = \text{MAX}(A-MPR_{\text{single}, NR}, A-MPR_{IM3})$$

- for the total configured transmission power,

$$A-MPR_{\text{tot}} = P_{\text{PowerClass}, EN-DC} - \min(P_{\text{PowerClass}, EN-DC}, 10 \cdot \log_{10}(10^{((P_{\text{PowerClass}, E-UTRA} - A-MPR_{E-UTRA})/10)} + 10^{((P_{\text{PowerClass}, NR} - A-MPR_{NR})/10)}))$$

where

$$A-MPR_{E-UTRA} = \text{MAX}(A-MPR_{\text{single}, E-UTRA} + MPR_{\text{single}, E-UTRA}, A-MPR_{IM3})$$

with

- $A\text{-MPR}_{\text{single, E-UTRA}}$ is the A-MPR defined for the E-UTRA transmission in TS 36.101 [4]
- $A\text{-MPR}_{\text{single, NR}}$ is the A-MPR defined for the NR transmission in TS 38.101-1 [2]
- $\text{MPR}_{\text{single, E-UTRA}}$ is the MPR defined for the E-UTRA transmission in TS 36.101 [4]

For UEs not supporting dynamic power sharing the following

- for the MCG,

$$A\text{-MPR}_c = \text{MAX}(A\text{-MPR}_{\text{single, E-UTRA}} + \text{MPR}_{\text{single, E-UTRA}}, A\text{-MPR}_{\text{IM3}})$$

- for the SCG,

$$A\text{-MPR}'_c = \text{MAX}(A\text{-MPR}_{\text{single, NR}}, A\text{-MPR}_{\text{IM3}})$$

where

- $A\text{-MPR}_{\text{single, E-UTRA}}$ is the A-MPR defined for the E-UTRA transmission in TS 36.101 [4]
- $A\text{-MPR}_{\text{single, NR}}$ is the A-MPR defined for the NR transmission in TS 38.101-1 [2]
- $\text{MPR}_{\text{single, E-UTRA}}$ is the MPR defined for the E-UTRA transmission in TS 36.101 [4]

The UE determines the Channel Configuration Case and the value of $A\text{-MPR}_{\text{IM3}}$ as follows:

If $F_{\text{IM3, low_block, low}} < 2490.5 \text{ MHz}$

Channel Configuration Case B. $A\text{-MPR}_{\text{IM3}}$ defined in Clause 6.2B.3.1.2.2

Else

Channel Configuration Case A. $A\text{-MPR}_{\text{IM3}}$ defined in Clause 6.2B.3.1.2.1

where

- $F_{\text{IM3, low_block, low}} = (2 * F_{\text{low_channel, low_edge}}) - F_{\text{high_channel, high_edge}}$
- $F_{\text{low_channel, low_edge}}$ is the lowermost frequency of lower transmission bandwidth configuration.
- $F_{\text{high_channel, high_edge}}$ is the uppermost frequency of upper transmission bandwidth configuration.

Where the transmission bandwidth configuration for NR is the maximum frequency span covering all the configured SCSSpecificCarrier for scenarios that carrier bandwidths with different SCS can be fully overlapped.

6.2B.3.1.2.1 $A\text{-MPR}_{\text{IM3}}$ for NS_04 to meet -13 dBm / 1MHz for 26dBm UE power

$A\text{-MPR}$ in this clause is relative to 26 dBm for a power class 2 Cell Group. The same $A\text{-MPR}$ is used relative to 23 dBm for a power class 3 Cell Group. For the UE is configured with channel configurations Case A or Case C (defined in Clause 6.2B.3.2.1), the allowed maximum output power reduction for IM3s applied to transmission on the MCG and the SCG with non-contiguous resource allocation is defined as follows:

$$A\text{-MPR}_{\text{IM3}} = M_A$$

Where M_A is defined as follows

3GPP

$$M_A = \begin{array}{ll} 15 & ; \quad 0 \leq B < 0.5 \\ 10 & ; \quad 0.5 \leq B < 1.0 \\ 8 & ; \quad 1.0 \leq B < 2.0 \\ 6 & ; \quad 2.0 \leq B \end{array}$$

Where:

For UEs supporting dynamic power sharing,

$$B = (L_{\text{CRB_alloc, E-UTRA}} * 12 * \text{SCS}_{\text{E-UTRA}} + L_{\text{CRB_alloc, NR}} * 12 * \text{SCS}_{\text{NR}}) / 1,000,000$$

For UEs not supporting dynamic power sharing,

For E-UTRA

$$B = (L_{\text{CRB_alloc, E-UTRA}} * 12 * \text{SCS}_{\text{E-UTRA}} + 12 * \text{SCS}_{\text{NR}}) / 1,000,000$$

Where $\text{SCS}_{\text{NR}} = 15,000$ Hz is assumed in calculation of B.

For NR

$$B = (12 * \text{SCS}_{\text{E-UTRA}} + L_{\text{CRB_alloc, NR}} * 12 * \text{SCS}_{\text{NR}}) / 1,000,000$$

Where $\text{SCS}_{\text{E-UTRA}} = 15,000$ Hz is assumed in calculation of B.

and M_A is reduced by 1 dB for $B < 2.0$.

6.2B.3.1.2.2 A-MPR for NS_04 to meet -25 dBm / 1MHz for 26 dBm UE power

A-MPR in this clause is relative to 26 dBm for a power class 2 Cell Group. The same A-MPR is used relative to 23 dBm for a power class 3 Cell Group. For the UE is configured with channel configurations Case B or Case D (defined in Clause 6.2B.3.2.1), the allowed maximum output power reduction for IM3s applied to transmission on the MCG and the SCG with non-contiguous resource allocation is defined as follows:

$$\text{A-MPR}_{\text{IM3}} = M_A$$

Where M_A is defined as follows

$$M_A = \begin{array}{ll} 15 & ; \quad 0 \leq B < 1.0 \\ 14 & ; \quad 1.0 \leq B < 2.0 \\ 13 & ; \quad 2.0 \leq B < 5.0 \\ 12 & ; \quad 5.0 \leq B \end{array}$$

Where:

For UEs supporting dynamic power sharing,

$$B = (L_{\text{CRB_alloc, E-UTRA}} * 12 * \text{SCS}_{\text{E-UTRA}} + L_{\text{CRB_alloc, NR}} * 12 * \text{SCS}_{\text{NR}}) / 1,000,000$$

For UEs not supporting dynamic power sharing,

For E-UTRA

$$B = (L_{\text{CRB_alloc, E-UTRA}} * 12 * \text{SCS}_{\text{E-UTRA}} + 12 * \text{SCS}_{\text{NR}}) / 1,000,000$$

Where $\text{SCS}_{\text{NR}} = 15,000$ Hz is assumed in calculation of B.

For NR

$$B = (12 * \text{SCS}_{\text{E-UTRA}} + L_{\text{CRB_alloc, NR}} * 12 * \text{SCS}_{\text{NR}}) / 1,000,000$$

Where $SCS_{E-UTRA} = 15,000$ Hz is assumed in calculation of B.

and M_A is reduced by 1 dB.

6.2B.3.2 Intra-band non-contiguous EN-DC

6.2B.3.2.0 General

For intra-band non-contiguous EN-DC band combinations with additional requirements the A-MPR allowed are specified in Table 6.2B.3.2.0-1 for UEs configured with EN-DC and combinations of network signalling values indicated in the E-UTRA and NR cell group(s). Unless otherwise stated the A-MPR specified in clause 6.2B.3.2 for intra-band non-contiguous EN-DC configurations is the total power reduction allowed including MPR. For UEs scheduled with single uplink transmission, AMPR in clause 6.2.4 of [4] and 6.2.3 of [2] apply.

Table 6.2B.3.2.0-1: Allowed power reduction for intra-band non-contiguous EN-DC

DC configuration	Requirement (clause)	E-UTRA network signalling value	NR network signalling value	A-MPR (clause)
DC_41A_n41A ¹	6.6.3.3.19 and 6.6.2.2.2 of [4] and 6.5.2.3.2 and 6.5.3.3.1 of [2]	NS_01 or NS_04	NS_04	6.2B.3.2.1
NOTE 1: Only applies to UEs that support dual UL transmission for this EN-DC combination. NOTE 2: The requirement applies when the combination of network signalling values in the two CGs is set (only for UEs configured with EN-DC).				

6.2B.3.2.1 A-MPR for NS_04

When the UE is configured for B41/n41 intra-band non-contiguous EN-DC and it receives IE NS_04, the UE determines the total allowed maximum output power reduction as specified in this clause. The A-MPR for EN-DC defined in this clause is used instead of MPR defined in 6.2B.2.2, not additively, so EN-DC MPR=0 when NS_04 is signaled.

For UE supporting dynamic power sharing the following:

- for the MCG, $A-MPR_c$ in accordance with TS 36.101 [4]
- for the SCG,

$$A-MPR'_c = A-MPR_{NR} = \text{MAX}(A-MPR_{\text{single},NR}, A-MPR_{EN-DC})$$

- for the total configured transmission power,

$$A-MPR_{\text{tot}} = P_{\text{PowerClass},EN-DC} - \min(P_{\text{PowerClass},EN-DC}, 10 \cdot \log_{10}(10^{((P_{\text{PowerClass},E-UTRA} - A-MPR_{E-UTRA})/10)} + 10^{((P_{\text{PowerClass},NR} - A-MPR_{NR})/10)}))$$

where

$$A-MPR_{E-UTRA} = \text{MAX}(A-MPR_{\text{single},E-UTRA} + MPR_{\text{single},E-UTRA}, A-MPR_{EN-DC})$$

$$A-MPR_{EN-DC} = \text{MAX}(A-MPR_{IM3}, A-MPR_{ACL\text{Roverlap}})$$

with

- $A-MPR_{\text{single},E-UTRA}$ is the A-MPR defined for the E-UTRA transmission in TS 36.101 [4]

- $A\text{-MPR}_{\text{single,NR}}$ is the A-MPR defined for the NR transmission in TS 38.101-1 [2]
- $\text{MPR}_{\text{single,E-UTRA}}$ is the MPR defined for the E-UTRA transmission in TS 36.101 [4]

For UEs not supporting dynamic power sharing the following

- for the MCG,

$$A\text{-MPR}_c = \text{MAX}(A\text{-MPR}_{\text{single, E-UTRA}} + \text{MPR}_{\text{single,E-UTRA}}, A\text{-MPR}_{\text{IM3}}, A\text{-MPR}_{\text{ACLRoverlap}})$$

- for the SCG,

$$A\text{-MPR}'_c = \text{MAX}(A\text{-MPR}_{\text{single,NR}}, A\text{-MPR}_{\text{IM3}}, A\text{-MPR}_{\text{ACLRoverlap}})$$

where

- $A\text{-MPR}_{\text{single, E-UTRA}}$ is the A-MPR defined for the E-UTRA transmission in TS 36.101 [4]
- $A\text{-MPR}_{\text{single,NR}}$ is the A-MPR defined for the NR transmission in TS 38.101-1 [2]
- $\text{MPR}_{\text{single,E-UTRA}}$ is the MPR defined for the E-UTRA transmission in TS 36.101 [4]

The UE determines the Channel Configuration Case and the value of $A\text{-MPR}_{\text{IM3}}$ as follows:

$$\text{If AND}(F_{\text{IM3,low_block,high}} < F_{\text{filter,low}}, \text{MAX}(\text{SEM}_{-13,\text{high}}, F_{\text{IM3,high_block,low}}) > F_{\text{filter,high}})$$

Channel Configuration Case C. $A\text{-MPR}_{\text{IM3}}$ defined in Clause 6.2B.3.1.2.1

Else

Channel Configuration Case D. $A\text{-MPR}_{\text{IM3}}$ defined in Clause 6.2B.3.1.2.2

where

- $F_{\text{IM3,low_block,high}} = (2 * F_{\text{low_channel,high_edge}}) - F_{\text{high_channel,low_edge}}$
- $F_{\text{IM3,high_block,low}} = (2 * F_{\text{high_channel,low_edge}}) - F_{\text{low_channel,high_edge}}$
- $F_{\text{low_channel,low_edge}}$ is the lowermost frequency of lower transmission bandwidth configuration.
- $F_{\text{low_channel,high_edge}}$ is the uppermost frequency of lower transmission bandwidth configuration.
- $F_{\text{high_channel,low_edge}}$ is the lowermost frequency of upper transmission bandwidth configuration.
- $F_{\text{high_channel,high_edge}}$ is the uppermost frequency of upper transmission bandwidth configuration.
- $F_{\text{filter,low}} = 2480 \text{ MHz}$
- $F_{\text{filter,high}} = 2745 \text{ MHz}$
- $\text{SEM}_{-13,\text{high}}$ = Threshold frequency where upper spectral emission mask for upper channel drops from -13 dBm / 1MHz to -25 dBm / 1MHz, as specified in Clause 6.6.2.2.2 in [4] and Clause 6.5.2.3.2 in [2] respectively.

Where the transmission bandwidth configuration for NR is the maximum frequency span covering all the configured SCSSpecificCarrier for scenarios that carrier bandwidths with different SCS can be fully overlapped.

The UE determines the value of $A\text{-MPR}_{\text{ACL}R\text{overlap}}$ as specified in Table 6.2B.3.2.1-1:

Table 6.2B.3.2.1-1: $A\text{-MPR}_{\text{ACL}R\text{overlap}}$

W_{gap}	$A\text{-MPR}_{\text{ACL}R\text{overlap}}$
$< BW_{\text{channel,E-UTRA}} + BW_{\text{channel,NR}}$	4 dB
$\geq BW_{\text{channel,E-UTRA}} + BW_{\text{channel,NR}}$	0 dB
NOTE 1: $W_{\text{gap}} = F_{\text{high_channel,low_edge}} - F_{\text{low_channel,high_edge}}$	

6.2B.3.3 Inter-band EN-DC within FR1

For inter-band EN-DC between E-UTRA and FR1 NR, UE additional maximum output power reduction specified in TS 36.101 [4] and TS 38.101-1 [2] apply for E-UTRA and NR respectively. Unless specified in Table 6.2B.3.3-1, for inter-band carrier aggregation with uplink assigned to LTE and NR bands, the requirements in [2] clause 6.2.3 apply for NR uplink component carrier and the requirements in [4] clause 6.2.4 apply for LTE uplink component carrier.

Unless otherwise stated, for inter-band EN-DC with uplink assigned to LTE and NR bands and specified in Table 6.2B.3.3-1, the combined requirements and allowed A-MPR are applicable on both LTE and NR bands when LTE and NR component carriers are active. The requirements in Table 6.2B.3.3-1 are specified in terms of an additional spectrum emission requirement. The emission requirements specified in Table 6.2B.3.3-1 also apply for the frequency ranges that are less than F_{OoB} (MHz) from the edge of the channel bandwidth specified in TS 36.101 [4] and TS 38.101-1 [2], respectively.

Table 6.2B.3.3-1: Additional Requirements for inter-band EN-DC (two-bands)

NR CA combination	Band	Applied NS	Requirements (clause) (TS 36.101 [4])	Requirements (clause) (TS 38.101-1 [2])	A-MPR (table/clause) (TS 36.101 [4])	A-MPR (table/clause) (TS 38.101-1 [2])	Note
DC_1_n28	1	05	6.6.3.3.1	N/A	Table 6.2.4-1 (NS_05)	N/A	2
	n28	17	N/A	6.5.3.3.2	N/A	N/A	
DC_1_n40	1	05	6.6.3.3.1	N/A	Table 6.2.4-1 (NS_05)	N/A	
DC_1_n77	1	05	6.6.3.3.1	N/A	Table 6.2.4-1 (NS_05)	N/A	1
	n84	05	N/A	6.5.3.3.4	N/A	Clause 6.2.3.4	
		05U	N/A	6.5.3.3.4	N/A	Clause 6.2.3.4	
DC_1_n78 DC_1_n84_ULSUP- TDM_n78	1	05	6.6.3.3.1	N/A	Table 6.2.4-1 (NS_05)	N/A	1
	n84	05	N/A	6.5.3.3.4	N/A	Clause 6.2.3.4	
		05U	N/A	6.5.3.3.4, 6.5.2.4.2	N/A	Clause 6.2.3.4	
DC_1_n79	1	05	6.6.3.3.1	N/A	Table 6.2.4-1 (NS_05)	N/A	1
	n84	05	N/A	6.5.3.3.4	N/A	Clause 6.2.3.4	
		05U	N/A	6.5.3.3.4, 6.5.2.4.2	N/A	Clause 6.2.3.4	
DC_3_n1	n1	05	N/A	6.5.3.3.4	N/A	Clause 6.2.3.4	1
		05U	N/A	6.5.3.3.4, 6.5.2.4.2	N/A	Clause 6.2.3.4	
DC_3_n28	n28	17	N/A	6.5.3.3.2	N/A	N/A	2
DC_19_n77	19	08	6.6.3.3.3	N/A	Table 6.2.4-1 (NS_08)	N/A	
DC_19_n78	19	08	6.6.3.3.3	N/A	Table 6.2.4-1 (NS_08)	N/A	

DC_19_n79	19	08	6.6.3.3.3	N/A	Table 6.2.4-1 (NS_08)	N/A	
DC_21_n77	21	09	6.6.3.3.4	N/A	Table 6.2.4-1 (NS_09)	N/A	
DC_21_n78	21	09	6.6.3.3.4	N/A	Table 6.2.4-1 (NS_09)	N/A	
DC_21_n79	21	09	6.6.3.3.4	N/A	Table 6.2.4-1 (NS_09)	N/A	
DC_28_n77	28	17	6.6.3.3.10	N/A	Table 6.2.4-1 (NS_17)	N/A	2
DC_28_n78	28	17	6.6.3.3.10	N/A	Table 6.2.4-1 (NS_17)	N/A	2
DC_28_n79	28	17	6.6.3.3.10	N/A	Table 6.2.4-1 (NS_17)	N/A	2

NOTE 1: NS_05U, NS_43U and NS_100 can be signalled for NR bands that have UTRA services deployed and protected range is specified in clause 6.5.2.4.2 of TS38.101-1[2] and the requirements in clause 6.5.2.4.2 are only applicable to the signalling band.
NOTE 2: Applicable when the assigned NR carrier is confined within 718 MHz and 748 MHz and when the channel bandwidth used is 5 or 10 MHz.

6.2B.3.3A Inter-band NE-DC within FR1

Unless specified in Table 6.2B.3.3A-1, for inter-band carrier aggregation with uplink assigned to LTE and NR bands, the requirements in [2] clause 6.2.3 apply for NR uplink component carrier and the requirements in [4] clause 6.2.4 apply for LTE uplink component carrier.

Unless otherwise stated, for inter-band EN-DC with uplink assigned to LTE and NR bands and specified in Table 6.2B.3.3A-1, the combined requirements and allowed A-MPR are applicable on both LTE and NR bands when LTE and NR component carriers are active. The requirements in Table 6.2B.3.3A-1 are specified in terms of an additional spectrum emission requirement. The emission requirements specified in Table 6.2B.3.3A-1 also apply for the frequency ranges that are less than F_{OoB} (MHz) from the edge of the channel bandwidth specified in TS 36.101 [4] and TS 38.101-1 [2], respectively.

Table 6.2B.3.3A-1: Additional Requirements for inter-band NE-DC (two-bands)

NR CA combination	Band	Applied NS	Requirements (clause) (TS 36.101 [4])	Requirements (clause) (TS 38.101-1 [2])	A-MPR (table/clause) (TS 36.101 [4])	A-MPR (table/clause) (TS 38.101-1 [2])	Note
DC_n1_28	n1	05	N/A	6.5.3.3.4	N/A	Clause 6.2.3.4	1, 2
		05U	N/A	6.5.3.3.4, 6.5.2.4.2	N/A	Clause 6.2.3.4	
	28	17	6.6.3.3.10	N/A	Table 6.2.4-1 (NS_17)	N/A	

NOTE 1: NS_05U can be signalled for NR bands that have UTRA services deployed and protected range is specified in clause 6.5.2.4.2 of TS38.101-1[2] and the requirements in clause 6.5.2.4.2 are only applicable to the signalling band.
NOTE 2: Applicable when the assigned NR carrier is confined within 718 MHz and 748 MHz and when the channel bandwidth used is 5 or 10 MHz.

6.2B.3.4 Inter-band EN-DC including FR2

UE additional maximum output power reduction requirement for E-UTRA single carrier and CA operation specified in clauses 6.2.4 and 6.2.4A of TS 36.101 [4] and for NR single carrier, CA operation and UL-MIMO specified in clause 6.2.3, 6.2A.3 and 6.2D.3 of TS 38.101-2 [3] apply.

6.2B.3.5 Inter-band EN-DC including both FR1 and FR2

UE additional maximum output power reduction requirement for E-UTRA single carrier and CA operation specified in clauses 6.2.4 and 6.2.4A of TS 36.101 [4] and for NR single carrier specified in clause 6.2.3 of TS 38.101-1 [2] and for NR single carrier, CA operation and UL-MIMO specified in clause 6.2.3, 6.2A.3, and 6.2D.3 of TS 38.101-2 [3] apply.

6.2B.4 Configured output power for DC

6.2B.4.1 Configured output power level

6.2B.4.1.1 Intra-band contiguous EN-DC

The following requirements apply for one component carrier per CG configured for synchronous DC.

For intra-band dual connectivity with one uplink serving cell per CG on E-UTRA and NR respectively, the UE is allowed to set its configured maximum output power $P_{\text{CMAX},c(i)}$ for serving cell $c(i)$ of CG i , $i = 1, 2$, and its total configured maximum transmission power for EN-DC operation = $10\log_{10}()$ with as specified in clause 7.6 of TS 38.213 [10].

The configured maximum output power $P_{\text{CMAX_E-UTRA},c}(p)$ in sub-frame p for the configured E-UTRA uplink carrier shall be set within the bounds:

$$P_{\text{CMAX_L_E-UTRA},c}(p) \leq P_{\text{CMAX_E-UTRA},c}(p) \leq P_{\text{CMAX_H_E-UTRA},c}(p)$$

where $P_{\text{CMAX_L_E-UTRA},c}$ and $P_{\text{CMAX_H_E-UTRA},c}$ are the limits for a serving cell c as specified in TS 36.101 [4] clause 6.2.5 modified by P_{LTE} as follows:

$$P_{\text{CMAX_L_E-UTRA},c} = \text{MIN} \{ \text{MIN}(P_{\text{EMAX},c}, P_{\text{EMAX, EN-DC}}, P_{\text{LTE}}) \cdot \frac{P_{\text{PowerClass, EN-DC}}}{P_{\text{PowerClass, E-UTRA}}} - \Delta P_{\text{PowerClass, EN-DC}}, (P_{\text{PowerClass, E-UTRA}} - \Delta P_{\text{PowerClass, E-UTRA}}) - \text{MAX}(\text{MPR}_c + \text{A-MPR}_c + \Delta T_{\text{IB},c}, \frac{P_{\text{PowerClass, E-UTRA}}}{P_{\text{PowerClass, E-UTRA}}}) \}$$

$$P_{\text{CMAX_H_E-UTRA},c} = \text{MIN} \{ P_{\text{EMAX},c}, P_{\text{EMAX, EN-DC}}, P_{\text{LTE}}, P_{\text{PowerClass, EN-DC}}, P_{\text{PowerClass, E-UTRA}} - \Delta P_{\text{PowerClass, E-UTRA}} \}$$

where

- $P_{\text{EMAX, EN-DC}}$ is the value given by the field *p-maxUE-FR1* of the *RRConnectionReconfiguration-v1530* IE as defined in TS 36.331 [8];
- P_{LTE} is the value given by the field *p-maxEUTRA-r15* of the *RRConnectionReconfiguration-v1510* IE as defined in TS 36.331 [8] which is the same as P_{LTE} in TS 38.213 [10];
- $\Delta T_{\text{C_EUTRA},c} = 1.5$ dB when NOTE 2 in Table 6.2.2-1 of TS 36.101 [4] applies; $\Delta T_{\text{C_EUTRA},c} = 0$ dB otherwise;

and whenever NS_01 is not indicated within CG 1:

- for a UE indicating support of dynamicPowerSharing, the MPR_c and the $A-MPR_c$ are determined in accordance with the DCI of serving cell c of the CG 1 and the specification in clause 6.2.4 of TS 36.101 [4];
- for a UE not indicating support of dynamicPowerSharing, the $A-MPR_c$ is determined in accordance with clause 6.2B.3.1 with parameters applicable for UEs not indicating support of dynamicPowerSharing and $MPR_c = 0$ dB;

and whenever NS_01 is indicated in CG 1:

- for a UE indicating support of dynamicPowerSharing, the MPR_c is determined in accordance with the DCI of serving cell c of the CG 1 and the specification in clause 6.2.4 of TS 36.101 [4];
- for a UE not indicating support of dynamicPowerSharing, the MPR_c is determined in accordance with clause 6.2B.2.1 with parameters applicable for UEs not indicating support of dynamicPowerSharing and $A-MPR_c = 0$ dB;

The configured maximum output power $P_{CMAX,f,c,NR}(q)$ in physical channel q for the configured NR carrier shall be set within the bounds:

$$P_{CMAX_L,f,c,NR}(q) \leq P_{CMAX,f,c,NR}(q) \leq P_{CMAX_H,f,c,NR}(q)$$

where $P_{CMAX_L,f,c,NR}$ and $P_{CMAX_H,f,c,NR}$ are the limits for a serving cell c as specified in clause 6.2.4 of TS 38.101-1 [2] modified as follows:

$$P_{CMAX_L,f,c,NR} = \min \left\{ \min(P_{EMAX,c}, P_{EMAX, EN-DC}, P_{NR}) - \left\lfloor \frac{F_0}{4.4} \right\rfloor T_{C_NR, c}, (P_{PowerClass, EN-DC} - \Delta P_{PowerClass, EN-DC}), (P_{PowerClass, NR} - \Delta P_{PowerClass, NR}) - \max(\max(MPR_c, A-MPR_c) + \Delta T_{IB, c} + \left\lfloor \frac{F_0}{4.4} \right\rfloor T_{C_NR, c} + \Delta T_{RxsRS}, P-MPR_c) \right\}$$

$$P_{CMAX_H,f,c,NR} = \min \{ P_{EMAX,c}, P_{EMAX, EN-DC}, P_{NR}, P_{PowerClass, EN-DC}, P_{PowerClass, NR} - \Delta P_{PowerClass, NR} \}$$

where

- $P_{EMAX, EN-DC}$ is the value given by the field *p-maxUE-FR1* of the *RRCConnectionReconfiguration-v1530* IE as defined in TS 36.331 [8];
- P_{LTE} signalled by RRC as *p-MaxEUTRA-r15* in TS 36.331 [8];
- P_{NR} is the value given by the field *p-NR-FR1* of the *PhysicalCellGroupConfig* IE as defined in [9] and signalled by RRC;
- $\Delta T_{C_E-UTRA, c} = 1.5$ dB when NOTE 2 in Table 6.2.2-1 in TS 36.101 [4] applies for a serving cell c , otherwise $\left\lfloor \frac{F_0}{4.4} \right\rfloor T_{C_E-UTRA, c} = 0$ dB;
- $\left\lfloor \frac{F_0}{4.4} \right\rfloor T_{C_NR, c} = 1.5$ dB when NOTE 3 in Table 6.2.1-1 in TS 38.101-1 [2] applies for a serving cell c , otherwise $\left\lfloor \frac{F_0}{4.4} \right\rfloor T_{C_NR, c} = 0$ dB;
- $\Delta T_{IB, c}$ specified in clause 6.2B.4.2.1 for EN-DC, the individual Power Class defined in table 6.2B.1.1 and any other additional power reductions parameters specified in clauses 6.2B.2 and 6.2B.3 for EN-DC are applicable to $P_{CMAX_E-UTRA, c}$ and $P_{CMAX,f,c,NR}$ evaluations.
- $P_{PowerClass, EN-DC}$ is defined in clause 6.2B.1.1 for intra-band contiguous EN-DC;

- $P_{\text{PowerClass,NR}}$ is the nominal UE power of the power class that the UE supports for the NR band of the EN-DC combination as defined in clause 6.2.1 of 38.101-1 [2];
- $P_{\text{PowerClass,E-UTRA}}$ is the nominal UE power of the power class that the UE supports for the E-UTRA band of the EN-DC combination as defined in clause 6.2.2 of 36.101 [4];
- $\Delta P_{\text{PowerClass,EN-DC}}$ is 3 dB for a power class 2 capable EN-DC UE when LTE UL/DL configuration is 0 or 6; or LTE UL/DL configuration is 1 and special subframe configuration is 0 or 5; $\Delta P_{\text{PowerClass,EN-DC}} = 3$ dB when the IE *p-maxUE-FR1* as defined in TS 36.331 [8] is provided and set to the maximum output power of the default power class or lower; otherwise $\Delta P_{\text{PowerClass,EN-DC}} = 0$ dB;

and whenever NS_01 is not indicated within CG 2:

- for a UE indicating support of dynamicPowerSharing, $A\text{-MPR}_c = A\text{-MPR}'_c$ with $A\text{-MPR}'_c$ determined in accordance with clause 6.2B.3.1 and $\text{MPR}_c = 0$ dB if transmission(s) in subframe p on CG 1 overlap in time with physical channel q on CG 2;
- for a UE indicating support of dynamicPowerSharing, $A\text{-MPR}_c$ is determined in accordance with TS 38.101-1 [2] if transmission(s) in subframe p on CG 1 does not overlap in time with physical channel q on CG 2;
- for a UE not indicating support of dynamicPowerSharing, the $A\text{-MPR}_c$ is determined in accordance with clause 6.2B.3.1 with parameters applicable for UEs not indicating support of dynamicPowerSharing and $\text{MPR}_c = 0$ dB;

and whenever NS_01 is indicated in CG 2.

- for a UE indicating support of dynamicPowerSharing, $\text{MPR}_c = \text{MPR}'_c$ with MPR'_c determined in accordance with clause 6.2B.2.1 and $A\text{-MPR}_c = 0$ dB if transmission(s) in subframe p on CG 1 overlap in time with physical channel q on CG 2;
- for a UE indicating support of dynamicPowerSharing, MPR_c is determined in accordance with TS 38.101-1 [2] if transmission(s) in subframe p on CG 1 does not overlap in time with physical channel q on CG 2;
- for a UE not indicating support of dynamicPowerSharing, the MPR_c is determined in accordance with clause 6.2B.2.1 with parameters applicable for UEs not indicating support of dynamicPowerSharing and $A\text{MPR}_c = 0$ dB;

If the transmissions from NR and E-UTRA do not overlap, then the complete clauses for configured transmitted power for E-UTRA and NR respectively from their own specifications apply with the modifications specified above. The lower value between $P_{\text{PowerClass, EN-DC}}$ or $P_{\text{EMAX, EN-DC}}$ shall not be exceeded at any time by UE.

If the EN-DC UE is not supporting dynamic power sharing, then the complete clauses for configured transmitted power for E-UTRA and NR respectively from their own specifications TS 36.101 [4] and TS 38.101-1 [2] respectively apply with the modifications specified above.

If the UE does not support dynamic power sharing,

$$= \text{MIN} \{ P_{\text{EMAX, EN-DC}}, P_{\text{PowerClass, EN-DC}} - \Delta P_{\text{PowerClass,EN-DC}} \} + 0.3 \text{ dB}$$

For UEs indicating support of dynamicPowerSharing in the *UE-MRDC-Capability* IE the UE can configure the total maximum transmission power within the range

$$P_{\text{EN-DC,tot_L}} \leq P_{\text{EN-DC,tot_H}}$$

where

$$P_{\text{EN-DC,tot_L}}(p,q) = \text{MIN}\{ P_{\text{PowerClass,EN-DC}} - \Delta P_{\text{PowerClass,EN-DC}} - \text{MAX}\{\text{MPR}_{\text{tot}}, \text{A-MPR}_{\text{tot}}\}, P_{\text{EMAX,EN-DC}}\}$$

$$P_{\text{EN-DC,tot_H}}(p,q) = \text{MIN}\{P_{\text{PowerClass,EN-DC}}, P_{\text{EMAX,EN-DC}}\}$$

for sub-frame p on CG 1 overlapping with physical channel q on CG 2 and with MPR_{tot} and $\text{A-MPR}_{\text{tot}}$ in accordance with 6.2B.2.1 and clause 6.2B.3.1, respectively.

The measured total maximum output power P_{UMAX} over both CGs/RATs, measured over the transmission reference time duration is

$$P_{\text{UMAX}} = 10 \log_{10} [p_{\text{UMAX},c,E\text{-UTRA}} + p_{\text{UMAX},f,c,NR}],$$

where $p_{\text{UMAX},c,E\text{-UTRA}}$ and $p_{\text{UMAX},f,c,NR}$ denotes the measured output power of serving cell c for *E-UTRA* and *NR* respectively, expressed in linear scale.

For UEs indicating support of dynamicPowerSharing, the measured total configured maximum output power P_{UMAX} shall be within the following bounds:

$$P_{\text{CMAX_L}} - T_{\text{LOW}}(P_{\text{CMAX_L}}) \leq P_{\text{UMAX}} \leq P_{\text{CMAX_H}} + T_{\text{HIGH}}(P_{\text{CMAX_H}})$$

with the tolerances $T_{\text{LOW}}(P_{\text{CMAX_L}})$ and $T_{\text{HIGH}}(P_{\text{CMAX_H}})$ for applicable values of $P_{\text{CMAX_L}}$ and $P_{\text{CMAX_H}}$ specified in Table 6.2B.4.1.1-2.

When an UL subframe transmission p from E-UTRA overlap with a physical channel q from the NR, then for P_{UMAX} evaluation, the E-UTRA subframe p is taken as reference period T_{REF} and always considered as the reference measurement duration and the following rules are applicable.

T_{REF} and T_{eval} are specified in Table 6.2B.4.1.1-1 when same or different subframes and physical channel durations are used in aggregated carriers. $P_{\text{PowerClass,EN-DC}}$ shall not be exceeded by the UE during any evaluation period of time.

Table 6.2B.4.1.1-1: P_{CMAX} evaluation window

transmission duration	T_{REF}	T_{eval}
Different transmission duration in different RAT carriers	E-UTRA Subframe	$\text{Min}(T_{\text{no_hopping}}, \text{Physical Channel Length})$

For each T_{REF} , the $P_{\text{CMAX_H}}$ is evaluated per T_{eval} and given by the maximum value over the transmission(s) within the T_{eval} as follows:

$$P_{\text{CMAX_H}} = \text{MAX}\{ P_{\text{CMAX_EN-DC_H}}(p,q), P_{\text{CMAX_EN-DC_H}}(p,q+1), \dots, P_{\text{CMAX_EN-DC_H}}(p,q+n) \}$$

where $P_{\text{CMAX_EN-DC_H}}$ are the applicable upper limits for each overlapping scheduling unit pairs (p,q) , $(p, q+1)$, up to $(p, q+n)$ for each applicable T_{eval} duration, where $q+n$ is the last NR UL physical channel overlapping with E-UTRA subframe p .

While $P_{\text{CMAX_L}}$ is computed as follows:

$$P_{\text{CMAX_L}} = \text{MIN} \{ P_{\text{CMAX_EN-DC_L}}(p,q), P_{\text{CMAX_EN-DC_L}}(p,q+1), \dots, P_{\text{CMAX_EN-DC_L}}(p,q+n) \}$$

where $P_{\text{CMAX_EN-DC_L}}$ are the applicable lower limits for each overlapping scheduling unit pairs (p,q) , $(p, q+1)$, up to $(p, q+n)$ for each applicable T_{eval} duration, where $q+n$ is the last NR UL physical channel overlapping with E-UTRA subframe p ,

With

$$P_{\text{CMAX_EN-DC_H}}(p,q) = \text{MIN} \{ 10 \log_{10} [p_{\text{CMAX_H_E-UTRA,c}}(p) + p_{\text{CMAX_H,f,c,NR}}(q)], P_{\text{EMAX, EN-DC}}, P_{\text{PowerClass, EN-DC}} \}$$

And:

$$a = 10 \log_{10} [p_{\text{CMAX_E-UTRA,c}}(p) + p_{\text{CMAX,f,c,NR}}(q)] > P_{\text{EN-DC,tot_L}}$$

$$b = 10 \log_{10} [p_{\text{CMAX_E-UTRA,c}}(p) + p_{\text{CMAX,f,c,NR}}(q) / X_{\text{scale}}] > P_{\text{EN-DC,tot_L}}$$

If $a = \text{FALSE}$ and the configured transmission power spectral density between the MCG and SCG differs by less than 6 dB

$$P_{\text{CMAX_EN-DC_L}}(p,q) = \text{MIN} \{ 10 \log_{10} [p_{\text{CMAX_L_E-UTRA,c}}(p) + p_{\text{CMAX_L,f,c,NR}}(q)], P_{\text{EMAX, EN-DC}}, P_{\text{PowerClass, EN-DC}} - \Delta P_{\text{PowerClass, EN-DC}} \}$$

ELSE If $(a=\text{TRUE})$ AND $(b=\text{FALSE})$ and the configured transmission power spectral density between the MCG and SCG differs by less than 6 dB

$$P_{\text{CMAX_EN-DC_L}}(p,q) = \text{MIN} \{ 10 \log_{10} [p_{\text{CMAX_L_E-UTRA,c}}(p) + p_{\text{CMAX_L,f,c,NR}}(q) / X_{\text{scale}}], P_{\text{EMAX, EN-DC}}, P_{\text{PowerClass, EN-DC}} - \Delta P_{\text{PowerClass, EN-DC}} \}$$

ELSE If $b = \text{TRUE}$ or the transmission power after power scaling spectral density between the MCG and SCG differs by more than 6 dB

$$P_{\text{CMAX_EN-DC_L}}(p,q) = \text{MIN} \{ 10 \log_{10} [p_{\text{CMAX_L_E-UTRA,c}}(p)], P_{\text{EMAX, EN-DC}}, P_{\text{PowerClass, EN-DC}} - \Delta P_{\text{PowerClass, EN-DC}} \}$$

where

- $p_{\text{CMAX_H_E-UTRA,c}}(p)$ is the E-UTRA higher limit of the maximum configured power expressed in linear scale;
- $p_{\text{CMAX_H,f,c,NR}}(q)$ is the NR higher limit of the maximum configured power expressed in linear scale;
- $p_{\text{CMAX_L_E-UTRA,c}}(p)$ is the E-UTRA lower limit of the maximum configured power expressed in linear scale;
- $p_{\text{CMAX_L,f,c,NR}}(q)$ is the NR lower limit of the maximum configured power expressed in linear scale;
- $P_{\text{PowerClass, EN-DC}}$ is defined in clause 6.2B.1.1 for intra-band EN-DC;
- X_{scale} is the linear value of X dB which is configured by RRC and can only take values $[0, 6]$ dB

- $P_{\text{CMAX_E-UTRA},c}(p)$ is the linear value of $P_{\text{CMAX_E-UTRA},c}(p)$, the real configured max power for E-UTRA
- $P_{\text{CMAX},f,c, NR}(q)$ is the linear value of $P_{\text{CMAX},f,c, NR}(q)$, the real configured max power of NR

Table 6.2B.4.1.1-2: P_{CMAX} tolerance for Dual Connectivity E-UTRA-NR

$P_{\text{CMAX}}(\text{dBm})$	Tolerance $T_{\text{LOW}}(P_{\text{CMAX_L}})$ (dB)	Tolerance $T_{\text{HIGH}}(P_{\text{CMAX_H}})$ (dB)
$23 \leq P_{\text{CMAX}} \leq 33$	3.0	2.0
$22 \leq P_{\text{CMAX}} < 23$	5.0	2.0
$21 \leq P_{\text{CMAX}} < 22$	5.0	3.0
$20 \leq P_{\text{CMAX}} < 21$	6.0	4.0
$16 \leq P_{\text{CMAX}} < 20$	5.0	
$11 \leq P_{\text{CMAX}} < 16$	6.0	
$-40 \leq P_{\text{CMAX}} < 11$	7.0	

If the UE supports dynamic power sharing, and when E-UTRA and NR transmissions overlap and the condition (If (a=TRUE) AND (b=FALSE)) is met, SCG shall be transmitted and the following supplementary minimum requirement apply for the measured SCG power, $P_{\text{UMAX},f,c, NR}(q)$, under nominal conditions and unless otherwise stated

$$10\log(p_{\text{CMAX_L},f,c, NR}(q)/X_{\text{scale}}) - T_{\text{LOW}}(10\log(p_{\text{CMAX_L},f,c, NR}(q)/X_{\text{scale}})) \leq P_{\text{UMAX},f,c, NR}(q) \leq 10\log(p_{\text{CMAX_H},f,c, NR}(q)) + T_{\text{HIGH}}(10\log(p_{\text{CMAX_H},f,c, NR}(q))).$$

with the tolerances T_{LOW} and T_{HIGH} for applicable values of P_{CMAX} specified in Table 6.2B.4.1.1-2.

If the UE supports dynamic power sharing, the measured maximum output power in subframe p on CG 1, $p_{\text{UMAX},c,E-UTRA}$, shall meet the requirements in clause 6.2.5 in TS 36.101 [4] with the limits $P_{\text{CMAX_L},c}$ and $P_{\text{CMAX_H},c}$ replaced by $P_{\text{CMAX_L_E-UTRA},c}$ and $P_{\text{CMAX_H_E-UTRA},c}$ as specified above, respectively.

If the configured transmission power spectral density between the MCG and SCG differs by more than 6 dB, then

$$P_{\text{UMAX},f,c, NR}(q) \leq 10\log(p_{\text{CMAX_H},f,c, NR}(q)) + T_{\text{HIGH}}(10\log(p_{\text{CMAX_H},f,c, NR}(q))).$$

6.2B.4.1.2 Intra-band non-contiguous EN-DC

The following requirements apply for one component carrier per CG configured for synchronous DC. The CG(s) are indexed by $j = 1$ for MCG and $j = 2$ for SCG.

The configured maximum output power $P_{\text{CMAX_E-UTRA},c}(p)$ in sub-frame p for the configured E-UTRA uplink carrier shall be set in accordance with clause 6.2B.4.1.1 but where

- for a UE not indicating support of dynamicPowerSharing, the A-MPR_c determined in accordance with clause 6.2B.3.2 with parameters applicable for UEs not indicating support of dynamicPowerSharing and MPR_c = 0 dB;

whenever NS_01 is not indicated within CG 1 while

- for a UE not indicating support of dynamicPowerSharing, the MPR_c determined in accordance with clause 6.2B.2.2 with parameters applicable for UEs not indicating support of dynamicPowerSharing and $A-MPR_c = 0$ dB;

whenever NS_01 is indicated in CG 1.

The configured maximum output power $P_{C_{MAX},f,c,NR}(q)$ in physical channel q for the configured NR carrier shall be set in accordance with clause 6.2B.4.1.1 but where

- for a UE indicating support of dynamicPowerSharing, $A-MPR_c = A-MPR'_c$ with $A-MPR'_c$ determined in accordance with clause 6.2B.3.2 and $MPR_c = 0$ dB if transmission(s) in subframe p on CG 1 overlap in time with physical channel q on CG 2;
- for a UE indicating support of dynamicPowerSharing, $A-MPR_c$ is determined in accordance with TS 38.101-1 [2] if transmission(s) in subframe p on CG 1 does not overlap in time with physical channel q on CG 2;
- for a UE not indicating support of dynamicPowerSharing, the $A-MPR_c$ is determined in accordance with clause 6.2B.3.2 with parameters applicable for UEs not indicating support of dynamicPowerSharing and $MPR_c = 0$ dB;

whenever NS_01 is not indicated in CG 2 while

- for a UE indicating support of dynamicPowerSharing, $MPR_c = MPR'_c$ with MPR'_c determined in accordance with clause 6.2B.2.2 and $A-MPR_c = 0$ dB if transmission(s) in subframe p on CG 1 overlap in time with physical channel q on CG 2;
- for a UE indicating support of dynamicPowerSharing, MPR_c is determined in accordance with TS 38.101-1 [2] if transmission(s) in subframe p on CG 1 does not overlap in time with physical channel q on CG 2;
- for a UE not indicating support of dynamicPowerSharing, the MPR_c is determined in accordance with clause 6.2B.2.2 with parameters applicable for UEs not indicating support of dynamicPowerSharing and $A-MPR_c = 0$ dB;

whenever NS_01 is indicated in CG 2.

For UEs indicating support of dynamicPowerSharing in the *UE-MRDC-Capability IE*, the UE can configure the total transmission power in accordance with clause 6.2B.4.1.1 but with $P_{powerclass,EN-DC}$ the EN-DC power class of the intra-band non-contiguous band combination configured and $A-MPR$ determined in accordance with clause 6.2B.3.2. The total maximum output power P_{UMAX} over both CGs is measured in accordance with clause 6.2B.4.1.1 and shall be within the limits specified in clause 6.2B.4.1.1 but with parameters applicable for the non-contiguous band combination configured.

The maximum output power levels $p_{UMAX,c,E-UTRA}$ and $p_{UMAX,f,c,NR}$ for the CGs are measured in accordance with clause 6.2B.4.1.1 and shall be within the limits specified in clause 6.2B.4.1.1 but with parameters applicable for the non-contiguous band combination configured.

6.2B.4.1.3 Inter-band EN-DC within FR1

For inter-band dual connectivity with one uplink serving cell per CG on E-UTRA and NR respectively, the UE is allowed to set its configured maximum output power $P_{\text{CMAX},c(i),i}$ for serving cell $c(i)$ of CG i , $i = 1, 2$, and its total configured maximum transmission power for EN-DC operation, $= 10\log_{10}()$ with as specified in clause 7.6 of TS 38.213 [10].

The configured maximum output power $P_{\text{CMAX_E-UTRA},c}(p)$ in sub-frame p for the configured E-UTRA uplink carrier shall be set within the bounds:

$$P_{\text{CMAX_L_E-UTRA},c}(p) \leq P_{\text{CMAX_E-UTRA},c}(p) \leq P_{\text{CMAX_H_E-UTRA},c}(p)$$

where $P_{\text{CMAX_L_E-UTRA},c}$ and $P_{\text{CMAX_H_E-UTRA},c}$ are the limits for a serving cell c as specified in TS 36.101 [4] clause 6.2.5 modified by P_{LTE} as follows:

$$P_{\text{CMAX_L_E-UTRA},c} = \text{MIN} \{ P_{\text{EMAX, EN-DC}}, (P_{\text{PowerClass, EN-DC}} - \Delta P_{\text{PowerClass, EN-DC}}), \text{MIN}(P_{\text{EMAX},c}, P_{\text{LTE}}) - \left\lceil \frac{F_0}{4.4} \right\rceil_{\text{C_E-UTRA},c}, (P_{\text{PowerClass, E-UTRA}} - \Delta P_{\text{PowerClass, E-UTRA}}) - \text{MAX}(\text{MPR}_c + \text{A-MPR}_c + \Delta T_{\text{IB},c} + \left\lceil \frac{F_0}{4.4} \right\rceil_{\text{C_E-UTRA},c} + \left\lceil \frac{F_0}{4.4} \right\rceil_{\text{ProSe}}, P\text{-MPR}_c) \}$$

$$P_{\text{CMAX_H_E-UTRA},c} = \text{MIN} \{ P_{\text{EMAX},c}, P_{\text{EMAX, EN-DC}}, (P_{\text{PowerClass, EN-DC}} - \Delta P_{\text{PowerClass, EN-DC}}), P_{\text{LTE}}, P_{\text{PowerClass, E-UTRA}} - \Delta P_{\text{PowerClass, E-UTRA}} \}$$

The configured maximum output power $P_{\text{CMAX},f,c,\text{NR}}(q)$ in physical-channel q for the configured NR carrier shall be set within the bounds:

$$P_{\text{CMAX_L},f,c,\text{NR}}(q) \leq P_{\text{CMAX},f,c,\text{NR}}(q) \leq P_{\text{CMAX_H},f,c,\text{NR}}(q)$$

where $P_{\text{CMAX_L},f,c,\text{NR}}$ and $P_{\text{CMAX_H},f,c,\text{NR}}$ are the limits for a serving cell c as specified in clause 6.2.4 of TS 38.101-1 [2] modified as follows:

$$P_{\text{CMAX_L},f,c,\text{NR}} = \text{MIN} \{ P_{\text{EMAX, EN-DC}}, (P_{\text{PowerClass, EN-DC}} - \Delta P_{\text{PowerClass, EN-DC}}), \text{MIN}(P_{\text{EMAX},c}, P_{\text{NR}}) - \left\lceil \frac{F_0}{4.4} \right\rceil_{\text{C_NR},c}, (P_{\text{PowerClass, NR}} - \Delta P_{\text{PowerClass, NR}}) - \text{MAX}(\text{MAX}(\text{MPR}_c, \text{A-MPR}_c) + \Delta T_{\text{IB},c} + \left\lceil \frac{F_0}{4.4} \right\rceil_{\text{C_NR},c} + \Delta T_{\text{RxsRS}}, P\text{-MPR}_c) \}$$

$$P_{\text{CMAX_H},f,c,\text{NR}} = \text{MIN} \{ P_{\text{EMAX},c}, P_{\text{EMAX, EN-DC}}, (P_{\text{PowerClass, EN-DC}} - \Delta P_{\text{PowerClass, EN-DC}}), P_{\text{NR}}, P_{\text{PowerClass, NR}} - \Delta P_{\text{PowerClass, NR}} \}$$

where

- $P_{\text{EMAX, EN-DC}}$ is the value given by the field *p-maxUE-FR1* of the *RRCCONNECTIONRECONFIGURATION-v1530* IE as defined in TS 36.331 [8];
- P_{LTE} is the value given by the field *p-maxEUTRA-r15* of the *RRCCONNECTIONRECONFIGURATION-v1510* IE as defined in TS 36.331 [8];
- P_{NR} is the value given by the field *p-NR-FR1* of the *PhysicalCellGroupConfig* IE as defined TS 38.331 [9];
- $\Delta T_{\text{C_E-UTRA},c} = 1.5$ dB when NOTE 2 in Table 6.2.2-1 in TS 36.101 [4] applies for a serving cell c , otherwise $\left\lceil \frac{F_0}{4.4} \right\rceil_{\text{C_E-UTRA},c} = 0$ dB;
- $\left\lceil \frac{F_0}{4.4} \right\rceil_{\text{C_NR},c} = 1.5$ dB when NOTE 3 in Table 6.2.1-1 in TS 38.101-1 [2] applies for a serving cell c , otherwise $\left\lceil \frac{F_0}{4.4} \right\rceil_{\text{C_NR},c} = 0$ dB;

- $\Delta T_{IB,c}$ specified in clause 6.2B.4.2.3 for EN-DC, the individual Power Class defined in table 6.2B.1.3 and any other additional power reductions parameters specified in clauses 6.2B.2 and 6.2B.3 for EN-DC are applicable to $P_{CMAX_E-UTRA,c}$ and $P_{CMAX,f,c,NR}$ evaluations.
- $P_{PowerClass, EN-DC}$ is defined in clause 6.2B.1.3 for inter-band EN-DC;
- $P_{PowerClass,NR}$ is the nominal UE power of the power class that the UE supports for the NR band of the EN-DC combination as defined in clause 6.2.1 of 38.101-1 [2];
- $P_{PowerClass,E-UTRA}$ is the nominal UE power of the power class that the UE supports for the E-UTRA band of the EN-DC combination as defined in clause 6.2.2 of 36.101 [4];
- $\Delta P_{PowerClass,EN-DC} = 3$ dB for a power class 2 capable EN-DC UE when IE p-maxUE-FR1, as defined in TS 38.331 [9], is provided and set to the maximum output power of the default power class or lower; otherwise $\Delta P_{PowerClass,EN-DC} = 0$ dB;

If the transmissions from NR and E-UTRA do not overlap, then the complete clauses for configured transmitted power for E-UTRA and NR respectively from their own specifications apply with the modifications specified above. The lower value between $P_{PowerClass, EN-DC}$ or $P_{EMAX, EN-DC}$ shall not be exceeded at any time by UE.

$= 10\log_{10}()$ with the configured maximum transmission power for EN-DC operation as specified in clause 7.6 of TS 38.213 [10].

The total configured maximum transmission power for both synchronous and non-synchronous operation is

$$= \text{MIN} \{ P_{EMAX, EN-DC}, P_{PowerClass, EN-DC} - \Delta P_{PowerClass, EN-DC} \}$$

If the UE does not support dynamic power sharing,

$$= \text{MIN} \{ P_{EMAX, EN-DC}, P_{PowerClass, EN-DC} - \Delta P_{PowerClass, EN-DC} \} + 0.3 \text{ dB}$$

If the EN-DC UE does not support dynamic power sharing, then the complete clauses for configured transmitted power for E-UTRA and NR respectively from their own specifications TS 36.101 [4] and TS 38.101-1 [2] respectively apply with the modifications specified above and applies.

When a UE supporting dynamic sharing is configured for overlapping E-UTRA uplink and NR uplink transmissions, the UE can set its configured maximum output power $P_{CMAX_E-UTRA,c}$ and $P_{CMAX,f,c,NR}$ for the configured E-UTRA and NR uplink carriers, respectively, and its configured maximum transmission power for EN-DC operation, , as specified above.

The measured total maximum output power P_{UMAX} over both CGs/RATs, measured over the transmission reference time duration is

$$P_{UMAX} = 10 \log_{10} [p_{UMAX,c,E-UTRA} + p_{UMAX,c,NR}],$$

where $p_{UMAX,c,E-UTRA}$ and $p_{UMAX,c,NR}$ denotes the measured output power of serving cell c for E-UTRA and NR respectively, expressed in linear scale.

The measured total configured maximum output power P_{UMAX} shall be within the following bounds:

$$P_{\text{CMAX_L}} - T_{\text{LOW}}(P_{\text{CMAX_L}}) \leq P_{\text{UMAX}} \leq P_{\text{CMAX_H}} + T_{\text{HIGH}}(P_{\text{CMAX_H}})$$

with the tolerances $T_{\text{LOW}}(P_{\text{CMAX_H}})$ and $T_{\text{HIGH}}(P_{\text{CMAX_H}})$ for applicable values of P_{CMAX} specified in Table 6.2B.4.1.3-2.

When an UL subframe transmission p from E-UTRA overlap with a physical-channel q from the NR, then for P_{UMAX} evaluation, the E-UTRA subframe p is taken as reference period T_{REF} and always considered as the reference measurement duration and the following rules are applicable.

T_{REF} and T_{eval} are specified in Table 6.2B.4.1.3-1 when same or different subframe and physical-channel durations are used in aggregated carriers. $P_{\text{PowerClass,EN-DC}}$ shall not be exceeded by the UE during any evaluation period of time.

Table 6.2B.4.1.3-1: P_{CMAX} evaluation window

transmission duration	T_{REF}	T_{eval}
Different transmission duration in different RAT carriers	E-UTRA Subframe	$\text{Min}(T_{\text{no_hopping}}, \text{Physical Channel Length})$

For each T_{REF} , the $P_{\text{CMAX_H}}$ is evaluated per T_{eval} and given by the maximum value over the transmission(s) within the T_{eval} as follows:

$$P_{\text{CMAX_H}} = \text{MAX} \{ P_{\text{CMAX_EN-DC_H}}(p,q), P_{\text{CMAX_EN-DC_H}}(p,q+1), \dots, P_{\text{CMAX_EN-DC_H}}(p,q+n) \}$$

where $P_{\text{CMAX_EN-DC_H}}$ are the applicable upper limits for each overlapping scheduling unit pairs (p,q) , $(p, q+1)$, up to $(p, q+n)$ for each applicable T_{eval} duration, where $q+n$ is the last NR UL physical-channel overlapping with E-UTRA subframe p .

While $P_{\text{CMAX_L}}$ is computed as follows:

$$P_{\text{CMAX_L}} = \text{MIN} \{ P_{\text{CMAX_EN-DC_L}}(p,q), P_{\text{CMAX_EN-DC_L}}(p,q+1), \dots, P_{\text{CMAX_EN-DC_L}}(p,q+n) \}$$

where $P_{\text{CMAX_EN-DC_L}}$ are the applicable lower limits for each overlapping scheduling unit pairs (p,q) , $(p, q+1)$, up to $(p, q+n)$ for each applicable T_{eval} duration, where $q+n$ is the last NR UL physical-channel overlapping with E-UTRA subframe p ,

With

$$P_{\text{CMAX_EN-DC_H}}(p,q) = \text{MIN} \{ 10 \log_{10} [p_{\text{CMAX_H_E-UTRA,c}}(p) + p_{\text{CMAX_H,f,c,NR}}(q)], P_{\text{EMAX, EN-DC}}, P_{\text{PowerClass, EN-DC}} \}$$

And:

$$a = 10 \log_{10} [p_{\text{CMAX_E-UTRA,c}}(p) + p_{\text{CMAX,f,c,NR}}(q)] >$$

$$b = 10 \log_{10} [p_{\text{CMAX_E-UTRA,c}}(p) + p_{\text{CMAX,f,c,NR}}(q) / X_{\text{scale}}] >$$

If $a = \text{FALSE}$

$$P_{\text{CMAX_EN-DC_L}}(p,q) = \text{MIN} \{ 10 \log_{10} [p_{\text{CMAX_L_E-UTRA,c}}(p) + p_{\text{CMAX_L,f,c,NR}}(q)], P_{\text{EMAX, EN-DC}}, P_{\text{PowerClass, EN-DC}} \}$$

ELSE If $(a=\text{TRUE})$ AND $(b=\text{FALSE})$

$$P_{\text{CMAX_EN-DC_L}}(p,q) = \text{MIN} \{10 \log_{10} [p_{\text{CMAX_L_E-UTRA,c}}(p) + p_{\text{CMAX_L,f,c,NR}}(q) / X_{\text{scale}}], P_{\text{EMAX, EN-DC}}, P_{\text{PowerClass, EN-DC}}\}$$

ELSE If b= TRUE

$$P_{\text{CMAX_EN-DC_L}}(p,q) = \text{MIN} \{10 \log_{10} [p_{\text{CMAX_L_E-UTRA,c}}(p)], P_{\text{EMAX, EN-DC}}, P_{\text{PowerClass, EN-DC}}\}$$

where

- $P_{\text{CMAX_H_E-UTRA,c}}(p)$ is the E-UTRA higher limit of the maximum configured power expressed in linear scale;
- $P_{\text{CMAX_H,f,c,NR}}(q)$ is the NR higher limit of the maximum configured power expressed in linear scale;
- $P_{\text{CMAX_L_E-UTRA,c}}(p)$ is the E-UTRA lower limit of the maximum configured power expressed in linear scale;
- $P_{\text{CMAX_L,f,c,NR}}(q)$ is the NR lower limit of the maximum configured power expressed in linear scale;
- $P_{\text{PowerClass, EN-DC}}$ is defined in clause 6.2B.1.3-1 for inter-band EN-DC;
- X_{scale} is the linear value of X dB which is configured by RRC and can only take values [0 , 6]
- $P_{\text{CMAX_E-UTRA,c}}(p)$ is the linear value of $P_{\text{CMAX_E-UTRA,c}}(p)$, the real configured max power for E-UTRA
- $P_{\text{CMAX,f,c,NR}}(q)$ is the linear value of $P_{\text{CMAX,f,c,NR}}(q)$, the real configured max power of NR

Table 6.2B.4.1.3-2: P_{CMAX} tolerance for Dual Connectivity E-UTRA-NR

$P_{\text{CMAX}}(\text{dBm})$	Tolerance $T_{\text{LOW}}(P_{\text{CMAX_L}})$ (dB)	Tolerance $T_{\text{HIGH}}(P_{\text{CMAX_H}})$ (dB)
$23 \leq P_{\text{CMAX}} \leq 33$	3.0	2.0
$22 \leq P_{\text{CMAX}} < 23$	5.0	2.0
$21 \leq P_{\text{CMAX}} < 22$	5.0	3.0
$20 \leq P_{\text{CMAX}} < 21$	6.0	4.0
$16 \leq P_{\text{CMAX}} < 20$	5.0	
$11 \leq P_{\text{CMAX}} < 16$	6.0	
$-40 \leq P_{\text{CMAX}} < 11$	7.0	
NOTE 1: For UEs not indicating support of dynamic power sharing, the upper tolerance T_{high} shall be reduced by 0.3 dB for $P \geq 20$ dBm.		

When E-UTRA and NR transmissions overlap and the condition (If (a=TRUE) AND (b=FALSE)) is met, SCG shall be transmitted and the following supplementary minimum requirement apply for the measured SCG power, $P_{\text{UMAX,f,c,NR}}(q)$, under nominal conditions.

$$10\log(p_{\text{CMAX_L,f,c,NR}}(q)/X_{\text{scale}}) - T_{\text{LOW}}(10\log(p_{\text{CMAX_L,f,c,NR}}(q)/X_{\text{scale}})) \leq P_{\text{UMAX,f,c,NR}}(q) \leq 10\log(p_{\text{CMAX_H,f,c,NR}}(q)) + T_{\text{HIGH}}(10\log(p_{\text{CMAX_H,f,c,NR}}(q))).$$

with the tolerances T_{LOW} and T_{HIGH} for applicable values of P_{CMAX} specified in Table 6.2B.4.1.3-2.

6.2B.4.1.3a Inter-band NE-DC within FR1

For inter-band dual connectivity with one uplink serving cell per CG on E-UTRA and NR respectively, the UE is allowed to set its configured maximum output power $P_{\text{CMAX},c(i)}$ for serving cell $c(i)$ of CG i , $i = 1, 2$, and its total configured maximum transmission power for NE-DC operation, = $10\log_{10}()$ with as specified in clause 7.6.1A of TS 38.213 [10].

The configured maximum output power $P_{\text{CMAX_E-UTRA},c}(p)$ in sub-frame p for the configured E-UTRA uplink carrier shall be set within the bounds:

$$P_{\text{CMAX_L_E-UTRA},c}(p) \leq P_{\text{CMAX_E-UTRA},c}(p) \leq P_{\text{CMAX_H_E-UTRA},c}(p)$$

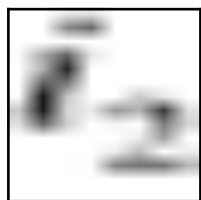
where $P_{\text{CMAX_L_E-UTRA},c}$ and $P_{\text{CMAX_H_E-UTRA},c}$ are the limits for a serving cell c as specified in TS 36.101 [4] clause 6.2.5 modified by P_{LTE} as follows:

$$P_{\text{CMAX_L_E-UTRA},c} = \text{MIN} \left\{ P_{\text{EMAX, NE-DC}}, (P_{\text{PowerClass, NE-DC}} - \Delta P_{\text{PowerClass, NE-DC}}), \text{MIN}(P_{\text{EMAX},c}, P_{\text{LTE}}) - \left[\frac{F_0}{4.4} \right]_{\text{C_E-UTRA}, c}, (P_{\text{PowerClass, E-UTRA}} - \Delta P_{\text{PowerClass, E-UTRA}}) - \text{MAX}(\text{MPR}_c + \text{A-MPR}_c + \Delta T_{\text{IB},c} + \left[\frac{F_0}{4.4} \right]_{\text{C_E-UTRA}, c} + \left[\frac{F_0}{4.4} \right]_{\text{ProSe}}, P\text{-MPR}_c) \right\}$$

$$P_{\text{CMAX_H_E-UTRA},c} = \text{MIN} \{ P_{\text{EMAX},c}, P_{\text{EMAX, NE-DC}}, (P_{\text{PowerClass, NE-DC}} - \Delta P_{\text{PowerClass, NE-DC}}), P_{\text{LTE}}, P_{\text{PowerClass, E-UTRA}} - \Delta P_{\text{PowerClass, E-UTRA}} \}$$

with exception that

- if no symbol of slot i_1 of the NR that is indicated as uplink or flexible by *TDD-UL-DL-ConfigurationCommon* or *TDD-UL-DL-ConfigDedicated* overlaps with subframe i_2 of the E-UTRA; or



if slot(s) that is indicated as downlink by *TDD-UL-DL-ConfigurationCommon* or *TDD-ConfigDedicated* does not overlap with subframe

A; then

$$P_{\text{CMAX_L_E-UTRA},c} = \text{MIN} \left\{ P_{\text{EMAX, NE-DC}}, (P_{\text{PowerClass, NE-DC}} - \Delta P_{\text{PowerClass, NE-DC}}), P_{\text{EMAX},c} - \left[\frac{F_0}{4.4} \right]_{\text{C_E-UTRA}, c}, (P_{\text{PowerClass, E-UTRA}} - \Delta P_{\text{PowerClass, E-UTRA}}) - \text{MAX}(\text{MPR}_c + \text{A-MPR}_c + \Delta T_{\text{IB},c} + \left[\frac{F_0}{4.4} \right]_{\text{C_E-UTRA}, c} + \left[\frac{F_0}{4.4} \right]_{\text{ProSe}}, P\text{-MPR}_c) \right\}$$

$$P_{\text{CMAX_H_E-UTRA},c} = \text{MIN} \{ P_{\text{EMAX},c}, P_{\text{EMAX, NE-DC}}, (P_{\text{PowerClass, NE-DC}} - \Delta P_{\text{PowerClass, NE-DC}}), P_{\text{PowerClass, E-UTRA}} - \Delta P_{\text{PowerClass, E-UTRA}} \}$$

The configured maximum output power $P_{\text{CMAX},f,c,\text{NR}}(q)$ in physical-channel q for the configured NR carrier shall be set within the bounds:

$$P_{\text{CMAX_L},f,c,\text{NR}}(q) \leq P_{\text{CMAX},f,c,\text{NR}}(q) \leq P_{\text{CMAX_H},f,c,\text{NR}}(q)$$

where $P_{\text{CMAX_L},f,c,\text{NR}}$ and $P_{\text{CMAX_H},f,c,\text{NR}}$ are the limits for a serving cell c as specified in clause 6.2.4 of TS 38.101-1 [2] modified by P_{NR} as follows:

$$P_{\text{CMAX_L},f,c,\text{NR}} = \text{MIN} \left\{ P_{\text{EMAX, NE-DC}}, (P_{\text{PowerClass, NE-DC}} - \Delta P_{\text{PowerClass, NE-DC}}), \text{MIN}(P_{\text{EMAX},c}, P_{\text{NR}}) - \left[\frac{F_0}{4.4} \right]_{\text{C_NR}, c}, (P_{\text{PowerClass, NR}} - \Delta P_{\text{PowerClass, NR}}) - \text{MAX}(\text{MPR}_c + \text{A-MPR}_c + \Delta T_{\text{IB},c} + \left[\frac{F_0}{4.4} \right]_{\text{C_NR}, c} + \Delta T_{\text{RxsRS}}, P\text{-MPR}_c) \right\}$$

$$P_{\text{CMAX_H},f,c,\text{NR}} = \text{MIN} \{ P_{\text{EMAX},c}, P_{\text{EMAX, NE-DC}}, (P_{\text{PowerClass, NE-DC}} - \Delta P_{\text{PowerClass, NE-DC}}), P_{\text{NR}}, P_{\text{PowerClass, NR}} - \Delta P_{\text{PowerClass, NR}} \}$$

- $P_{\text{EMAX, NE-DC}}$ signalled by RRC as *p-UE-FR1* in TS 38.331 [9];
- P_{LTE} signalled by RRC as *p-MaxEUTRA* in TS 36.331 [8];
- P_{NR} signalled by RRC as *p-NR-FR1* defined in TS 38.331 [9];

- $-\Delta T_{C_E-UTRA,c} = 1.5\text{dB}$ when NOTE 2 in Table 6.2.2-1 in TS 36.101 [4] applies for a serving cell c , otherwise $\left[\frac{F_0}{4.4}\right] T_{C_E-UTRA,c} = 0\text{dB}$;
- $\left[\frac{F_0}{4.4}\right] T_{C_NR,c} = 1.5\text{dB}$ when NOTE 3 in Table 6.2.1-1 in TS 38.101-1 [2] applies for a serving cell c , otherwise $\left[\frac{F_0}{4.4}\right] T_{C_NR,c} = 0\text{dB}$;
 - $\Delta T_{IB,c}$ specified in clause 6.2B.4.2.3 for NE-DC, the individual Power Class defined in table 6.2B.1.3a and any other additional power reductions parameters specified in clauses 6.2B.2.3a for NE-DC are applicable to $P_{\text{CMAX_E-UTRA},c}$ and $P_{\text{CMAX},f,c,NR}$ evaluations.
 - $P_{\text{PowerClass, NE-DC}}$ is defined in clause 6.2B.1.3a for inter-band NE-DC;
 - $P_{\text{PowerClass,NR}}$ is the nominal UE power of the power class that the UE supports for the NR band of the NE-DC combination as defined in clause 6.2.1 of 38.101-1 [2];
 - $P_{\text{PowerClass,E-UTRA}}$ is the nominal UE power of the power class that the UE supports for the E-UTRA band of the NE-DC combination as defined in clause 6.2.2 of 36.101 [4];
 - $\Delta P_{\text{PowerClass,NE-DC}} = 3\text{ dB}$ for a power class 2 capable NE-DC UE when the IE *p-UE-FR1* as defined in TS 38.331 [9] is provided and set to the maximum output power of the default power class or lower; otherwise $\Delta P_{\text{PowerClass,NE-DC}} = 0\text{ dB}$;

If the transmissions from NR and E-UTRA do not overlap, then the complete clauses for configured transmitted power for E-UTRA and NR respectively from their own specifications apply with the modifications specified above. The lower value between $P_{\text{PowerClass, NE-DC}}$ or $P_{\text{EMAX, NE-DC}}$ shall not be exceeded at any time by UE.

$= 10\log_{10}()$ with the configured maximum transmission power for NE-DC operation as specified in clause 7.6 of TS 38.213 [10].

The total configured maximum transmission power for both synchronous and non-synchronous operation is

$$= \text{MIN} \{ P_{\text{EMAX, NE-DC}}, P_{\text{PowerClass, NE-DC}} - \Delta P_{\text{PowerClass, NE-DC}} \}$$

If the UE does not support dynamic power sharing,

$$= \text{MIN} \{ P_{\text{EMAX, NE-DC}}, P_{\text{PowerClass, NE-DC}} - \Delta P_{\text{PowerClass, NE-DC}} \} + 0.3\text{ dB}$$

If the NE-DC UE does not support dynamic power sharing, then the complete clauses for configured transmitted power for E-UTRA and NR respectively from their own specifications TS 36.101 [4] and TS 38.101-1 [2] respectively apply with the modifications specified above and applies.

When a UE supporting dynamic sharing is configured for overlapping E-UTRA uplink and NR uplink transmissions, the UE can set its configured maximum output power $P_{\text{CMAX_E-UTRA},c}$ and $P_{\text{CMAX},f,c,NR}$ for the configured E-UTRA and NR uplink carriers, respectively, and its configured maximum transmission power for NE-DC operation, as specified above.

The measured total maximum output power P_{UMAX} over both CGs/RATs, measured over the transmission reference time duration is

$$P_{\text{UMAX}} = 10 \log_{10} [p_{\text{UMAX},c,E-UTRA} + p_{\text{UMAX},c,NR}],$$

where $p_{\text{UMAX},c,E-UTRA}$ and $p_{\text{UMAX},c,NR}$ denotes the measured output power of serving cell c for E-UTRA and NR respectively, expressed in linear scale.

The measured total configured maximum output power P_{UMAX} shall be within the following bounds:

$$P_{\text{CMAX_L}} - T_{\text{LOW}}(P_{\text{CMAX_L}}) \leq P_{\text{UMAX}} \leq P_{\text{CMAX_H}} + T_{\text{HIGH}}(P_{\text{CMAX_H}})$$

with the tolerances $T_{\text{LOW}}(P_{\text{CMAX_L}})$ and $T_{\text{HIGH}}(P_{\text{CMAX_H}})$ for applicable values of P_{CMAX} specified in Table 6.2B.4.1.3a-2.

When an UL subframe transmission p from E-UTRA overlap with a physical-channel q from the NR, then for P_{UMAX} evaluation, the E-UTRA subframe p is taken as reference period T_{REF} and always considered as the reference measurement duration and the following rules are applicable.

T_{REF} and T_{eval} are specified in Table 6.2B.4.1.3a-1 when same or different subframe and physical-channel durations are used in aggregated carriers. $P_{\text{PowerClass, NE-DC}}$ shall not be exceeded by the UE during any evaluation period of time.

Table 6.2B.4.1.3a-1: P_{CMAX} evaluation window

transmission duration	T_{REF}	T_{eval}
Different transmission duration in different RAT carriers	LTE Subframe	$\text{Min}(T_{\text{no_hopping}}, \text{Physical Channel Length})$

For each T_{REF} , the $P_{\text{CMAX_H}}$ is evaluated per T_{eval} and given by the maximum value over the transmission(s) within the T_{eval} as follows:

$$P_{\text{CMAX_H}} = \text{MAX} \{ P_{\text{CMAX_NE-DC_H}}(p, q), P_{\text{CMAX_NE-DC_H}}(p, q+1), \dots, P_{\text{CMAX_NE-DC_H}}(p, q+n) \}$$

where $P_{\text{CMAX_NE-DC_H}}$ are the applicable upper limits for each overlapping scheduling unit pairs (p, q) , $(p, q+1)$, up to $(p, q+n)$ for each applicable T_{eval} duration, where $q+n$ is the last NR UL physical-channel overlapping with LTE subframe p .

While $P_{\text{CMAX_L}}$ is computed as follows:

$$P_{\text{CMAX_L}} = \text{MIN} \{ P_{\text{CMAX_NE-DC_L}}(p, q), P_{\text{CMAX_NE-DC_L}}(p, q+1), \dots, P_{\text{CMAX_NE-DC_L}}(p, q+n) \}$$

where $P_{\text{CMAX_NE-DC_L}}$ are the applicable lower limits for each overlapping scheduling unit pairs (p, q) , $(p, q+1)$, up to $(p, q+n)$ for each applicable T_{eval} duration, where $q+n$ is the last NR UL physical-channel overlapping with LTE subframe p .

With

$$P_{\text{CMAX_NE-DC_H}}(p, q) = \text{MIN} \{ 10 \log_{10} [p_{\text{CMAX_H_E-UTRA}, c}(p) + p_{\text{CMAX_H}, f, c, NR}(q)], P_{\text{EMAX, NE-DC}}, P_{\text{PowerClass, NE-DC}} \}$$

And:

$$a = 10 \log_{10} [p_{\text{CMAX_E-UTRA}, c}(p) + p_{\text{CMAX}, f, c, NR}(q)] >$$

If $a = \text{TRUE}$

$$P_{\text{CMAX_NE-DC_L}}(p, q) = \text{MIN} \{ 10 \log_{10} [p_{\text{CMAX_L_E-UTRA}, c}(p)], P_{\text{EMAX, NE-DC}}, P_{\text{PowerClass, NE-DC}} \}$$

Else

$$P_{\text{CMAX_NE-DC_L}}(p, q) = \text{MIN} \{ 10 \log_{10} [p_{\text{CMAX_L_E-UTRA}, c}(p) + p_{\text{CMAX_L}, f, c, NR}(q)], P_{\text{EMAX, NE-DC}}, P_{\text{PowerClass, NE-DC}} \}$$

where

- $p_{\text{CMAX_H_E-UTRA}, c}(p)$ is the E-UTRA higher limit of the maximum configured power expressed in linear scale;
- $p_{\text{CMAX_H}, f, c, NR}(q)$ is the NR higher limit of the maximum configured power expressed in linear scale;
- $p_{\text{CMAX_L_E-UTRA}, c}(p)$ is the E-UTRA lower limit of the maximum configured power expressed in linear scale;

- $P_{\text{CMAX_L},f,c,\text{NR}}(q)$ is the NR lower limit of the maximum configured power expressed in linear scale;
- $P_{\text{PowerClass, NE-DC}}$ is defined in clause 6.2B.1.3a for inter-band NE-DC;
- $P_{\text{CMAX_E-UTRA},c}(p)$ is the linear value of $P_{\text{CMAX_E-UTRA},c}(p)$, the real configured max power for E-UTRA
- $P_{\text{CMAX},f,c,\text{NR}}(q)$ is the linear value of $P_{\text{CMAX},f,c,\text{NR}}(q)$, the real configured max power of NR

Table 6.2B.4.1.3a-2: P_{CMAX} tolerance for Dual Connectivity E-UTRA-NR

P _{CMAX} (dBm)	Tolerance T _{LOW} (P _{CMAX_L}) (dB)	Tolerance T _{HIGH} (P _{CMAX_H}) (dB)
23 ≤ P _{CMAX} ≤ 33	3.0	2.0
22 ≤ P _{CMAX} < 23	5.0	2.0
21 ≤ P _{CMAX} < 22	5.0	3.0
20 ≤ P _{CMAX} < 21	6.0	4.0
16 ≤ P _{CMAX} < 20	5.0	
11 ≤ P _{CMAX} < 16	6.0	
-40 ≤ P _{CMAX} < 11	7.0	
NOTE 1: For UEs not indicating support of dynamic power sharing, the upper tolerance T _{high} shall be reduced by 0.3 dB for P ≥ 20 dBm.		

When E-UTRA and NR transmissions overlap and the condition $a = \text{TRUE}$, $P_{\text{UMAX},f,c,\text{NR}}(q)$ for MCG, under nominal conditions, shall meet

$$P_{\text{UMAX},f,c,\text{NR}}(q) \leq 10\log(P_{\text{CMAX_H},f,c,\text{NR}}(q)) + T_{\text{HIGH}}(10\log(P_{\text{CMAX_H},f,c,\text{NR}}(q))).$$

with the tolerances T_{LOW} and T_{HIGH} for applicable values of P_{CMAX} specified in Table 6.2B.4a.1.3-2.

When LTE and NR transmissions overlap and the condition $a = \text{FALSE}$), then P_{UMAX} , under nominal conditions, shall be within the following bounds:

$$P_{\text{CMAX_L}} - T_{\text{LOW}}(P_{\text{CMAX_L}}) \leq P_{\text{UMAX}} \leq P_{\text{CMAX_H}} + T_{\text{HIGH}}(P_{\text{CMAX_H}})$$

where $P_{\text{CMAX_L}}$, $P_{\text{CMAX_H}}$, and P_{UMAX} are specified above with the tolerances T_{LOW} and T_{HIGH} specified in Table 6.2B.4a.1.3-2 for applicable values of $P_{\text{CMAX_L}}$ and $P_{\text{CMAX_H}}$.

6.2B.4.1.4 Inter-band EN-DC including FR2

For inter-band dual connectivity with one uplink serving cell per CG on E-UTRA and NR respectively, with NR configured in FR2, the UE is allowed to set its configured maximum output power $P_{\text{CMAX},c(i),i}$ for serving cell $c(i)$ of CG i , $i = 1, 2$.

The UE maximum configured power $P_{\text{CMAX},c(i)}$, on E-UTRA for the subframe i shall be set according to clause 6.2.5 from TS 36.101 [4]. Applicable inter-band $\Delta T_{\text{IB},c}$ parameters shall be used according to the clauses 6.2B.4.2.4 or 6.2B.4.2.5.

The UE maximum configured power $P_{\text{CMAX},c(j)}$, on NR for the slot j shall be set according to subclass 6.2.4 from TS 38.101-2 [3].

For the configured power measurements TS 36.101 [4] clause 6.2.5 and TS 38.101-2 [3] clause 6.2.4 are applicable.

6.2B.4.1.5 Inter-band EN-DC including both FR1 and FR2

For inter-band dual connectivity with one uplink serving cell per CG on E-UTRA and NR respectively, with both CGs configured in FR1, the requirements specified in clause 6.2B.4.1.3 apply.

For inter-band dual connectivity with one uplink serving cell per CG on E-UTRA and NR respectively, with NR configured in FR2, the requirements specified in clause 6.2B.4.1.4 apply.

For inter-band dual connectivity with one uplink serving cell in first CG on E-UTRA and two uplink serving cells in second CG on NR FR1 and NR FR2 respectively, the UE is allowed to set its configured maximum output power $P_{\text{CMAX},c(i),i}$ for serving cell $c(i)$, $i = 1, 2, 3$ with $i=1$ for E-UTRA, $i=2$ for NR FR1 and $i=3$ for NR FR2.

- For serving cell on FR2, the requirements specified in clause 6.2.4 in TS 38.101-2 [3] apply to the UE maximum configured power $P_{\text{CMAX},c(3),3}$ and the measured maximum configured power.
- For remaining inter-band dual connectivity involving CG1 and CG2, the requirements specified in clause 6.2B.4.1.3 apply.

6.2B.4.2 $\Delta T_{\text{IB},c}$ for DC

6.2B.4.2.0 General

For the UE which supports inter-band EN-DC or NE-DC configuration, $\Delta T_{\text{IB},c}$ in Tables below applies where unless otherwise stated, the same $\Delta T_{\text{IB},c}$ is applicable to NR band(s) part for DC configurations which have the same NR operating band combination. Unless otherwise stated, $\Delta T_{\text{IB},c}$ is set to zero.

Unless $\Delta T_{\text{IB},c}$ is specified for the NE-DC configuration, the specified $\Delta T_{\text{IB},c}$ for the EN-DC configuration including same bands as the corresponding NE-DC configuration is applicable for the NE-DC configuration.

6.2B.4.2.1 Intra-band contiguous EN-DC

$\Delta T_{\text{IB},c}$ is not applicable for intra-band contiguous EN-DC.

6.2B.4.2.2 Intra-band non-contiguous EN-DC

$\Delta T_{\text{IB},c}$ is not applicable for intra-band non-contiguous EN-DC.

6.2B.4.2.3 Inter-band EN-DC within FR1

6.2B.4.2.3.1 $\Delta T_{\text{IB},c}$ for EN-DC two bands

Table 6.2B.4.2.3.1-1: $\Delta T_{\text{IB},c}$ due to EN-DC(two bands)

Inter-band EN-DC configuration	E-UTRA or NR Band	$\Delta T_{\text{IB},c}$ (dB)
DC_1_n28	1	0.3
	n28	0.6
DC_1_n40	1	0.5
	n40	0.5
DC_1_n51	1	0.6
	n51	0.6
DC_1_n77	1	0.6
	n77	0.8
DC_1_n78	1	0.3
	n78	0.8
DC_2_n5	2	0.3

	n5	0.3
DC_2_n66	2	0.5
	n66	0.5
DC_2_n71	2	0.3
	n71	0.3
DC_2_n78	2	0.6
	n78	0.8
DC_3_n7	3	0.5
	n7	0.5
DC_3_n28	3	0.3
	n28	0.3
DC_3_n40	3	0.5
	n40	0.5
DC_3_n51	3	0.3
	n51	0.3
DC_3_n77	3	0.6
	n77	0.8
DC_3_n78	3	0.6
	n78	0.8
DC_5_n40	5	0.3
	n40	0.3
DC_5_n66	5	0.3
	n66	0.3
DC_5_n78	5	0.6
	n78	0.8
DC_7_n28	7	0.3
	n28	0.3
DC_7_n51	7	0.3
	n51	0.3
DC_7_n78 DC_7-7_n78	7	0.5
	n78	0.8
DC_8_n40	8	0.3
	n40	0.3
DC_8_n77	8	0.6
	n77	0.8
DC_8_n78	8	0.6
	n78	0.8
DC_11_n77	11	0.4
	n77	0.8
DC_11_n78	11	0.4
	n78	0.8
DC_12_n5	12	0.4
	n5	0.8
DC_12_n66	12	0.8
	n66	0.3
DC_18_n77	18	0.3
	n77	0.8
DC_18_n78	18	0.3
	n78	0.8

DC_19_n77	19	0.3
	n77	0.8
DC_19_n78	19	0.3
	n78	0.8
DC_20_n8	20	0.4
	n8	0.4
DC_20_n28	20	0.5
	n28	0.5
DC_20_n51	20	0.5
	n51	0.5
DC_20_n77	20	0.6
	n77	0.8
DC_20_n78	20	0.6
	n78	0.8
DC_21_n77	21	0.4
	n77	0.8
DC_21_n78	21	0.4
	n78	0.8
DC_25_n41	25	0.5
	n41	0.4 ¹
		0.9 ²
DC_26_n41	26	0.3
	n41	0.3
DC_26_n77	26	0.3
	n77	0.8
DC_26_n78	26	0.3
	n78	0.8
DC_28_n51	28	0.5
	n51	0.5
DC_28_n77	28	0.5
	n77	0.8
DC_28_n78	28	0.5
	n78	0.8
DC_30_n5	30	0.3
	n5	0.3
DC_30_n66	30	0.5
	n66	0.8
DC_38_n78	n78	0.5
DC_39_n78	39	0.3
	n78	0.8
DC_39_n79	39	0.3
	n79	0.8
DC_40_n77	n77	0.5
DC_41_n77	41	0.3
	n77	0.8
DC_41_n78	41	0.3
	n78	0.8
DC_41_n79	41	0.3
	n79	0.8

DC_42_n51	42	0.6
	n51	0.8
DC_66_n5	66	0.3
	n5	0.3
DC_66_n71	66	0.3
	n71	0.3
DC_66_n78	66	0.6
	n78	0.8
NOTE 1: The requirement is applied for UE transmitting on the frequency range of 2545-2690 MHz.		
NOTE 2: The requirement is applied for UE transmitting on the frequency range of 2496-2545 MHz.		

6.2B.4.2.3.2 $\Delta T_{IB,c}$ for EN-DC three bands

Table 6.2B.4.2.3.2-1: $\Delta T_{IB,c}$ due to EN-DC (three bands)

Inter-band EN-DC configuration	E-UTRA or NR Band	$\Delta T_{IB,c}$ (dB)
DC_1-3_n28	1	0.3
	3	0.3
	n28	0.6
DC_1-3_n77	1	0.6
	3	0.6
	n77	0.8
DC_1-3_n78	1	0.6
	3	0.6
	n78	0.8
DC_1-3_n79	1	0.3
	3	0.3
DC_1-5_n78	1	0.3
	5	0.6
	n78	0.8
DC_1-7_n28	1	0.5
	7	0.6
	n28	0.6
DC_1-7_n78 DC_1-7-7_n78	1	0.6
	7	0.6
	n78	0.8
DC_1-8_n78	1	0.3
	8	0.6
	n78	0.8
DC_1-18_n77	1	0.3
	18	0.3
	n77	0.8
DC_1-18_n78	1	0.3
	18	0.3
	n78	0.8
DC_1-19_n77	1	0.3
	19	0.3
	n77	0.8
DC_1-19_n78	1	0.3
	19	0.3
	n78	0.8
DC_1-19_n79	1	0.3
	19	0.3

DC_1-20_n28	1	0.3
	20	0.6
	n28	0.6
DC_1-20_n78	1	0.3
	20	0.3
	n78	0.8
DC_1-21_n77	1	0.3
	21	0.3
	n77	0.8
DC_1-21_n78	1	0.6
	21	0.4
	n78	0.8
DC_1-21_n79	1	0.3
	21	0.3
DC_1-28_n77	1	0.3
	28	0.6
	n77	0.8
DC_1-28_n78	1	0.3
DC_1_n28-n78	28 or n28	0.6
	n78	0.8
DC_1_n28-n79	1	0.3
	n28	0.3
DC_1-41_n77	1	0.5
	41	0.5
	n77	0.8
DC_1-41_n78	1	0.5
	41	0.5
	n78	0.8
DC_1-41_n79	1	0.5
	41	0.5
DC_1-42_n77	1	0.6
	42	0.8
	n77	0.8
DC_1-42_n78	1	0.3
	42	0.8
	n78	0.8
DC_1-42_n79	1	0.3
	42	0.8
DC_1_n77-n79	1	0.6
	n77	0.8
DC_1_SUL_n78-n84	1	0.3
	n78	0.8
	n84	0.3
DC_1_n78-n79	1	0.3
	n78	0.8
	n79	0.5
DC_2-5_n66	2	0.5
	5	0.3
	n66	0.5
DC_2-30_n66	2	0.5
	30	0.3
	n66	0.5
DC_2-66_n71	2	0.5
	66	0.5
	n71	0.3
DC_2-(n)71	2	0.3

	71	0.3
	n71	
DC_3_n3-n77	3	0.6
	n3	0.6
	n77	0.8
DC_3_n3-n78	3	0.6
	n3	0.6
	n78	0.8
DC_3-5_n78	3	0.6
	5	0.6
	n78	0.8
DC_3-7_n28	3	0.5
	7	0.5
	n28	0.3
DC_3-7_n78, DC_3-7-7_n78	3	0.6
	7	0.6
	n78	0.8
DC_3-8_n78	3	0.6
	8	0.6
	n78	0.8
DC_3-19_n77	3	0.6
	19	0.3
	n77	0.8
DC_3-19_n78	3	0.6
	19	0.3
	n78	0.8
DC_3-19_n79	3	0.3
	19	0.3
DC_3-20_n28	3	0.3
	20	0.5
	n28	0.5
DC_3-20_n78	3	0.5
	20	0.3
	n78	0.8
DC_3-21_n77	3	0.8
	21	0.9
	n77	0.8
DC_3-21_n78	3	0.8
	21	0.9
	n78	0.8
DC_3-21_n79	3	0.8
	21	0.9
DC_3-28_n78	3	0.5
	28	0.3
	n78	0.8
DC_3_n28-n78	3	0.5
	n28	0.3
	n78	0.8
DC_3-38_n78	3	0.6
	n78	0.8
DC_3-41_n78	3	0.6
	41	0.3 ¹
		0.8 ²

	n78	0.8
DC_3-42_n77	3	0.6
	42	0.8
	n77	0.8
DC_3-42_n78	3	0.6
	42	0.8
	n78	0.8
DC_3-42_n79	3	0.6
	42	0.8
DC_3_n77-n79	3	0.6
	n77	0.8
DC_3_n78-n79	3	0.6
	n78	0.8
	n79	0.5
DC_3_SUL_n78-n80	3	0.6
	n78	0.8
	n80	0.6
DC_3_SUL_n78-n82	3	0.5
	n78	0.8
	n82	0.3
DC_5-7_n78, DC_5-7-7_n78	5	0.6
	7	0.6
	n78	0.8
DC_5-30_n66	5	0.3
	30	0.3
	n66	0.5
DC_7-20_n28	7	0.3
	20	0.6
	n28	0.6
DC_7-20_n78	7	0.3
	20	0.3
	n78	0.8
DC_7-28_n78	7	0.3
	28	0.3
	n78	0.8
DC_7_n28-n78	7	0.3
	n28	0.3
	n78	0.8
DC_7-46_n78	7	0.5
	n78	0.8
DC_8_SUL_n78- n81	8	0.6
	n78	0.8
	n81	0.6
DC_18-28_n77	18	0.5
	28	0.5
	n77	0.8
DC_18-28_n78	18	0.5
	28	0.5
	n78	0.8
DC_18-28_n79	18	0.5
	28	0.5
DC_19-21_n77	19	0.3
	21	0.4
	n77	0.8
DC_19-21_n78	19	0.3

	21	0.4
	n78	0.8
DC_19-21_n79	19	0.3
	21	0.4
DC_19-42_n77	19	0.3
	42	0.8
	n77	0.8
DC_19-42_n78	19	0.3
	42	0.8
	n78	0.8
DC_19-42_n79	19	0.3
	42	0.8
DC_19_n77-n79	19	0.3
	n77	0.8
DC_19_n78-n79	19	0.3
	n78	0.8
	n79	0.5
DC_20_n8-n75	20	0.4
	n8	0.4
DC_20_n28-n75	20	0.5
	n28	0.7
DC_20_n28-n78	20	0.6
	n28	0.6
	n78	0.8
DC_20_n75-n78	20	0.5
	n78	0.8
DC_20_n76-n78	20	0.5
	n78	0.8
DC_20_SUL_n78-n82	20	0.6
	n78	0.8
	n82	0.6
DC_20_SUL_n78-n83	20	0.8
	n78	0.8
	n83	0.8
DC_21-42_n77	21	0.4
	42	0.8
	n77	0.8
DC_21-42_n78	21	0.4
	42	0.8
	n78	0.8
DC_21-42_n79	21	0.4
	42	0.8
DC_21_n77-n79	21	0.4
	n77	0.8
DC_21_n78-n79	21	0.4
	n78	0.8
	n79	0.5
DC_28-42_n77	28	0.5
	42	0.8
	n77	0.8
DC_28-42_n78	28	0.5
	42	0.8
	n78	0.8

DC_28-42_n79	28	0.5
	42	0.8
DC_28_SUL_n78-n83	28	0.5
	n78	0.8
	n83	0.5
DC_41-42_n77	41	0.5
	42	0.8
	n77	0.8
DC_41-42_n78	41	0.5
	42	0.8
	n78	0.8
DC_41-42_n79	41	0.3
	42	0.8
DC_66_(n)71	66	0.3
	71	0.3
	n71	0.3
DC_66_SUL_n78-n86	66	0.6
	n78	0.8
	n86	0.6
NOTE 1: The requirement is applied for UE transmitting on the frequency range of 2545-2690 MHz.		
NOTE 2: The requirement is applied for UE transmitting on the frequency range of 2496-2545 MHz.		

6.2B.4.2.3.3 $\Delta T_{IB,c}$ for EN-DC four bands

Table 6.2B.4.2.3.3-1: $\Delta T_{IB,c}$ due to EN-DC(four bands)

Inter-band EN-DC configuration	E-UTRA or NR Band	$\Delta T_{IB,c}$ (dB)
DC_1-3-5_n78	1	0.6
	3	0.6
	5	0.3
	n78	0.8
DC_1-3-7_n28	1	0.6
	3	0.6
	7	0.6
	n28	0.6
DC_1-3-7_n78 DC_1-3-7-7_n78	1	0.7
	3	0.7
	7	0.7
	n78	0.8
DC_1-3-8_n78	1	0.6
	3	0.6
	8	0.6
	n78	0.8
DC_1-3-19_n78	1	0.6
	3	0.6
	19	0.3
	n78	0.8
DC_1-3-19_n79	1	0.3
	3	0.3
	19	0.3
DC_1-3-20_n28	1	0.3
	3	0.3
	20	0.6
	n28	0.6

DC_1-3-20_n78	1	0.6
	3	0.6
	20	0.3
	n78	0.8
DC_1-3-21_n77	1	0.6
	3	0.8
	21	0.9
	n77	0.8
DC_1-3-21_n78	1	0.6
	3	0.8
	21	0.9
	n78	0.8
DC_1-3-21_n79	1	0.3
	3	0.8
	21	0.9
DC_1-3-28_n77	1	0.6
	3	0.6
	28	0.6
	n77	0.8
DC_1-3-28_n78 DC_1-3_n28-n78	1	0.6
	3	0.6
	28 or n28	0.6
	n78	0.8
DC_1-3-28_n79	1	0.6
	3	0.6
	28	0.6
DC_1-3-42_n77	1	0.6
	3	0.6
	42	0.8
	n77	0.8
DC_1-3-42_n78	1	0.6
	3	0.6
	42	0.8
	n78	0.8
DC_1-3-42_n79	1	0.6
	3	0.6
	42	0.8
DC_1-5-7_n78 DC_1-5-7-7_n78	1	0.6
	5	0.6
	7	0.6
	n78	0.8
DC_1-7-20_n28	1	0.5
	7	0.6
	20	0.6
	n28	0.6
DC_1-7-20_n78	1	0.6
	7	0.7
	20	0.4
	n78	0.8
DC_1-7_n28-n78	1	0.6
	7	0.6
	n28	0.6
	n78	0.8

DC_1-18-28_n77	1	0.3
	18	0.5
	28	0.5
	n77	0.8
DC_1-18-28_n78	1	0.3
	18	0.5
	28	0.5
	n78	0.8
DC_1-18-28_n79	1	0.3
	18	0.5
	28	0.5
DC_1-19-42_n77	1	0.6
	19	0.3
	42	0.8
	n77	0.8
DC_1-19-42_n78	1	0.3
	19	0.3
	42	0.8
	n78	0.8
DC_1-19-42_n79	1	0.3
	19	0.3
	42	0.8
DC_1-20_n28-n78	1	0.3
	20	0.6
	n28	0.6
	n78	0.8
DC_1-21-28_n77	1	0.6
	21	0.4
	28	0.6
	n77	0.8
DC_1-21-28_n78	1	0.3
	21	0.4
	28	0.6
	n78	0.8
DC_1-21-28_n79	1	0.3
	21	0.4
	28	0.6
DC_1-21-42_n77	1	0.6
	21	0.4
	42	0.8
	n77	0.8
DC_1-21-42_n78	1	0.3
	21	0.4
	42	0.8
	n78	0.8
DC_1-21-42_n79	1	0.3
	21	0.4
	42	0.8
DC_1-28-42_n77	1	0.6
	28	0.6
	42	0.8
	n77	0.8
DC_1-28-42_n78	1	0.3
	28	0.6
	42	0.8
	n78	0.8
DC_1-28-42_n79	1	0.3

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	28	0.6
	42	0.8
DC_1-41-42_n77	1	0.5
	41	0.5
	42	0.8
	n77	0.8
DC_1-41-42_n78	1	0.5
	41	0.5
	42	0.8
	n78	0.8
DC_1-41-42_n79	1	0.5
	41	0.5
	42	0.8
DC_2-66-(n)71	2	0.5
	66	0.5
	71	0.3
	n71	
DC_3-5-7_n78, DC_3-5-7-7_n78	3	0.6
	5	0.6
	7	0.6
	n78	0.8
DC_3-7-20_n28	3	0.5
	7	0.5
	20	0.6
	n28	0.5
DC_3-7-20_n78	3	0.6
	7	0.6
	20	0.3
	n78	0.8
DC_3-7-28_n78	3	0.6
	7	0.6
	28	0.6
	n78	0.8
DC_3-7_n28-n78	3	0.6
	7	0.6
	n28	0.6
	n78	0.8
DC_3-19-21_n77	3	0.8
	19	0.3
	21	0.9
	n77	0.8
DC_3-19-21_n78	3	0.8
	19	0.3
	21	0.9
	n78	0.8
DC_3-19-21_n79	3	0.8
	19	0.3
	21	0.9
DC_3-19-42_n77	3	0.6
	19	0.3
	42	0.8
	n77	0.8
DC_3-19-42_n78	3	0.6
	19	0.3

	42	0.8
	n78	0.8
DC_3-19-42_n79	3	0.6
	19	0.3
	42	0.8
DC_3-20_n28-n78	3	0.6
	20	0.6
	n28	0.6
	n78	0.8
DC_3-21-42_n77	3	0.8
	21	0.9
	42	0.8
	n77	0.8
DC_3-21-42_n78	3	0.8
	21	0.9
	42	0.8
	n78	0.8
DC_3-21-42_n79	3	0.8
	21	0.9
	42	0.8
DC_3-28-42_n77	3	0.6
	28	0.5
	42	0.8
	n77	0.8
DC_3-28-42_n78	3	0.6
	28	0.5
	42	0.8
	n78	0.8
DC_3-28-42_n79	3	0.6
	28	0.5
	42	0.8
DC_7-20_n28-n78	7	0.3
	20	0.6
	n28	0.6
	n78	0.8
DC_19-21-42_n77	19	0.3
	21	0.4
	42	0.8
	n77	0.8
DC_19-21-42_n78	19	0.3
	21	0.4
	42	0.8
	n78	0.8
DC_19-21-42_n79	19	0.3
	21	0.4
	42	0.8
DC_21-28-42_n77	21	0.4
	28	0.5
	42	0.8
	n77	0.8
DC_21-28-42_n78	21	0.4
	28	0.5
	42	0.8
	n78	0.8
DC_21-28-42_n79	21	0.4
	28	0.5

	42	0.8
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6.2B.4.2.3.4 $\Delta T_{IB,c}$ for EN-DC five bands

Table 6.2B.4.2.3.4-1: $\Delta T_{IB,c}$ due to EN-DC (five bands)

Inter-band EN-DC configuration	E-UTRA or NR Band	$\Delta T_{IB,c}$ (dB)
DC_1-3-5-7_n78, DC_1-3-5-7-7_n78	1	0.6
	3	0.6
	5	0.6
	7	0.6
	n78	0.8
DC_1-3-7-20_n28	1	0.6
	3	0.6
	7	0.6
	20	0.6
	n28	0.6
DC_1-3-7-20_n78	1	0.6
	3	0.6
	7	0.6
	20	0.6
	n78	0.6
DC_1-3-7_n28-n78	1	0.7
	3	0.7
	7	0.7
	n28	0.6
	n78	0.8
DC_1-3-19-21_n77	1	0.6
	3	0.8
	19	0.3
	21	0.9
	n77	0.8
DC_1-3-19-21_n78	1	0.6
	3	0.8
	19	0.3
	21	0.9
	n78	0.8
DC_1-3-19-21_n79	1	0.3
	3	0.8
	19	0.3
	21	0.9
DC_1-3-19-42_n77	1	0.6
	3	0.6
	19	0.3
	42	0.8
	n77	0.8
DC_1-3-19-42_n78	1	0.6
	3	0.6
	19	0.3
	42	0.8
	n78	0.8
DC_1-3-19-42_n79	1	0.6
	3	0.6

	19	0.3
	42	0.8
DC_1-3-20_n28-n78	1	0.6
	3	0.6
	20	0.6
	n28	0.6
	n78	0.8
DC_1-3-21-42_n77	1	0.6
	3	0.8
	21	0.9
	42	0.8
	n77	0.6
DC_1-3-21-42_n78	1	0.6
	3	0.8
	21	0.9
	42	0.8
	n78	0.6
DC_1-3-21-42_n79	1	0.6
	3	0.8
	21	0.9
	42	0.8
DC_1-3-28-42_n77	1	0.6
	3	0.6
	28	0.6
	42	0.8
	n77	0.8
DC_1-3-28-42_n78	1	0.6
	3	0.6
	28	0.6
	42	0.8
	n78	0.8
DC_1-3-28-42_n79	1	0.6
	3	0.6
	28	0.6
	42	0.8
DC_1-7-20_n28-n78	1	0.6
	7	0.7
	20	0.6
	n28	0.6
	n78	0.8
DC_1-19-21-42_n77	1	0.3
	19	0.3
	21	0.4
	42	0.8
	n77	0.8
DC_1-19-21-42_n78	1	0.3
	19	0.3
	21	0.4
	42	0.8
	n78	0.8
DC_1-19-21-42_n79	1	0.3
	19	0.3
	21	0.4
	42	0.8
DC_1-21-28-42_n77	1	0.6
	21	0.4

	28	0.6
	42	0.8
	n77	0.8
DC_1-21-28-42_n78	1	0.3
	21	0.4
	28	0.6
	42	0.8
	n78	0.8
DC_1-21-28-42_n79	1	0.3
	21	0.4
	28	0.6
	42	0.8
DC_3-7-20_n28-n78	3	0.6
	7	0.6
	20	0.6
	n28	0.6
	n78	0.8

6.2B.4.2.3.5 $\Delta T_{IB,c}$ for EN-DC six bands

Table 6.2B.4.2.3.5-1: $\Delta T_{IB,c}$ due to EN-DC (six bands)

Inter-band EN-DC configuration	E-UTRA or NR Band	$\Delta T_{IB,c}$ (dB)
DC_1-3-7-20_n28-n78	1	0.7
	3	0.7
	7	0.7
	20	0.6
	n28	0.6
	n78	0.8

6.2B.4.2.3a Inter-band NE-DC within FR1

Unless $\Delta T_{IB,c}$ is specified in this clause, the value of $\Delta T_{IB,c}$ for the correspondingly specified EN-DC configuration in clause 6.2B.4.2.3 is applicable.

6.2B.4.2.4 Inter-band EN-DC including FR2

6.2B.4.2.4.1 $\Delta T_{IB,c}$ for EN-DC two bands

Unless otherwise stated, $\Delta T_{IB,c}$ for E-UTRA and FR2 NR bands of inter-band EN-DC combinations defined in table 5.5B.5.1-1 is set to zero.

Table 6.2B.4.2.4.1-1: Void

6.2B.4.2.4.2 $\Delta T_{IB,c}$ for EN-DC three bands

Unless otherwise stated, $\Delta T_{IB,c}$ for FR2 NR bands is set to zero, and $\Delta T_{IB,c}$ for constituent E-UTRA bands for inter-band EN-DC defined in table 5.5B.5.2-1 is the same as those for the corresponding E-UTRA CA configuration specified in TS 36.101 [4], without the FR2 NR bands.

Table 6.2B.4.2.4.2-1: Void

6.2B.4.2.4.3 $\Delta T_{IB,c}$ for EN-DC four bands

Unless otherwise stated, $\Delta T_{IB,c}$ for FR2 NR bands is set to zero, and $\Delta T_{IB,c}$ for constituent E-UTRA bands for inter-band EN-DC defined in table 5.5B.5.3-1 is the same as those for the corresponding E-UTRA CA configuration specified in TS 36.101 [4], without the FR2 NR bands.

Table 6.2B.4.2.4.3-1: Void

6.2B.4.2.4.4 $\Delta T_{IB,c}$ for EN-DC five bands

Unless otherwise stated, $\Delta T_{IB,c}$ for FR2 NR bands is set to zero, and $\Delta T_{IB,c}$ for constituent E-UTRA bands for inter-band EN-DC defined in table 5.5B.5.4-1 is the same as those for the corresponding E-UTRA CA configuration specified in TS 36.101 [4], without the FR2 NR bands.

Table 6.2B.4.2.4.4-1: Void

6.2B.4.2.4.5 Void

6.2B.4.2.5 Inter-band EN-DC including both FR1 and FR2

6.2B.4.2.5.1 $\Delta T_{IB,c}$ for EN-DC three bands

Unless otherwise stated, for inter-band EN-DC configurations defined in table 5.5B.6.2-1, $\Delta T_{IB,c}$ for constituent FR2 NR bands is set to zero, and $\Delta T_{IB,c}$ for constituent E-UTRA and FR1 NR bands is the same as those for the corresponding inter band EN-DC configuration without the FR2 bands specified in 6.2B.4.2.3.

Table 6.2B.4.2.5.1-1: Void

6.2B.4.2.5.2 $\Delta T_{IB,c}$ for EN-DC four bands

Unless otherwise stated, for inter-band EN-DC configurations defined in table 5.5B.6.3-1, $\Delta T_{IB,c}$ for constituent FR2 NR bands is set to zero, and $\Delta T_{IB,c}$ for constituent E-UTRA and FR1 NR bands is the same as those for the corresponding inter band EN-DC configuration without the FR2 bands specified in 6.2B.4.2.3.

6.2B.4.2.5.3 $\Delta T_{IB,c}$ for EN-DC five bands

Unless otherwise stated, for inter-band EN-DC configurations defined in table 5.5B.6.4-1, $\Delta T_{IB,c}$ for constituent FR2 NR bands is set to zero, and $\Delta T_{IB,c}$ for constituent E-UTRA and FR1 NR bands is the same as those for the corresponding inter band EN-DC configuration without the FR2 bands specified in 6.2B.4.2.3.

6.2B.4.2.5.4 $\Delta T_{IB,c}$ for EN-DC six bands

Unless otherwise stated, for inter-band EN-DC configurations defined in table 5.5B.6.5-1, $\Delta T_{IB,c}$ for constituent FR2 NR bands is set to zero, and $\Delta T_{IB,c}$ for constituent E-UTRA and FR1 NR bands is the same as those for the corresponding inter band EN-DC configuration without the FR2 bands specified in 6.2B.4.2.3.

6.2B.5 Configured output power for NR-DC

6.2B.5.1 Configured output power level

6.2B.5.1.1 Inter-band NR-DC between FR1 and FR2

For synchronous inter-band NR-DC [12] with MCG in FR1 and SCG in FR2 combined with one uplink serving cell per CG, the UE is allowed to set its configured maximum output power $P_{\text{CMAX},c(i),i}$ for serving cell $c(i)$ of CG i , $i = 1, 2$ as specified in clause 6.2.4 of TS 38.101-1 [2] and clause 6.2.4 TS 38.101-2 [3] independently.

6.3 Output power dynamics

Output power dynamics for EN-DC operations in FR1 and FR2 as specified in TS 38.101-1 [2] and TS 38.101-2 [3], respectively. E-UTRA as specified in TS 36.101 [4]. For intra-band contiguous EN-DC operation in FR1, minimum output power requirements specified in clause 6.3.1 of TS 38.101-1 [2] and clause 6.3.2 of TS 36.101 [4] shall only apply when the power of all NR and E-UTRA carriers are set to minimum value. Similarly, OFF power requirements specified in clause 6.3.2 of TS 38.101-1 [2] and clause 6.3.3 of TS 36.101 [4] shall only apply when the power of all NR and E-UTRA carriers are OFF. The OFF power condition in transmit ON/OFF time mask requirements specified in clause 6.3.3 of TS 38.101-1 [2] and clause 6.3.4 of TS 36.101 [4] is applicable only when all NR and E-UTRA carriers are OFF. If both E-UTRA and NR transition between ON and OFF states simultaneously, the longer transient time shall apply to both. If either E-UTRA or NR is OFF and the other carrier transitions from OFF to ON, then the transient time associated with that carrier applies.

6.3A Output power dynamics for CA

For inter-band NR CA between FR1 and FR2, output power dynamics as specified in TS 38.101-1 [2] and TS 38.101-2 [3] apply for FR1 and FR2 respectively.

6.3B Output power dynamics for DC

6.3B.0 General

The E-UTRA and NR switching time mask defines the observation period between E-UTRA subframe and NR slot/mini-slot boundary. Both E-UTRA subframe and NR slot/mini-slot have ON power transmissions. The ON power is defined as the mean power over the symbol duration excluding any transient period. For E-UTRA subframe or NR slot/mini-slot having OFF power transmission, the general time mask for E-UTRA or NR shall apply.

For inter-band EN-DC, output power dynamics requirement for E-UTRA single carrier and CA operation specified in clauses 6.3 and 6.3A of TS 36.101 [4] and for NR single carrier and CA operation specified in clause 6.3 and 6.3A of TS 38.101-1 [2] and for NR single carrier, CA operation and UL-MIMO specified in clause 6.3, 6.3A and 6.3D of TS 38.101-2 [3] apply.

6.3B.1 Output power dynamics for EN-DC with UL sharing from UE perspective

6.3B.1.1 E-UTRA and NR switching time mask for TDM based UL sharing from UE perspective

The E-UTRA and NR switching time mask is applicable for non-simultaneous transmissions between E-UTRA and NR in TDM based UL sharing from the UE perspective in the same channel, which is shared by E-UTRA and NR .

For UEs reporting E-UTRA and NR switching time capability of type 1 with switching time < 0.5 us for TDM based UL sharing from UE perspective within FR1 time masks in Figure 6.3B.1.1-1 and Figure 6.3B.1.1-2 shall apply. For UEs reporting E-UTRA and NR switching time capability of type 2 with switching time < 20 us for TDM based UL sharing from UE perspective within FR1, time masks in Figure 6.3B.1.1-3 and Figure 6.3B.1.1-4 shall apply.

Figure 6.3B.1.1-1: E-UTRA to NR switching time mask for type 1 for TDM based UL sharing from UE perspective within FR1

Figure 6.3B.1.1-2: NR to E-UTRA switching time mask for type 1 for TDM based UL sharing from UE perspective within FR1

Figure 6.3B.1.1-3: E-UTRA to NR switching time mask for type 2 for TDM based UL sharing from UE perspective within FR1

Figure 6.3B.1.1-4: NR to E-UTRA switching time mask for type 2 for TDM based UL sharing from UE perspective within FR1

6.3B.1a Output power dynamics for NE-DC with UL sharing from UE perspective

The E-UTRA and NR switching time mask is applicable for non-simultaneous transmissions between E-UTRA and NR in TDM based UL sharing from the UE perspective in the same channel, which is shared by E-UTRA and NR. Unless otherwise specified, the 6.3B.1.1 clauses for NE-DC are applicable.

6.3B.2 Output power dynamics for intra-band EN-DC without dual PA capability

For intra-band contiguous EN-DC configurations DC_(n)41 and DC_(n)71, and all intra-band non-contiguous EN-DC configurations without dual PA capability, maximum UL switching time is defined as 120 us and DL reception interruption is allowed during UL switching. Time masks in Figure 6.3B.2-1 and Figure 6.3B.2-2 shall apply. Unless otherwise stated, for other intra-band contiguous EN-DC configurations, the switching time in 6.3B.1.1 applies.

Figure 6.3B.2-1: E-UTRA to NR switching time mask for intra-band EN-DC without dual PA capability when single UL is allowed

Figure 6.3B.2-2: NR to E-UTRA switching time mask for intra-band EN-DC without dual PA capability when single UL is allowed

6.3B.3 Output power dynamics for intra-band EN-DC with dual PA capability

For both intra-band contiguous and non-contiguous EN-DC with dual PA capability, time masks in Figure 6.3B.3-1 and Figure 6.3B.3-2 shall apply.

Figure 6.3B.3-1: E-UTRA to NR switching time mask for intra-band EN-DC with dual PA capability

Figure 6.3B.3-2: NR to E-UTRA switching time mask for intra-band EN-DC with dual PA capability

6.4 Void

6.4A Transmit signal quality for CA

6.4A.1 Frequency error for CA

For inter-band NR CA between FR1 and FR2, frequency error as specified in TS 38.101-1 [2] and TS 38.101-2 [3] apply for FR1 and FR2 respectively.

6.4A.2 Transmit modulation quality for CA

For inter-band NR CA between FR1 and FR2, transmit modulation quality as specified in TS 38.101-1 [2] and TS 38.101-2 [3] apply for FR1 and FR2 respectively.

6.4B Transmit signal quality for DC

6.4B.1 Frequency error for DC

6.4B.1.1 Frequency error for Intra-band contiguous EN-DC

For intra-band contiguous EN-DC, the requirement shall apply on each component carrier as defined in clause 6.5.1 in TS 36.101 [4] and in clause 6.4.1 in TS 38.101-1 [2], respectively.

6.4B.1.2 Frequency error for Intra-band non-contiguous EN-DC

For intra-band non-contiguous EN-DC, the requirement shall apply on each component carrier as defined in clause 6.5.1 in TS 36.101 [4] and in clause 6.4.1 in TS 38.101-1 [2], respectively.

6.4B.1.3 Frequency error for inter-band EN-DC within FR1

For inter-band EN-DC with uplink assigned to one E-UTRA band and one NR band, the requirements shall apply on each component carrier as defined in clause 6.5.1 in TS 36.101 [4] and in clause 6.4.1 in TS 38.101-1 [2], respectively, with all component carriers active. If multiple component carriers are assigned to one E-UTRA band, the requirements in clauses 6.5.1A in TS 36.101 [4] apply for those component carriers.

6.4B.1.3a Frequency error for inter-band NE-DC within FR1

For inter-band NE-DC with uplink assigned to one E-UTRA band and one NR band, the requirements shall apply on each component carrier as defined in clause 6.5.1 in TS 36.101 [4] and in clause 6.4.1 in TS 38.101-1 [2], respectively, with all component carriers active. If multiple component carriers are assigned to one E-UTRA band, the requirements in clauses 6.5.1A in TS 36.101 [4] apply for those component carriers, and if multiple component carriers are assigned to one NR band, the requirements in clauses 6.4A.1 in TS 38.101-1 [2] apply for those component carriers.

6.4B.1.4 Frequency error for inter-band EN-DC including FR2

Frequency error requirement for E-UTRA single carrier and CA operation specified in clauses 6.5.1 and 6.5.1A of TS 36.101 [4] and for NR single carrier, CA operation and UL-MIMO specified in clause 6.4.1, 6.4A.1 and 6.4D.1 of TS 38.101-2 [3] apply.

6.4B.1.5 Frequency error for inter-band EN-DC including both FR1 and FR2

Frequency error requirement for E-UTRA single carrier and CA operation specified in clauses 6.5.1 and 6.5.1A of TS 36.101 [4] and for NR single carrier and CA operation specified in clause 6.4.1 and 6.4A.1 of TS 38.101-1 [2] and for NR single carrier, CA operation and UL-MIMO specified in clause 6.4.1, 6.4A.1 and 6.4D.1 of TS 38.101-2 [3] apply.

6.4B.2 Transmit modulation quality for DC

6.4B.2.1 Transmit modulation quality for Intra-band contiguous EN-DC

6.4B.2.1.1 Error Vector Magnitude

For the intra-band contiguous EN-DC with one component carrier per CG the EVM requirement applies with PRB allocation in one of the CG and the other CG unallocated. The EVM requirements for each CG are according to clause 6.5.2 of TS 36.101 [4] for the MCG and 6.4.2 of TS 38.101-1 [2] for the SCG with EN-DC configured.

6.4B.2.1.2 Carrier leakage

The carrier leakage requirements for each CG are according to clause 6.5.2 of TS 36.101 [4] for the MCG and 6.4.2 of TS 38.101-1 [2] for the SCG with EN-DC configured.

6.4B.2.1.3 In-band emissions

For the MCG the in-band emission requirements in Table 6.5.2A.3.1-1 and 6.5.2A.3.1-2 in TS 36.101 [4] apply within the aggregated transmission bandwidth configuration of

the EN-DC bandwidth with the carriers of both CGs active and one single contiguous PRB allocation of bandwidth L_{CRB} within the MCG at the edge of the said aggregated transmission bandwidth configuration.

For the SCG the in-band emission requirements in Table 6.4.2.3-1 in TS 38.101-1 [2] apply within the aggregated transmission bandwidth configuration of the EN-DC bandwidth with the carriers of both CGs active and one single contiguous PRB allocation of bandwidth L_{CRB} within the SCG at the edge of the aggregated transmission bandwidth configuration.

6.4B.2.2 Transmit modulation quality for Intra-band non-contiguous EN-DC

6.4B.2.2.1 Error Vector Magnitude

For the intra-band non-contiguous EN-DC with one component carrier per CG the EVM requirement applies with PRB allocation in one of the CG and the other CG unallocated. The EVM requirements for each CG are according to clause 6.5.2 of TS 36.101 [4] for the MCG and 6.4.2 of TS 38.101-1 [2] for the SCG with EN-DC configured.

6.4B.2.2.2 Carrier leakage

The carrier leakage requirements for each CG are according to clause 6.5.2 of TS 36.101 [4] for the MCG and 6.4.2 of TS 38.101-1 [2] for the SCG with EN-DC configured and PRB allocation only in the CG being measured.

6.4B.2.2.3 In-band emissions

For the MCG the in-band emission requirements in Table 6.5.2A.3.1-1 and 6.5.2A.3.1-2 in TS 36.101 [4] apply within the transmission bandwidth configuration of the MCG with the carriers of both CGs active and one single contiguous PRB allocation of bandwidth L_{CRB} within the MCG at the edge of the transmission bandwidth configuration.

For the SCG the in-band emission requirements in Table 6.4.2.3-1 in TS 38.101-1 [2] apply within the transmission bandwidth configuration of the SCG with the carriers of both CGs active and one single contiguous PRB allocation of bandwidth L_{CRB} within the SCG at the edge of the transmission bandwidth configuration.

6.4B.2.3 Transmit modulation quality for Inter-band EN-DC within FR1

For inter-band EN-DC with uplink assigned to one E-UTRA band and one NR band, the requirements shall apply on each component carrier as defined in clause 6.5.2 in TS 36.101 [4] and in clause 6.4.2 in TS 38.101-1 [2], respectively, with all component carriers active, applies with PRB allocation in one of the CG and the other CG unallocated. If multiple component carriers are assigned to one E-UTRA band, the requirements in subclauses 6.5.2A in TS 36.101 [4] apply for those component carriers.

6.4B.2.3a Transmit modulation quality for Inter-band NE-DC within FR1

For inter-band NE-DC with uplink assigned to one E-UTRA band and one NR band, the requirements shall apply on each component carrier as defined in clause 6.5.2 in TS

36.101 [4] and in clause 6.4.2 in TS 38.101-1 [2], respectively, with all component carriers active, applies with PRB allocation in one of the CG and the other CG unallocated. If multiple component carriers are assigned to one E-UTRA band, the requirements in subclauses 6.5.2A in TS 36.101 [4] apply for those component carriers, and if multiple component carriers are assigned to one NR band, the requirements in subclauses 6.4A.2 in TS 38.101-1 [2] apply for those component carriers.

6.4B.2.4 Transmit modulation quality for Inter-band EN-DC including FR2

Transmit modulation quality requirement for E-UTRA single carrier and CA operation specified in clauses 6.5.2 and 6.5.2A of TS 36.101 [4] and for NR single carrier, CA operation and UL-MIMO specified in clause 6.4.2, 6.4A.2 and 6.4D.2 of TS 38.101-2 [3] apply.

6.4B.2.5 Transmit modulation quality for inter-band EN-DC including both FR1 and FR2

Transmit modulation quality requirement for E-UTRA single carrier and CA operation specified in clauses 6.5.2 and 6.5.2A of TS 36.101 [4] and for NR single carrier and CA operation specified in clause 6.4.2 and 6.4A.2 of TS 38.101-1 [2] and for NR single carrier, CA operation and UL-MIMO specified in clause 6.4.2, 6.4A.2 and 6.4D.2 of TS 38.101-2 [3] apply.

6.5 Void

6.5A Output RF spectrum emissions for CA

6.5A.1 Occupied bandwidth for CA

For inter-band NR CA between FR1 and FR2, occupied bandwidth specified in TS 38.101-1 [2] and TS 38.101-2 [3] apply for each frequency range respectively.

6.5A.2 Out-of-band emissions for CA

For inter-band NR CA between FR1 and FR2, out-of-band emissions specified in TS 38.101-1 [2] and TS 38.101-2 [3] apply for each frequency range respectively.

6.5A.3 Spurious emissions for CA

6.5A.3.1 Inter-band CA between FR1 and FR2

Unless otherwise stated, for inter-band CA between FR1 and FR2, spurious emission and UE co-existence requirements specified in TS 38.101-1 [2] and TS 38.101-2 [3] apply for each component carrier respectively.

Table 6.5A.3.1-1: Void

6.5A.4 Transmit intermodulation for CA

For inter-band NR CA between FR1 and FR2, transmit intermodulation specified in TS 38.101-1 [2] apply for each component carrier for NR FR1.

6.5B Output RF spectrum emissions for DC

6.5B.1 Occupied bandwidth for EN-DC

6.5B.1.1 Intra-band contiguous EN-DC

For intra-band contiguous EN-DC the occupied bandwidth is a measure of the bandwidth containing 99% of the total integrated power of the transmitted spectrum. The OBW shall be less than the aggregated channel bandwidth for EN-DC, denoted as ENBW in clause 5.3B.

6.5B.1.2 Intra-band non-contiguous EN-DC

For intra-band non-contiguous EN-DC, occupied bandwidth requirement for E-UTRA single carrier and CA operation specified in clauses 6.6.1 and 6.6.1A of TS 36.101 [4] and for NR single carrier specified in clause 6.5.1 of TS 38.101-1 [2] apply.

1.3 Inter-band EN-DC within FR1

Occupied bandwidth requirement for E-UTRA single carrier and CA operation specified in clauses 6.6.1 and 6.6.1A of TS 36.101 [4] and for NR single carrier specified in clause 6.5.1 of TS 38.101-1 [2] apply.

6.5B.1.4 Inter-band EN-DC including FR2

Occupied bandwidth requirement for E-UTRA single carrier and CA operation specified in clauses 6.6.1 and 6.6.1A of TS 36.101 [4] and for NR single carrier, CA operation and UL-MIMO specified in clause 6.5.1, 6.5A.1 and 6.5D.1 of TS 38.101-2 [3] apply.

6.5B.1.5 Inter-band EN-DC including both FR1 and FR2

Occupied bandwidth requirement for E-UTRA single carrier and CA operation specified in clauses 6.6.1 and 6.6.1A of TS 36.101 [4] and for NR single carrier and CA operation specified in clause 6.5.1 and 6.5A.1 of TS 38.101-1 [2] and for NR single carrier, CA operation and UL-MIMO specified in clause 6.5.1, 6.5A.1 and 6.5D.1 of TS 38.101-2 [3] apply.

6.5B.2 Out-of-band emissions for DC

6.5B.2.1 Intra-band contiguous EN-DC

The out of band emissions are unwanted emissions immediately outside the EN-DC aggregated channel bandwidth resulting from the modulation process and non-linearity in the transmitter but excluding spurious emissions. This out of band emission limit is specified in terms of a spectrum emission mask and an adjacent channel leakage power ratio.

Unless otherwise stated, the OOB limits specified for the DC combination in this clause supercede any OOB requirements specified for each sub-block in the respective TS 36.101 [4] and TS 38.101-1 [2].

The requirements apply to the sum of transmissions across all antenna connectors.

6.5B.2.1.1 Spectrum emissions mask

The spectrum emission mask of the UE applies to frequencies (Δf_{OOB}) starting from the $\pm$ edge of the EN-DC aggregated channel bandwidth. For frequencies offset greater than Δf_{OOB} as specified in Table 6.5B.2.1.1-1 the spurious requirements in clause 6.5B.3 are applicable.

The general spectrum emission for intra-band contiguous EN-DC is specified in Table 6.5B.2.1.1-1.

The power of any UE emission shall not exceed the levels specified in Table 6.5B.2.1.1-1 for the specified EN-DC aggregated channel bandwidth.

Table 6.5B.2.1.1-1. General spectrum emission mask for intra-band contiguous EN-DC

Δf_{OOB} (MHz)	Spectrum emission limit (dBm)	Measurement bandwidth
$\pm 0 - 1$	$\text{Max}(\text{Round}(10 \cdot \log(0.15/\text{ENBW})), -24)$	30 kHz
$\pm 1 - 5$	-10	1 MHz
$\pm 5 - \text{ENBW}$	-13	1 MHz
$\pm \text{ENBW} - (\text{ENBW} + 5)$	-25	1 MHz
NOTE: ENBW refers to the aggregated channel bandwidth in MHz as defined in clause 5.3B.		

6.5B.2.1.2 Additional spectrum emissions mask

6.5B.2.1.2.1 Requirements for network signalled value "NS_35"

When NS_35 is indicated in the MCG and NS_35 is indicated in the SCG, the requirements in Table 6.5B.2.1.2.1-1 apply in the frequency ranges immediately adjacent and outside the aggregated sub-blocks of the EN-DC configuration for DC_(n)71AA.

Table 6.5B.2.1.2.1-1: Additional requirements

Δf_{OOB} (MHz)	Frequency offset of measurement filter centre frequency, f_{offset}	Minimum requirement (dBm)	Measurement bandwidth
0 MHz $f < 0.1$ MHz	0.015 MHz $f_{\text{offset}} < 0.085$ MHz	-13	30 kHz
0.1 MHz $f < \text{ENBW}$	0.15 MHz $f_{\text{offset}} < \text{ENBW} - 0.05$ MHz	-13	100 kHz
$\text{ENBW} \leq f < \text{ENBW} + 5$ MHz	$\text{ENBW} + 0.5$ MHz $f_{\text{offset}} < \text{ENBW} + 4.5$ MHz	-25	1 MHz
NOTE 1: ENBW is the aggregated bandwidth of an E-UTRA sub-block and an adjacent NR sub-block; there is no frequency separation between the said sub-blocks. The sub-block bandwidths include any internal guard bands.			

6.5B.2.1.2.2 Requirements for network signalled value "NS_04"

Additional spectrum emission requirements are signalled by the network to indicate that the UE shall meet an additional requirement for a specific deployment scenario as part of the cell handover/broadcast message.

The Band 41/n41 SEM transition point from -13 dBm/MHz to -25 dBm/MHz is based on the emission bandwidth. The emission bandwidth is defined as the width of the signal between two points, one below the carrier center frequency and one above the carrier center frequency, outside of which all emissions are attenuated at least 26 dB below the transmitter power. Since the 26 dB emission bandwidth is implementation dependent, the transmission bandwidths occupied by RBs is used for the SEM. The emission bandwidth for E-UTRA carriers is document in TS 36.101 [4], and

the emission bandwidth for NR carriers is documented in TS 38.101-1 [2]. The total emission bandwidth for contiguous intra-band EN-DC is the sum of the emission bandwidth for each CC plus the guard band between contiguous CCs.

When "NS_04" is indicated in the cell, the power of any UE emission shall not exceed the levels specified in Table 6.5B.2.1.2.2-1.

Table 6.5B.2.1.2.2-1: DC_(n)41 SEM with NS_04

	Spectrum emission limit (dBm) / measurement bandwidth for each ENBW					
Δf_{OoB} MHz	15 MHz	20 MHz	40 MHz	50 MHz	> 50 MHz	Measurement bandwidth
$\pm 0 - 1$	-10	-10	-10			2 % ENBW
				-10		1 MHz
$\pm 1 - 5$	-10					1 MHz
$\pm 5 - X$	-13					
$\pm X - (\text{ENBW} + 5 \text{ MHz})$	-25					
NOTE: X is defined as the sum of the emission bandwidth of the component carriers plus the guard band between contiguous CCs.						

6.5B.2.1.3 Adjacent channel leakage ratio

For EN-DC operation with an E-UTRA sub-block immediately adjacent to an NR sub-block, the ACLR is defined as the ratio of the filtered mean power centred on the aggregated sub-block bandwidth ENBW to the filtered mean power centred on an adjacent bandwidth of the same size ENBW at nominal channel spacing. The UE shall meet the ACLR minimum requirement EN-DC_{ACLR} specified in Table 6.5B.2.1.3-1 with ENBW the sum of the sub-block bandwidths.

The assigned channel power and adjacent channel power are measured with rectangular filters with measurement bandwidths specified in 6.5B.2.1.3-1.

Table 6.5B.2.1.3-1: ACLR for intra-band EN-DC (contiguous sub-blocks)

Parameter	Unit	Value
EN-DC _{ACLR} for PC3	dBc	30
EN-DC _{ACLR} for PC2	dBc	31
Measurement bandwidth of EN-DC channel		1.00*ENBW
Measurement bandwidth of adjacent channel		0.95*ENBW
Frequency offset of adjacent channel		ENBW / -ENBW
NOTE 1: ENBW is the aggregated bandwidth in MHz as defined in clause 5.3B.		
NOTE 2: The frequency offset is that in between the centre frequencies of the measurement filters		

6.5B.2.2 Intra-band non-contiguous EN-DC

6.5B.2.2.1 Spectrum emissions mask

The spectral emission mask for intra-band non-contiguous EN-DC is a composite of the emission mask for each CC with the level set to the maximum value from each mask for each frequency outside of the transmission bandwidth of either carrier. A composite spectrum emission mask is a combination of individual CC spectrum emissions masks. Where two masks overlap the most relaxed limit is used. Composite

spectrum emission mask applies to frequencies up to Δf_{OoB} starting from the edges of the sub-blocks. If for some frequency an individual CC spectrum emission mask overlaps with the bandwidth of another CC then the emission mask does not apply for that frequency.

6.5B.2.2.2 Additional spectrum emissions mask

When additional spectrum emission mask or masks apply, the additional SEM(s) shall be used to calculate the composite SEM described in 6.5B.2.2.1.

6.5B.2.2.3 Adjacent channel leakage ratio

For intra-band non-contiguous EN-DC, the EN-DC Adjacent Channel Leakage power Ratio (EN-DC_{ACLR}) is the ratio of the sum of the filtered mean powers centred on the assigned E-UTRA and NR sub-block frequencies to the filtered mean power centred on an adjacent channel frequency at nominal channel spacing. In case the sub-block gap bandwidth W_{gap} is smaller than a E-UTRA or NR sub-block bandwidth, no EN-DC_{ACLR} requirement is set for the corresponding sub-block for the gap. The assigned EN-DC sub-block power and adjacent channel power are measured with rectangular filters with measurement bandwidths specified in TS 36.101 [4] for the E-UTRA sub-block, and TS 38.101-1 [2] for the NR sub-block. If the measured adjacent channel power is greater than -50dBm then the EN-DC_{ACLR} shall be higher than the value specified in for E-UTRA_{ACLR} and NR_{ACLR}.

6.5B.2.3 Inter-band EN-DC within FR1

Unless otherwise stated, the OOB requirements specified in clause 6.6.2.1 of TS 36.101 [4], sub-clause 6.6.2 of TS 36.101 [4] and clause 6.5.2 of TS 38.101-1 [2] apply for each component carrier.

6.5B.2.3a Inter-band NE-DC within FR1

Unless otherwise stated, the OOB requirements specified in clause 6.6.2.1 of TS 36.101 [4], sub-clause 6.6.2 of TS 36.101 [4] and clause 6.5.2 of TS 38.101-1 [2] apply for each component carrier.

6.5B.2.4 Inter-band EN-DC including FR2

Unless otherwise stated, out-of-band emissions requirement for E-UTRA single carrier and CA operation specified in clauses 6.6.2 of TS 36.101 [4] and for NR single carrier, CA operation and UL-MIMO specified in clause 6.5.2, 6.5A.2 and 6.5D.2 of TS 38.101-2 [3] apply.

6.5B.2.5 Inter-band EN-DC including both FR1 and FR2

Unless otherwise stated, out-of-band emissions requirement for E-UTRA single carrier and CA operation specified in clauses 6.6.2 of TS 36.101 [4] and for NR single carrier and CA operation specified in clause 6.5.2 and 6.5A.2 of TS 38.101-1 [2] and for NR single carrier, CA operation and UL-MIMO specified in clause 6.5.2, 6.5A.2 and 6.5D.2 of TS 38.101-2 [3] apply.

6.5B.3 Spurious emissions for DC

6.5B.3.1 Intra-band contiguous EN-DC

6.5B.3.1.1 General spurious emissions

The general spurious emissions requirements specified in clause 6.6.3.1 of TS 36.101 [4] and clause 6.5.3.1 of TS 38.101-1 [2] apply beyond any frequencies for which the out-of-band emissions requirements in clause 6.5B.2.1 apply.

6.5B.3.1.2 Spurious emission band UE co-existence

The requirements in Table 6.5B.3.1.2-1 apply on each component carrier with all component carriers are active.

Table 6.5B.3.1.2-1: Requirements for intra-band contiguous EN-DC

EN-DC Configuration	Spurious emission					
	Protected band	Frequency range (MHz)		Maximum Level (dBm)	MBW (MHz)	NOTE
DC_(n)71	E-UTRA Band 4, 5, 12, 13, 14, 17, 24, 26, 30, 48, 66	F_{DL_low}	-	F_{DL_high}	-50	1
	E-UTRA Band 2, 25, 41, 70	F_{DL_low}	-	F_{DL_high}	-50	1
	E-UTRA Band 29	F_{DL_low}	--	F_{DL_high}	-38	1
	E-UTRA Band 71	F_{DL_low}	-	F_{DL_high}	-50	1
DC_(n)41	E-UTRA Band 1, 2, 3, 4, 5, 8, 10, 11, 12, 13, 14, 17, 18, 19, 21, 24, 25, 26, 27, 28, 29, 30, 34, 39, 42, 44, 45, 48, 50, 51, 66, 70, 71, 73, 74 NR Band n77, n78 and n79	F_{DL_low}	-	F_{DL_high}	-50	1
	Frequency range	1884.5	-	1915.7	-41	0.3
	NR Band n79	F_{DL_low}	-	F_{DL_high}	-50	1
NOTE 1: F_{DL_low} and F_{DL_high} refer to each frequency band specified in Table 5.5-1 in 3GPP TS 36.101 [4] or in Table 5.2-1 in 3GPP TS 38.101-1 [2].						
NOTE 2: As exceptions, measurements with a level up to the applicable requirements defined in Table 6.6.3.1-2 in 3GPP TS 36.101 [4] and Table 6.5.3.1-2 in 3GPP TS 38.101-1 [2] are permitted for each assigned carrier used in the measurement due to 2 nd , 3 rd , 4 th or 5 th harmonic spurious emissions. Due to spreading of the harmonic emission the exception is also allowed for the first 1 MHz frequency range immediately outside the harmonic emission on both sides of the harmonic emission. This results in an overall exception interval centred at the harmonic emission of (2 MHz + N x L_{CRB} x 180 kHz), where N is 2, 3, 4, 5 for the 2 nd , 3 rd , 4 th or 5 th harmonic respectively. The exception is allowed if the measurement bandwidth (MBW) totally or partially overlaps the overall exception interval						
NOTE 3: These requirements also apply for the frequency ranges that are less than F_{OoB} (MHz) in Table 6.6.3.1-1, Table 6.6.3.1A-1 in 3GPP TS 36.101 [4] or in Table 6.5.3.1-1 in 3GPP TS 38.101-1 [2] from the edge of the channel bandwidth.						
NOTE 4: Applicable when co-existence with PHS system operating in 1884.5 - 1915.7 MHz.						

NOTE: To simplify the above Table, E-UTRA band numbers are listed for bands which are specified only for E-UTRA operation or both E-UTRA and NR operation. NR band numbers are listed for bands which are specified only for NR operation.

6.5B.3.2 Intra-band non-contiguous EN-DC

6.5B.3.2.1 General spurious emissions

The general spurious emissions requirements specified in clause 6.6.3.1 of TS 36.101 [4] and clause 6.5.3.1 of TS 38.101-1 [2] apply beyond any frequencies for which the out-of-band emissions requirements in clause 6.5B.2.2 apply. If for some frequency an individual CC spurious emission requirement overlaps with the general spectrum emission mask or the bandwidth of another CC then it does not apply.

6.5B.3.2.2 Spurious emission band UE co-existence

The requirements in Table 6.5B.3.2.2-1 apply with all component carriers are active.

Table 6.5B.3.2.2-1: Requirements for intra-band non-contiguous EN-DC

EN-DC Configuration	Spurious emission					
	Protected band	Frequency range (MHz)			Maximum Level (dBm)	MBW (MHz)
DC_41_n41	E-UTRA Band 1, 2, 3, 4, 5, 8, 10, 11, 12, 13 , 14, 17, 18, 19, 21, 24, 25, 26, 27, 28, 29, 34, 39, 42, 44, 45, 48, 50, 51, 66, 70, 71, 73, 74 NR Band n77, n78 and n79	F _{DL_low}	-	F _{DL_high}	-50	1
	Frequency range	1884.5	-	1915.7	-41	0.3
	E-UTRA Band 30	F _{DL_low}	-	F _{DL_high}	-40	1
NOTE 1: F _{DL_low} and F _{DL_high} refer to each frequency band specified in Table 5.5-1 in 3GPP TS 36.101 [4] or in Table 5.2-1 in 3GPP TS 38.101-1 [2].						
NOTE 2: As exceptions, measurements with a level up to the applicable requirements defined in Table 6.6.3.1-2 in 3GPP TS 36.101 [4] and Table 6.5.3.1-2 in 3GPP TS 38.101-1 [2] are permitted for each assigned carrier used in the measurement due to 2 nd , 3 rd , 4 th or 5 th harmonic spurious emissions. Due to spreading of the harmonic emission the exception is also allowed for the first 1 MHz frequency range immediately outside the harmonic emission on both sides of the harmonic emission. This results in an overall exception interval centred at the harmonic emission of (2MHz + N x L _{CRB} x 180kHz), where N is 2, 3, 4, 5 for the 2 nd , 3 rd , 4 th or 5 th harmonic respectively. The exception is allowed if the measurement bandwidth (MBW) totally or partially overlaps the overall exception interval						
NOTE3: Applicable when co-existence with PHS system operating in 1884.5 - 1915.7 MHz.						

NOTE: To simplify the above Table, E-UTRA band numbers are listed for bands which are specified only for E-UTRA operation or both E-UTRA and NR operation. NR band numbers are listed for bands which are specified only for NR operation.

6.5B.3.3 Inter-band EN-DC within FR1

6.5B.3.3.1 General spurious emissions

The general spurious emissions requirements specified in clause 6.6.3.1 of TS 36.101 [4], clause 6.5.3.1 of TS 38.101-1 [2] and TS 38.101-2 [3] apply for each component carrier. For the case of inter-band EN-DC with a single carrier per cell group, the general spurious emissions requirements also apply with both downlink carrier and both uplink carriers active. Limits on configured maximum output power for the uplink according to clause 6.2B.4 apply. If for some frequency a spurious emission requirement of an individual component carrier overlaps with the spectrum emission mask or channel bandwidth of another component carrier then it does not apply.

NOTE: The general spurious emission requirements with both uplink carriers active are allowed to be verified for only a single inter-band EN-DC configuration per NR band. Furthermore, the requirements are allowed to be verified by measuring spurious emissions at the specific frequencies where second and third order intermodulation products generated by the two transmitted carriers can occur.

Table 6.5B.3.3.1-1: (Void)

6.5B.3.3.2 Spurious emission band UE co-existence

This clause specifies additional the requirements for uplink EN-DC coexistence with protected bands with the single CC uplink assigned to E-UTRA and NR bands for the specified uplink carrier aggregation configurations in Table 6.5B.3.3.2-1. The intersection of the requirements for the individual bands specified in clause 6.5.3.2 of [2] and clause 6.6.3.2 of [4] shall also apply for the specified uplink EN-DC

configurations. Intersection of a requirement means that both UL constituent bands have the same protected band requirement specified and if one or both protected bands have note(s) associated those note(s) also apply.

The requirements in Table 6.5B.3.3.2-1 apply on each component carrier with all component carriers are active.

NOTE: For inter-band EN-DC with uplink assigned to one LTE band and one NR band the requirements in Table 6.5B.3.3.2-1 could be verified by measuring spurious emissions at the specific frequencies where second and third order intermodulation products generated by the two transmitted carriers can occur; in that case, the requirements for remaining applicable frequencies in Table 6.5B.3.3.2-1 and in 6.5.3.2 of [2] and clause 6.6.3.2 of [4] would be considered to be verified by the measurements verifying the uplink single carrier UE to UE co-existence requirements..

Table 6.5B.3.3.2-1: Requirements

EN-DC Configuration	Spurious emission						
	Protected band	Frequency range (MHz)			Maximum Level (dBm)	MBW (MHz)	NOTE
DC_1_n28	Frequency range	470	-	694	-42	8	5, 17
	Frequency range	470	-	710	-26.2	6	14
	Frequency range	758	-	773	-32	1	5
	Frequency range	773	-	803	-50	1	
	Frequency range	662	-	694	-26.2	6	5
DC_1_n40	Frequency range	1884.5	-	1915.7	-41	0.3	3
DC_3_n28	Frequency range	1884.5	-	1915.7	-41	0.3	13
	Frequency range	470	-	710	-26.2	6	14
	Frequency range	758	-	773	-32	1	5
	Frequency range	773	-	803	-50	1	
	Frequency range	1884.5	-	1915.7	-41	0.3	3, 9
DC_3_n40	Frequency range	1884.5	-	1915.7	-41	0.3	3
DC_3_n77	Frequency range	1884.5	-	1915.7	-41	0.3	3
DC_3_n78 DC_3_n80_ULSUP -TDM_n78	Frequency range	1884.5	-	1915.7	-41	0.3	3
DC_3_n79 DC_3_n80_ULSUP -TDM_n79	Frequency range	1884.5	-	1915.7	-41	0.3	3
DC_5_n40	Frequency range	1884.5	-	1915.7	-41	0.3	3
DC_5_n78	Frequency range	859	-	869	-27	1	
DC_7_n28	Frequency range	758	-	773	-32	1	5
	Frequency range	773	-	803	-50	1	
DC_8_n40	Frequency range	1884.5	-	1915.7	-41	0.3	3
DC_8_n77	Frequency range	860	-	890	-40	1	5, 12
	Frequency range	1884.5	-	1915.7	-41	0.3	3, 12
	Frequency range	860	-	890	-40	1	5, 12
DC_8_n78 DC_8_n81_ULSUP -TDM_n78	Frequency range	1884.5	-	1915.7	-41	0.3	3, 12
DC_8_n79 DC_8_n81_ULSUP -TDM_n79	Frequency range	860	-	890	-40	1	5, 12
	Frequency range	1884.5	-	1915.7	-41	0.3	3
DC_11_n77	Frequency range	945	-	960	-50	1	
	Frequency range	1884.5	-	1915.7	-41	0.3	3
	Frequency range	2545	-	2575	-50	1	

	Frequency range	2595	-	2645	-50	1	
DC_11_n78	Frequency range	945	-	960	-50	1	
	Frequency range	1884.5	-	1915.7	-41	0.3	3
	Frequency range	2545	-	2575	-50	1	
	Frequency range	2595	-	2645	-50	1	
DC_11_n79	Frequency range	945	-	960	-50	1	
	Frequency range	1884.5	-	1915.7	-41	0.3	3
	Frequency range	2545	-	2575	-50	1	
	Frequency range	2595	-	2645	-50	1	
DC_18_n77	Frequency range	945	-	960	-50	1	
	Frequency range	1884.5	-	1915.7	-41	0.3	3
	Frequency range	2545	-	2575	-50	1	
	Frequency range	2595	-	2645	-50	1	
DC_18_n78	Frequency range	945	-	960	-50	1	
	Frequency range	1884.5	-	1915.7	-41	0.3	3
	Frequency range	2545	-	2575	-50	1	
	Frequency range	2595	-	2645	-50	1	
DC_18_n79	Frequency range	945	-	960	-50	1	
	Frequency range	1884.5	-	1915.7	-41	0.3	3
	Frequency range	2545	-	2575	-50	1	
	Frequency range	2595	-	2645	-50	1	
DC_19_n77	Frequency range	945	-	960	-50	1	
	Frequency range	1884.5	-	1915.7	-41	0.3	3
	Frequency range	2545	-	2575	-50	1	
	Frequency range	2595	-	2645	-50	1	
DC_19_n78	Frequency range	945	-	960	-50	1	
	Frequency range	1884.5	-	1915.7	-41	0.3	3
	Frequency range	2545	-	2575	-50	1	
	Frequency range	2595	-	2645	-50	1	
DC_19_n79	Frequency range	945	-	960	-50	1	
	Frequency range	1884.5	-	1915.7	-41	0.3	3
	Frequency range	2545	-	2575	-50	1	
	Frequency range	2595	-	2645	-50	1	
DC_20_n51	Frequency range	758	-	788	-50	1	
DC_21_n77	Frequency range	945	-	960	-50	1	
	Frequency range	1884.5	-	1915.7	-41	0.3	3
	Frequency range	2545	-	2575	-50	1	
	Frequency range	2595	-	2645	-50	1	
DC_21_n78	Frequency range	945	-	960	-50	1	
	Frequency range	1884.5	-	1915.7	-41	0.3	3
	Frequency range	2545	-	2575	-50	1	
	Frequency range	2595	-	2645	-50	1	
DC_21_n79	Frequency range	945	-	960	-50	1	
	Frequency range	1884.5	-	1915.7	-41	0.3	3
	Frequency range	2545	-	2575	-50	1	
	Frequency range	2595	-	2645	-50	1	
DC_26_n41	Frequency range	1884.5		1915.7	-41	0.3	3
	Frequency range	703	-	799	-50	1	
	Frequency range	799	-	803	-40	1	5
	Frequency range	945	-	960	-50	1	
DC_26_n77	Frequency range	703	-	799	-50	1	
	Frequency range	799	-	803	-40	1	5
	Frequency range	945	-	960	-50	1	
	Frequency range	1884.5	-	1915.7	-41	0.3	3
	Frequency range	2545	-	2575	-50	1	2

	Frequency range	2595	-	2645	-50	1	
DC_26_n78	Frequency range	703	-	799	-50	1	
	Frequency range	799	-	803	-40	1	5
	Frequency range	945	-	960	-50	1	
	Frequency range	1884.5	-	1915.7	-41	0.3	3
	Frequency range	2545	-	2575	-50	1	2
	Frequency range	2595	-	2645	-50	1	
DC_26_n79	Frequency range	703	-	799	-50	1	
	Frequency range	799	-	803	-40	1	5
	Frequency range	945	-	960	-50	1	
	Frequency range	1884.5	-	1915.7	-41	0.3	3
	Frequency range	2545	-	2575	-50	1	2
	Frequency range	2595	-	2645	-50	1	
DC_28_n51	Frequency range	470	-	694	-42	8	5, 17
	Frequency range	470	-	710	-26.2	6	14
	Frequency range	662	-	694	-26.2	6	5
	Frequency range	758	-	773	-32	1	5
	Frequency range	773	-	803	-50	1	
DC_28_n77	Frequency range	758	-	773	-32	1	
	Frequency range	773	-	803	-50	1	
	Frequency range	1884.5	-	1915.7	-41	0.3	3, 9
DC_28_n78 DC_28_n83_ULSU P-TDM_n78	Frequency range	758	-	773	-32	1	
	Frequency range	773	-	803	-50	1	
	Frequency range	1884.5	-	1915.7	-41	0.3	3, 9
DC_28_n79	Frequency range	758	-	773	-32	1	
	Frequency range	773	-	803	-50	1	
	Frequency range	1884.5	-	1915.7	-41	0.3	3, 9
DC_38_n78	N/A						
DC_40_n77	N/A						
DC_41_n77	Frequency range	1884.5		1915.7	-41	0.3	3
DC_41_n78	Frequency range	1884.5	-	1915.7	-41	0.3	3
DC_41_n79	Frequency range	1884.5	-	1915.7	-41	0.3	3
DC_42_n77	N/A						
DC_42_n78	N/A						
DC_42_n79	N/A						
NOTE 1: Void							
NOTE 2: Void							
NOTE 3: Applicable when co-existence with PHS system operating in 1884.5 - 1915.7 MHz							
NOTE 4: Void							
NOTE 5: These requirements also apply for the frequency ranges that are less than F _{OoB} (MHz) in Table 6.6.3.1-1, Table 6.6.3.1A-1 in 3GPP TS 36.101 [4] or in Table 6.5.3.1-1 in 3GPP TS 38.101-1 [2] from the edge of the channel bandwidth.							
NOTE 6: Void							
NOTE 7: Void							
NOTE 8: Void							
NOTE 9: Applicable when the assigned E-UTRA or NR carrier is confined within 718 MHz and 748 MHz and when the channel bandwidth used is 5 or 10 MHz.							
NOTE 10: Void							
NOTE 11: Void							
NOTE 12: This requirement is applicable only for the following cases: A: for carriers of 5 MHz channel bandwidth when carrier centre frequency (Fc) is within the range 902.5 MHz ≤ Fc < 907.5 MHz with an uplink transmission bandwidth less than or equal to 20 RB; B: for carriers of 5 MHz channel bandwidth when carrier centre frequency (Fc) is within the range 907.5 MHz ≤ Fc ≤ 912.5 MHz without any restriction on uplink transmission bandwidth; C: for carriers of 10 MHz channel bandwidth when carrier centre frequency (Fc) is Fc = 910 MHz with an uplink transmission bandwidth less than or equal to 32 RB with RB _{start} > 3.							
NOTE 13: Void							
NOTE 14: This requirement is applicable for 5 and 10 MHz E-UTRA or NR channel bandwidth allocated within 718-728MHz. For carriers of 10 MHz bandwidth, this requirement applies for an uplink transmission bandwidth less than or equal to 30 RB with RB _{start} > 1 and RB _{start} < 48.							

NOTE 15: Void

NOTE 16: This requirement is applicable for any channel bandwidths within the range 1920 - 1980 MHz with the following restriction: for carriers of 15 MHz bandwidth when carrier centre frequency is within the range 1927.5 - 1929.5 MHz and for carriers of 20 MHz bandwidth when carrier centre frequency is within the range 1930 - 1938 MHz the requirement is applicable only for an uplink transmission bandwidth less than or equal to 54 RB.

NOTE 17: This requirement is applicable in the case of a 10 MHz E-UTRA or NR carrier confined within 703 MHz and 733 MHz, otherwise the requirement of -25 dBm with a measurement bandwidth of 8 MHz applies.

NOTE 18: Void

NOTE 19: Void

NOTE: To simplify the above Table, E-UTRA band numbers are listed for bands which are specified only for E-UTRA operation or both E-UTRA and NR operation. NR band numbers are listed for bands which are specified only for NR operation.

6.5B.3.3a Inter-band NE-DC within FR1

6.5B.3.3a.1 General spurious emissions

The general spurious emissions requirements specified in clause 6.6.3.1 of TS 36.101 [4], clause 6.5.3.1 of TS 38.101-1 [2] and TS 38.101-2 [3] apply for each component carrier.

6.5B.3.3a.2 Spurious emission band UE co-existence

This clause specifies the requirements for the specified NE-DC configurations that do not have a corresponding defined EN-DC, for coexistence with protected bands. For the NE-DC configurations that have a corresponding specified EN-DC configuration, the requirements in Table 6.5B.3.3.2-1 apply on each component carrier with all component carriers are active.

6.5B.3.4 Inter-band EN-DC including FR2

General spurious requirement for E-UTRA single carrier and CA operation specified in clauses 6.6.3.1 and 6.6.3.1A of TS 36.101 [4] and for NR single carrier, CA operation and UL-MIMO specified in clause 6.5.3, 6.5A.3 and 6.5D.3 of TS 38.101-2 [3] apply.

6.5B.3.4.1 Spurious emission band UE co-existence

This clause specifies the requirements for the specified EN-DC, for coexistence with protected bands. Unless otherwise stated, for inter-band EN-DC configurations defined in table 5.5B.5.1-1, no requirements for FR2 NR bands to protect E-UTRA and FR1 NR bands are applied to the constituent FR2 NR bands. Spurious emission band UE co-existence requirements specified in TS 36.101 [4] are applied to the constituent E-UTRA bands for the EN-DC configuration.

Spurious emission band UE co-existence requirement for E-UTRA single carrier and CA operation specified in clauses 6.6.3.2 and 6.6.3.2A of TS 36.101 [4] and for NR single carrier, CA operation and UL-MIMO specified in clause 6.5.3.1, 6.5A.3.1 and 6.5D.3.1 of TS 38.101-2 [3] apply.

Table 6.5B.3.4.1-1: Void

6.5B.3.5 Inter-band EN-DC including both FR1 and FR2

General spurious requirement for E-UTRA single carrier and CA operation specified in clauses 6.6.3.1 and 6.6.3.1A of TS 36.101 [4] and for NR single carrier and CA

operation specified in clause 6.5.3.1 and 6.5A.3.1 of TS 38.101-1 [2] and for NR single carrier, CA operation and UL-MIMO specified in clause 6.5.3, 6.5A.3 and 6.5D.3 of TS 38.101-2 [3] apply.

6.5B.3.5.1 Spurious emission band UE co-existence

This clause specifies the requirements for the specified EN-DC, for coexistence with protected bands. Unless otherwise stated, for inter-band EN-DC configurations defined in clause 5.5B.6, no requirements for FR2 NR bands to protect E-UTRA and FR1 NR bands are applied to the constituent FR2 NR bands. Spurious emission band UE co-existence requirements for constituent E-UTRA and FR1 NR bands for the inter-band EN-DC are the same as those for the corresponding EN-DC configuration without the FR2 bands specified in 6.5B.3.2.2.

Spurious emission band UE co-existence requirement for E-UTRA single carrier and CA operation specified in clauses 6.6.3.2 and 6.6.3.2A of TS 36.101 [4] and for NR single carrier and CA operation specified in clause 6.5.3.2 and 6.5A.3.2 of TS 38.101-1 [2] and for NR single carrier, CA operation and UL-MIMO specified in clause 6.5.3.1, 6.5A.3.1 and 6.5D.3.1 of TS 38.101-2 [3] apply.

Table 6.5B.3.5.1-1: Void

6.5B.4 Additional spurious emissions

6.5B.4.1 General

These requirements are specified in terms of an additional spectrum emission requirement. Additional spurious emission requirements are signalled by the network to indicate that the UE shall meet an additional requirement for a specific deployment scenario as part of the cell handover/broadcast message.

NOTE: For measurement conditions at the edge of each frequency range, the lowest frequency of the measurement position in each frequency range should be set at the lowest boundary of the frequency range plus MBW/2. The highest frequency of the measurement position in each frequency range should be set at the highest boundary of the frequency range minus MBW/2. MBW denotes the measurement bandwidth defined for the protected band.

6.5B.4.1.1 Void

6.5B.4.2 Intra-band contiguous EN-DC

6.5B.4.2.1 Minimum requirement (network signalled value "NS_04")

When "NS 04" is indicated in the cell, the power of any UE emission shall not exceed the levels specified in Table 6.5B.4.1.1-1. This requirement also applies for the frequency ranges that are less than F_{OoB} (MHz) in Table 6.6.3.1-1 from the edge of the channel bandwidth.

Table 6.5B.4.1.1-1: Additional requirements

Frequency band (MHz)	Channel bandwidth / Spectrum emission limit (dBm)	Measurement bandwidth
$2495 \leq f < 2496$	-13	1 % of Channel BW for contiguous BW up to 100 MHz, 1 MHz for contiguous BW > 100 MHz
$2490.5 \leq f < 2495$	-13	1 MHz

$0 < f < 2490.5$	-25	1 MHz
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6.5B.4.3 Intra-band non-contiguous EN-DC

6.5B.4.3.1 Minimum requirement (network signalled value "NS_04")

When "NS 04" is indicated in the cell, the power of any UE emission shall not exceed the levels specified in Table 6.5B.4.1.1-1. This requirement also applies for the frequency ranges that are less than F_{OOB} (MHz) in Table 6.6.3.1-1 from the edge of the channel bandwidth.

Table 6.5B.4.1.1-1: Additional requirements

Frequency band (MHz)	Channel bandwidth / Spectrum emission limit (dBm)	Measurement bandwidth
$2495 \leq f < 2496$	-13	1 % of Channel BW for contiguous BW up to 100 MHz, 1 MHz for contiguous BW > 100 MHz
$2490.5 \leq f < 2495$	-13	1 MHz
$0 < f < 2490.5$	-25	1 MHz

6.5B.4.4 Inter-band EN-DC within FR1

The additional spurious emissions requirements specified for E-UTRA in clause 6.6.3.3 and 6.6.3.3A of TS 36.101 [4] and for NR single carrier, CA operation and UL-MIMO specified in clause 6.5.3.3, 6.5A.3.3 and 6.5D.3 of TS 38.101-1 [2] apply for each component carrier.

6.5B.4.4a Inter-band NE-DC within FR1

The additional spurious emissions requirements specified for E-UTRA in clause 6.6.3.3 and 6.6.3.3A of TS 36.101 [4] and for NR single carrier, CA operation and UL-MIMO specified in clause 6.5.3.3, 6.5A.3.3 and 6.5D.3 of TS 38.101-1 [2] apply for each component carrier.

6.5B.4.5 Inter-band EN-DC including FR2

The additional spurious emissions requirements specified for E-UTRA in clause 6.6.3.3 and 6.6.3.3A of TS 36.101 [4] and for NR single carrier, CA operation and UL-MIMO specified in clause 6.5.3.3, 6.5A.3.3 and 6.5D.3 of TS 38.101-2 [3] apply for each component carrier.

6.5B.4.6 Inter-band EN-DC including both FR1 and FR2

The additional spurious emissions requirements specified for E-UTRA in clause 6.6.3.3 and 6.6.3.3A of TS 36.101 [4] and for NR single carrier, CA operation and UL-MIMO specified in clause 6.5.3.3, 6.5A.3.3 and 6.5D.3 of TS 38.101-1 [2] and in clause 6.5.3.3, 6.5A.3.3 and 6.5D.3 of TS 38.101-2 [3] apply for each component carrier.

6.5B.5 Transmit intermodulation for DC

6.5B.5.1 Intra-band contiguous EN-DC

Unless otherwise stated, no transmit intermodulation requirements are applied for intraband contiguous ENDC.

6.5B.5.2 Intra-band non-contiguous EN-DC

Unless otherwise stated, no transmit intermodulation requirements are applied for intra band noncontiguous EN DC.

6.5B.5.3 Inter-band EN-DC within FR1

The transmit intermodulation requirement specified in clauses 6.7.1 of TS 36.101 [4] and clauses 6.5.4 and 6.5A.4 of TS 38.101-1 [2] apply for each component carrier in E-UTRA bands and NR bands, respectively.

6.5B.5.3a Inter-band NE-DC within FR1

The transmit intermodulation requirement specified in clauses 6.7.1 and 6.7.1A of TS 36.101 [4] and clauses 6.5.4 and 6.5A.4 of TS 38.101-1 [2] apply for each component carrier in E-UTRA bands and NR bands, respectively.

6.5B.5.4 Inter-band EN-DC including FR2

Transmit intermodulation requirements specified in clause 6.7.1 and 6.7.1A of TS 36.101 [4] apply for each component carrier in E-UTRA bands.

6.5B.5.5 Inter-band EN-DC including both FR1 and FR2

Transmit intermodulation requirement specified in clauses 6.7.1 and 6.7.1A of TS 36.101 [4] and clauses 6.5.4 and 6.5A.4 of TS 38.101-1 [2] apply for each component carrier in E-UTRA bands and NR bands, respectively.

6.6B Beam correspondence for DC

6.6B.1 Void

6.6B.2 Void

6.6B.3 Void

6.6B.4 Inter-band EN-DC including FR2

Beam correspondence requirement specified in clause 6.6 and 6.6A of TS 38.101-2 [3] apply for NR FR2 bands.

6.6B.5 Inter-band EN-DC including both FR1 and FR2

Beam correspondence requirement specified in clause 6.6 and 6.6A of TS 38.101-2 [3] apply for NR FR2 bands.

7 Receiver characteristics

7.1 General

Unless otherwise stated the receiver characteristics are specified at the antenna connector(s) of the UE for the bands operating on frequency range 1 and over the air

of the UE for the bands operating on frequency range 2. The requirements for frequency range 1 and frequency range 2 can be verified separately. For the carrier in frequency range 1, requirements can be verified with NR FR2 link disabled. For the carrier in frequency range 2, requirements can be verified in OTA mode with E-UTRA or NR FR1 connecting to the network by OTA without calibration.

The requirements defined in this clause are the extra requirements compared with the single carrier requirements defined in TS 38.101-1 [2] and TS 38.101-2 [3].

Unless otherwise stated, the UL and DL reference measurement channels are the same with the configurations specified in TS 38.101-1 [2] and TS 38.101-2 [3].

Unless otherwise stated, requirements for NR receiver written in TS 38.101-1 [2] and TS 38.101-2 [3] apply and are assumed anchor agnostic. Requirements are verified under conditions where anchor resources do not interfere NR operation.

For intra-band EN-DC, the output power is configured as follows:

- One E-UTRA uplink carrier with the output power set to 29 dB below $P_{\text{CMAX_L,c}}$ and the NR band whose downlink is being tested has its uplink carrier output power set to 4 dB below $P_{\text{CMAX_L,f,c}}$.
- One NR uplink carrier with the output power set to 29 dB Below $P_{\text{CMAX_L,f,c}}$ and the E-UTRA band whose downlink is being tested has its uplink carrier output power set to 4 dB below $P_{\text{CMAX_L,c}}$.

For the additional requirements for intra-band non-contiguous EN-DC of two sub-blocks, an in-gap test refers to the case when the interfering signal is located at a negative offset with respect to the assigned lowest channel frequency of the highest sub-block and located at a positive offset with respect to the assigned highest channel frequency of the lowest sub-block.

For the additional requirements for intra-band non-contiguous EN-DC of two sub-blocks, an out-of-gap test refers to the case when the interfering signal(s) is (are) located at a positive offset with respect to the assigned channel frequency of the highest carrier frequency or located at a negative offset with respect to the assigned channel frequency of the lowest carrier frequency.

For the additional requirements for intra-band non-contiguous EN-DC of two sub-blocks with channel bandwidth larger than or equal to 5 MHz, the existing adjacent channel selectivity requirements, in-band blocking requirements (for each case), and narrow band blocking requirements apply for in-gap tests only if the corresponding interferer frequency offsets with respect to the two measured carriers satisfy the following condition in relation to the sub-block gap size W_{gap} for at least one of the E-UTRA or NR sub-blocks, so that the interferer frequency position does not change the nature of the core requirement tested:

$$W_{\text{gap}} \geq 2 \cdot |F_{\text{Interferer (offset)}}| - BW_{\text{Channel}}$$

For the E-UTRA sub-block, the $F_{\text{Interferer (offset)}}$, for a sub-block with a single component carrier is the interferer frequency offset with respect to carrier as specified in clause 7.5.1, clause 7.6.1 and clause 7.6.3 for the respective requirement in TS 36.101 [4] and $BW_{\text{Channel}} \cdot F_{\text{Interferer (offset)}}$ for the E-UTRA sub-block with two or more contiguous component carriers is the interference frequency offset with respect to the carrier adjacent to the gap is specified in clause 7.5.1A, 7.6.1A and 7.6.3A in TS 36.101 [4].

For the NR sub-block, the $F_{\text{Interferer (offset)}}$, for a sub-block with a single component carrier is the interferer frequency offset with respect to carrier as specified in clause 7.5.1, clause 7.6.1 and clause 7.6.3 for the respective requirement in TS 38.101-1 [2] and BW_{Channel} .

The interferer frequency offsets for adjacent channel selectivity, each in-band blocking case and narrow-band blocking shall be tested separately with a single in-gap interferer at a time.

For the requirements of FR1 in this clause, the UE shall be verified with four or eight Rx antenna ports and skip two Rx antenna ports requirements in operating bands where the UE is equipped with four or eight Rx antenna ports, otherwise, the UE shall be verified with two Rx antenna ports.

7.2 Void

7.3 Void

7.3A Reference sensitivity for CA

7.3A.1 General

For NR CA operation NR single carrier REFSENS requirements defined in TS 38.101-1 [2] and TS 38.101-2 [3] apply to all downlink bands part of NR CA configurations listed in Table 5.2A.1-1 unless sensitivity degradation is allowed as defined in clause 7.3A.

7.3A.2 Reference sensitivity power level for CA

7.3A.3 $\Delta R_{IB,c}$ for CA

For the UE which supports inter-band NR CA configuration, the minimum requirement for reference sensitivity in clause 7.3.2 in TS 38.101-1 [2] and clause 7.3.2, 7.3A.2 in TS 38.101-2 [3] shall be increased by the amount given in $\Delta R_{IB,c}$ in Tables below. Unless otherwise stated, $\Delta R_{IB,c}$ is set to zero.

In case the UE supports more than one of band combinations for CA, SUL or DC, and an operating band belongs to more than one band combinations then

- When the operating band frequency range is ≤ 1 GHz, the applicable additional $\Delta R_{IB,c}$ shall be the average value for all band combinations defined in clause 7.3A, 7.3B, 7.3C in TS 38.101-1 [2] and 7.3A, 7.3B in this specification, truncated to one decimal place that apply for that operating band among the supported band combinations. In case there is a harmonic relation between low band UL and high band DL, then the maximum $\Delta R_{IB,c}$ among the different supported band combinations involving such band shall be applied
- When the operating band frequency range is > 1 GHz, the applicable additional $\Delta R_{IB,c}$ shall be the maximum value for all band combinations defined in clause 7.3A, 7.3B, 7.3C in TS 38.101-1 [2] and 7.3A, 7.3B in this specification for the applicable operating bands.

7.3A.3.1 $\Delta R_{IB,c}$ for Inter-band CA between FR1 and FR2

Unless otherwise stated, $\Delta R_{IB,c}$ for NR FR1 band and FR2 band of inter-band CA defined in table 5.5A.1-1 is set to zero.

Table 7.3A.3.1-1: Void

7.3A.4 Void

7.3B Reference sensitivity level for DC

7.3B.1 General

For EN-DC, E-UTRA and NR single carrier, CA, and MIMO operation of REFSENS requirements defined in TS 38.101-1 [2], TS 38.101-2 [3] and TS 36.101 [4] apply to all downlink bands of EN-DC configurations listed in clause 5.5B, unless sensitivity degradation exception is allowed in this clause of this specification, clause 7.3 in TS 38.101-1 [2], clause 7.3 in TS 38.101-2 [3] or clause 7.3 in TS 36.101 [4]. Allowed exceptions specified in this clause also apply to any higher order EN-DC configuration combination containing one of the band combinations that exception is allowed for. Reference sensitivity exceptions are specified by applying maximum sensitivity degradation (MSD) into applicable REFSENS requirement. EN-DC REFSENS requirements shall be met for NR uplink transmissions using QPSK DFT-s-OFDM waveforms as defined in clause 7.3.2 [2]. Unless otherwise specified UL allocation uses the lowest SCS allowable for a given channel BW. Limits on configured maximum output power for the uplink according to clause 6.2B.4 shall apply. In case of interband EN-DC the receiver REFSENS requirements in this clause do not apply for 1.4 and 3 MHz E-UTRA carriers. For the case of inter-band EN-DC with a single carrier per cell group and multicarrier per cell group, in addition to the E-UTRA and NR single carrier, CA, and MIMO operation of REFSENS requirements defined in TS 38.101-1 [2], TS 38.101-2 [3], and TS 36.101 [4], the REFSENS requirements specified therein also apply with both downlink carriers and both uplink carriers active unless sensitivity exceptions are allowed in this clause of this specification, clause 7.3 in TS 38.101-1 [2] or clause 7.3 in TS 36.101 [4]. For reference sensitivity exception test points where the specified carrier frequency does not correspond to a valid NR-ARFCN, the closest NR-ARFCN as specified in clause 5.4.2 applies.

For operations with 4 or 8 Rx antenna ports in an E-UTRA band or 4 Rx antenna ports in an NR band, the MSD in the applicable bands shall be increased by the absolute value of $\Delta R_{IB,4R}$ in Table 7.3.1-1a or $\Delta R_{IB,8R}$ in Table 7.3.1-1aa of TS 36.101[4] for the E-UTRA band or $\Delta R_{IB,4R}$ in Table 7.3.2-2 of TS 38.101-1 for the NR band when $MSD > 0$.

NOTE: For inter-band EN-DC, the reference sensitivity requirement with both uplink carriers active is allowed to be verified for only a single inter-band EN-DC configuration per NR band.

7.3B.2 Reference sensitivity for DC

7.3B.2.1 Intra-band contiguous EN-DC

For intra-band contiguous EN-DC configurations, the reference sensitivity power level REFSENS is the minimum mean power applied to each one of the UE antenna ports at which the throughput for the carrier(s) of the E-UTRA and NR CGs shall meet or exceed the requirements for the specified E-UTRA and NR reference measurement channels. The reference sensitivity requirements apply with all uplink carriers and all downlink carriers active for EN-DC configuration and Uplink EN-DC configuration listed in Table 5.5B.2-1 and Table 5.5B.3-1, as supported by the UE. For EN-DC

configurations where uplink is not available in either the MCG or the SCG or for EN-DC configurations where the UE only supports single uplink operation, reference sensitivity requirements apply with single uplink transmission. The downlink carrier(s) from the cell group with uplink shall be configured closer to the uplink operating band than any of the downlink carriers from the cell group without uplink. Sensitivity degradation is allowed for Intra-band contiguous EN-DC configurations listed in Table 7.3B.2.1-1 the reference sensitivity is defined only for the specific uplink and downlink test points which are specified in Table 7.3B.2.1-1 and E-UTRA and NR single carrier requirements do not apply.

Table 7.3B.2.1-1: Reference sensitivity (MSD) for intra-band contiguous EN-DC

EN-DC configuration / channel allocations /MSD							
EN-DC configuration	E-UTRA/NR band	F _C (UL) (MHz)	Channel bandwidth (MHz)	UL allocation (LCRB)	F _C (DL) (MHz)	MSD (dB)	Duplex mode
DC_(n)71AA	71	665.5	5	5 (RB _{end} =24)	619.5	0	FDD
	n71	675.5	15	15 (RB _{start} = 0)	629.5	1.8	
DC_(n)71AA	71	670.5	15	15 (RB _{end} = 74)	624.5	0	
	n71	680.5	5	5 (RB _{start} = 0)	634.5	1.6	
DC_(n)71AA	71	668	10	10 (RB _{end} = 49)	622	0	
	n71	678	10	10 (RB _{start} = 0)	632	1.7	
DC_(n)71AA	71	668	10	10 (RB _{start} = 0)	622	17.2	
	n71	678	10	10 (RB _{end} = 51)	632	29.4	
NOTE 1: The transmitters powers shall be set to P _{UMAX} , as defined in TS 38.101-1 [2], TS 38.101-2 [3], and TS 36.101 [4], with additional limits on configured maximum output power for the uplink according to clause 6.2B.4.							

7.3B.2.2 Intra-band non-contiguous EN-DC

For intra-band non-contiguous EN-DC configurations, the reference sensitivity power level REFSENS is the minimum mean power applied to each one of the UE antenna ports at which the throughput for the carrier(s) of the E-UTRA and NR CGs shall meet or exceed the requirements for the specified E-UTRA and NR reference measurement channels.

For DC_3A_n3A intra-band non-contiguous EN-DC combination, only single switched UL is supported in rel.15 therefore, no MSD is specified and E-UTRA and NR single carrier requirements apply.

7.3B.2.3 Inter-band EN-DC within FR1

Reference sensitivity exceptions are specified for the condition when there is uplink transmission only in the aggressor band.

7.3B.2.3.1 Reference sensitivity exceptions due to UL harmonic interference for EN-DC in NR FR1

Sensitivity degradation is allowed for a band if it is impacted by UL harmonic interference from another band part of the same EN-DC configuration. Reference sensitivity exceptions for the victim band (high) are specified in Table 7.3B.2.3.1-1 with uplink configuration of the aggressor band (low) specified in Table 7.3B.2.3.1-2.

Table 7.3B.2.3.1-1: Reference sensitivity exceptions (MSD) due to UL harmonic for EN-DC in NR FR1

E-UTRA or NR Band / Channel bandwidth of the affected DL band / MSD													
UL band	DL band	5 MHz (dB)	10 MHz (dB)	15 MHz (dB)	20 MHz (dB)	25 MHz (dB)	30 MHz (dB)	40 MHz (dB)	50 MHz (dB)	60 MHz (dB)	80 MHz (dB)	90 MHz (dB)	100 MHz (dB)
1, 3	n77 ^{2,13}		23.9	22.1	20.9			17.9	16.8	16.0	14.8	14.3	13.8
	n77 ³		1.1	0.8	0.3								
2	n78 ^{2,13}		23.9	22.1	20.9			17.9	16.8	16.0	14.8	14.3	13.8
	n78 ³		1.1	0.8	0.3								
3	n78 ^{2,13}		23.9	22.1	20.9			17.9	16.8	16.0	14.8	14.3	13.8
	n78 ³		1.1	0.8	0.3								
5	n78 ^{6,7}		10.5	8.9	7.8			5.4	4.2	3.5	2.3	2.1	1.4
8	n77 ^{6,7} n78 ^{6,7}		10.8	9.1	8			5.1	4.2	3.5	2.3	2.1	1.4
8	n79 ^{4,5}							6.8	6.2	5.6	4.9		4.4
12	n66 ^{8,9,10}	10	7.5	6.2	5.5			2.4					
18	n77 ^{4,5} n78 ^{4,5}		10.4	8.9	7.8			4.7	3.7	3	1.7	1.2	0.7
19	n77 ^{4,5}		10.4	8.9	7.8			4.7	3.7	3	1.7	1.2	0.7
	n77 ^{6,7} n78 ^{6,7}		10.5	8.9	7.8			5.4	4.2	3.5	2.3	2.1	1.4
28	n51 ^{2,13}	27.8											
	n51 ³	1.9											
28	n77 ^{4,5} n78 ^{4,5}		10.4	8.9	7.8			4.7	3.7	3	1.7	1.2	0.7
20	n77 ^{6,7} n78 ^{6,7}		10.8	9.1	8			6	4.0	3.2	2.0	1.5	1.0
26	n41 ^{8,9}		10.3	8.4	7.4			5	4.3	3.9	3.1	2.9	2.7
26	n77 ^{6,7} n78 ^{6,7}		10.8	9.1	8			6	4.0	3.2	2.0	1.5	1.0
n28	1 ^{8,9,10}	10.2	7.6	6.2	5.3								
n71	2 ¹¹	4.6	1.0	0.7	0.6								
	2 ¹²	1.7	1.0	0.7	0.6								
66	n78 ^{2,13}		23.9	22.1	20.9			17.9	16.8	16.0	14.8	14.3	13.8
	n78 ³		1.1	0.8	0.3								

- NOTE 1: Void
- NOTE 2: The requirements should be verified for UL EARFCN or NR ARFCN of the aggressor (lower) band (superscript LB) such that in MHz and with carrier frequency in the victim (higher) band in MHz and the channel bandwidth configured in the lower band.
- NOTE 3: The requirements are only applicable to channel bandwidths no larger than 20 MHz and with a carrier frequency at MHz offset from in the victim (higher band) with , whereandare the channel bandwidths configured in the aggressor (lower) and victim (higher) bands in MHz, respectively.
- NOTE 4: These requirements apply when there is at least one individual RE within the uplink transmission bandwidth of the aggressor (lower) band for which the 5th transmitter harmonic is within the downlink transmission bandwidth of a victim (higher) band.
- NOTE 5: The requirements should be verified for UL EARFCN of the aggressor (lower) band (superscript LB) such that in MHz and with carrier frequency in the victim (higher) band in MHz and the channel bandwidth configured in the lower band.
- NOTE 6: These requirements apply when there is at least one individual RE within the uplink transmission bandwidth of the aggressor (lower) band for which the 4th transmitter harmonic is within the downlink transmission bandwidth of a

	victim (higher) band.
NOTE 7:	The requirements should be verified for UL EARFCN of the aggressor (lower) band (superscript LB) such that in MHz and with carrier frequency in the victim (higher) band in MHz and the channel bandwidth configured in the lower band.
NOTE 8:	These requirements apply when there is at least one individual RE within the uplink transmission bandwidth of the aggressor (lower) for which the 3rd transmitter harmonic is within the downlink transmission bandwidth of a victim (higher) band.
NOTE 9:	The requirements should be verified for UL EARFCN of the aggressor (lower) band (superscript LB) such that in MHz and with the carrier frequency in the victim (higher) band in MHz and the channel bandwidth configured in the low band.
NOTE 10:	Void.
NOTE 11:	These requirements apply when the lower edge frequency of the 5 MHz uplink channel in Band 71 is located at or below 668 MHz and the downlink channel in Band 2 is located with its upper edge at 1990 MHz.
NOTE 12:	These requirements apply when the lower edge frequency of the 10 MHz, 15 MHz, or 20 MHz uplink channel in Band 71 is located at or below 668 MHz and the downlink channel in Band 2 is located with its upper edge at 1990 MHz.
NOTE 13:	These requirements apply when there is at least one individual RE within the uplink transmission bandwidth of the aggressor (lower) band for which the 2nd transmitter harmonic is within the downlink transmission bandwidth of a victim (higher) band and a range ΔF_{HD} above and below the edge of this downlink transmission bandwidth. The value ΔF_{HD} depends on the EN-DC band combination: $\Delta F_{HD} = 10$ MHz for DC_1_n77, DC_2_n77, DC_66_n77, DC_3_n77, DC_3_n78, DC_28_n51, DC_66_n78.
NOTE 14:	MSD test point can be chosen according to supported BW and lowest SCS supported by the UE.

Table 7.3B.2.3.1-2: Uplink configuration for reference sensitivity exceptions due to UL harmonic interference for EN-DC in NR FR1

E-UTRA or NR Band / Channel bandwidth of the affected DL band / UL RB allocation of the aggressor band														
UL band	DL band	SCS of UL band (kHz)	5 MHz (L_{CRB})	10 MHz (L_{CRB})	15 MHz (L_{CRB})	20 MHz (L_{CRB})	25 MHz (L_{CRB})	30 MHz (L_{CRB})	40 MHz (L_{CRB})	50 MHz (L_{CRB})	60 MHz (L_{CRB})	80 MHz (L_{CRB})	90 MHz (L_{CRB})	100 MHz (L_{CRB})
1	n77	15		25	36	50			100	100	100	100	100	100
2	n78	15		25	36	50			50	50	50	50	50	50
3	n77, n78	15		25	36	50			50	50	50	50	50	50
5	n78	15		16	25	25			25	25	25	25	25	25
8	n77 n78	15		16	25	25			25	25	25	25	25	25
8	n79	15							25	25	25	25		25
12	n66	15	8	16	20	20			20					
18	n77, n78	15		16	25	25			25	25	25	25	25	25
19	n77, n78	15		16	25	25			25	25	25	25	25	25
20	n77, n78	15		16	25	25			25	25	25	25	25	25
26	n41	15		16	25	25			25	25	25	25	25	25
26	n77, n78	15		16	25	25			25	25	25	25	25	25
n28	1	15	8	16	25	25								
28	n51	15	12											
28	n77, n78	15		10	15	20			25	25	25	25	25	25
66	n78	15		25	36	50			100	100	100	100	100	100
n71	2	15	25^4 8^5	25^4 8^5	20^4 8^5	20^4 8^5								
NOTE 1: The UL configuration applies regardless of the channel bandwidth of the UL band unless the UL resource blocks exceed that specified in Table 7.3.1-2 in TS 36.101 [4] or Table 7.3.2-3 in TS 38.101-1 [2] for the uplink bandwidth in which case the allocation according to Table 7.3.1-2 in TS 36.101 [4] or Table 7.3.2-3 in TS 38.101-1 [2] applies														
NOTE 2: Void														
NOTE 3: Unless stated otherwise, UL resource blocks shall be centred within the transmission bandwidth configuration for the channel bandwidth.														

NOTE 4: These requirements apply when the lower edge frequency of the 5 MHz uplink channel in Band 71 is located at or below 668 MHz and the downlink channel in Band 2 is located with its upper edge at 1990 MHz.
NOTE 5: These requirements apply when the lower edge frequency of the 10 MHz, 15 MHz, or 20 MHz uplink channel in Band 71 is located at or below 668 MHz and the downlink channel in Band 2 is located with its upper edge at 1990 MHz.
NOTE 6: If the aggressor band is NR band, the test SCS and UL RB can be adjusted according to supported BW and lowest SCS supported by the UE

7.3B.2.3.2 Reference sensitivity exceptions due to receiver harmonic mixing for EN-DC in NR FR1

Sensitivity degradation is allowed for a band if it is impacted by receiver harmonic mixing due to another band part of the same EN-DC configuration. Reference sensitivity exceptions for the victim band (low) are specified in Table 7.3B.2.3.2-1 with uplink configuration of the aggressor band (high) specified in Table 7.3B.2.3.2-2.

Table 7.3B.2.3.2-1: Reference sensitivity exceptions (MSD) due to receiver harmonic mixing for EN-DC in NR FR1

E-UTRA or NR Band / Channel bandwidth of the affected DL band / MSD												
UL band	DL band	5 MHz (dB)	10 MHz (dB)	15 MHz (dB)	20 MHz (dB)	25 MHz (dB)	40 MHz (dB)	50 MHz (dB)	60 MHz (dB)	80 MHz (dB)	90 MHz (dB)	100 MHz (dB)
2	n71 ⁴	26.8	23.6	21.2	15.6							
n41	26 ⁴	24.3	24.3	22.5	N/A							
n77	3	5.7	4.0	3.0	2.7							
n78	3	5.7	4.0	3.0	2.7							
n77	19	7.2	5.0	3.8								
n77	41 ⁸	10.4	10.4	10.4	10.4							
n77	28 ²	28	25	23.2	22							
n78	19	7.2	5.0	3.8								
n78	41 ⁸	10.4	10.4	10.4	10.4							
n79	11 ⁴	39.3	36.3	34.5								
n79	19 ²	29.5	26.5	24.7								
n79	21 ⁴	39.3	36.3	34.5								
n79	26 ²	27	24	22.2								
n79	8 ²	25	22									

NOTE 1: These requirements apply when there is at least one individual RE within the uplink transmission bandwidth of the aggressor (higher) band for which the mixing product due to harmonic of victim (lower) band LO with leakage of aggressor (higher) band is within the downlink transmission bandwidth of a victim (lower) band.
NOTE 2: The requirements should be verified for DL EARFCN of the victim (lower) band (superscript LB) such that and with the DL carrier frequency in the lower band and the UL carrier frequency and the channel bandwidth configured in the higher band, both in MHz.
NOTE 3: Void
NOTE 4: The requirements should be verified for DL EARFCN or NRARFCN of the victim (lower) band (superscript LB) such that and with the DL carrier frequency in the lower band and the UL carrier frequency and the channel bandwidth configured in the higher band, both in MHz.
NOTE 5: Void
NOTE 6: Void
NOTE 7: Void
NOTE 8: The requirements should be verified for DL EARFCN of the victim (lower) band (superscript LB) such that and with the DL carrier frequency in the lower band and the UL carrier frequency and the channel bandwidth configured in the higher band, both in MHz.
NOTE 9: MSD test point can be chosen according to supported BW and lowest SCS supported by the UE.

Table 7.3B.2.3.2-2: Uplink configuration for reference sensitivity exceptions due to receiver harmonic mixing for EN-DC in NR FR1

E-UTRA or NR Band / SCS / Channel bandwidth of the affected DL band / UL RB allocation of the agressor band													
UL band	DL band	SCS of UL band (kHz)	5 MHz (L _{CRB})	10 MHz (L _{CRB})	15 MHz (L _{CRB})	20 MHz (L _{CRB})	25 MHz (L _{CRB})	40 MHz (L _{CRB})	50 MHz (L _{CRB})	60 MHz (L _{CRB})	80 MHz (L _{CRB})	90 MHz (L _{CRB})	100 MHz (L _{CRB})
2	n71	15	25	50	50	50							
n41	26	15	25	50	75								
n77	3	15	25	50	75	100							
n77	19	15	25	50	75								
n78	3	15	25	50	75	100							
n77	28	15	25	50	75	100							
n78	19	15	25	50	75								
n77	41	15	12	25	36	50							
n78	41	15	12	25	36	50							
n79	11	15	25	50	75								
n79	19	15	25	50	75								
n79	21	15	25	50	75								
n79	26	15	25	50	75								
n79	8	15	25	50									
NOTE 1: Void													
NOTE 2: Void													
NOTE 3: The UL configuration applies regardless of the channel bandwidth of the UL band. UL resource blocks allocation in the table shall be further limited to that specified in Table 7.3.1-2 in TS 36.101 [4] or Table 7.3.2-3 in TS 38.101-1 [2].													
NOTE 4: Unless otherwise stated, the UL resource blocks allocation is applied at the center of the channel bandwidth. The note applies to the entire table.													
NOTE 5: If the aggressor band is NR band, the test SCS and UL RB can be adjusted according to supported BW and lowest SCS supported by the UE.													

7.3B.2.3.3 Void

7.3B.2.3.4 Reference sensitivity exceptions due to cross band isolation for EN-DC in NR FR1

Sensitivity degradation is allowed for a band if it is impacted by UL of another band part of the same EN-DC configuration due to cross band isolation issues. Reference sensitivity exceptions for the victim band are specified in Table 7.3B.2.3.4-1 with uplink configuration of the agressor band specified in Table 7.3B.2.3.4-2.

Table 7.3B.2.3.4-1: Reference sensitivity exceptions (MSD) due to cross band isolation for EN-DC in NR FR1

E-UTRA or NR Band / Channel bandwidth of the affected DL band / MSD												
UL band	DL band	5 MHz (dB)	10 MHz (dB)	15 MHz (dB)	20 MHz (dB)	25 MHz (dB)	30 MHz (dB)	40 MHz (dB)	50 MHz (dB)	60 MHz (dB)	80 MHz (dB)	90 MHz (dB)
1	n40	6.6	6.6	6.6	6.6	6.6	6.6	6.6	6.6	6.6	6.6	
n40	1	8.3	8.3	8.3	8.3							
n41	25	0.6	0.6	0.6	0.6							

n77	41 ¹	4.5	4.5	4.5	4.5								
41	n77		8.3	8.3	8.3			6.3	5.3	4.5	4.0	3.9	
3	n51	6.4											
30	n66	8.3	8.3	8.3	8.3			8.3					
n78	7 ¹	4.5	4.5	4.5	4.5								
n78	38	3.3	3.3	3.3	3.3								
n78	41 ¹	4.5	4.5	4.5	4.5								
n78	46				7								
41	n78		8.3	8.3	8.3			6.3	5.3	4.5	4.0	3.9	
n79	42 ⁴	2.8	2.8	2.8	2.8								

NOTE 1: Applicable only when harmonic mixing MSD for this combination is not applied.
NOTE 2: The DL victim band should be configured using the lowest SCS that is compatible with the highest CBW for which an MSD is supported by the UE.
NOTE 3: MSD test point can be chosen according to supported BW and lowest SCS supported by the UE.
NOTE 4: The requirements only apply for UEs supporting inter-band DC_42_n79 ENDC with simultaneous Rx/Tx capability. Simultaneous Rx/Tx capability does not apply for UEs supporting band 42 with a n77 implementation only. These restrictions are applicable to related higher order configurations.

Table 7.3B.2.3.4-2: Uplink configuration for reference sensitivity exceptions due to cross band isolation for EN-DC in NR FR1

		E-UTRA or NR Band / SCS / Channel bandwidth of the affected DL band / UL RB allocation of the aggressor band												
UL band	DL band	SCS of UL band (kHz)	5 MHz (L _{CRB})	10 MHz (L _{CRB})	15 MHz (L _{CRB})	20 MHz (L _{CRB})	25 MHz (L _{CRB})	30 MHz (L _{CRB})	40 MHz (L _{CRB})	50 MHz (L _{CRB})	60 MHz (L _{CRB})	80 MHz (L _{CRB})	90 MHz (L _{CRB})	100 MHz (L _{CRB})
1	n40	15	25	50	75	100	100	100	100	100	100	100		
n40	1	15	25	50	75	100								
n41	25	30	160	160	160	160								
n77	41	30	270	270	270	270								
41	n77	15		100	100	100			100	100	100	100	100	100
3	n51	15	25											
30	n66	15	25	25	25	25			25					
n78	7	30	270	270	270	270								
n78	38	30	270	270	270	270								
n78	41	30	270	270	270	270								
n78	46	30				270								
41	n78	15		100	100	100			100	100	100	100	100	100
n79	42	30	270 ⁴	270 ⁴	270 ⁴	270 ⁴								

NOTE 1: The UL configuration applies regardless of the channel bandwidth of the UL band. UL resource blocks allocation in the table shall be further limited to that specified in Table 7.3.1-2 in TS 36.101 [4] or Table 7.3.2-3 in TS 38.101-1 [2].
NOTE 2: When the maximum UL RB allocation "L_{CRB}" value is less than the maximum transmission bandwidth configuration "N_{RB}" defined in Table 5.3.2-1 in 38.101-1 [2] for the specified UL band SCS, the UL band should be configured using the lowest CBW that is compatible with the maximum specified L_{CRB} value.
NOTE 3: If the aggressor band is NR band, the test SCS and UL RB can be adjusted according to supported BW and lowest SCS supported by the UE.
NOTE 4: The requirements only apply for UEs supporting inter-band ENDC with simultaneous Rx/Tx capability. Simultaneous Rx/Tx capability does not apply for UEs supporting band 42 with a n77 implementation only. These restrictions are applicable to related higher order configurations.

7.3B.2.3.5 MSD for intermodulation interference due to dual uplink operation for EN-DC in NR FR1

For EN-DC configurations in NR FR1 the UE may indicate capability of not supporting simultaneous dual uplink operation due to possible intermodulation interference overlapping in frequency to its own primary downlink channel bandwidth if

- the intermodulation order is 2;
- the intermodulation order is 3 when both operating bands are between 450 MHz – 960 MHz or between 1427 MHz – 2690 MHz

In the case for EN-DC configurations in NR FR1 for which the intermodulation products caused by dual uplink operation do not interfere with its own primary downlink channel bandwidth as defined in Annex I the UE is mandated to operate in dual and triple uplink mode.

For EN-DC configurations in NR FR1 with uplink and downlink assigned to E-UTRA and NR FR1 bands given in Table 7.3B.2.3.5.1-1, Table 7.3B.2.3.5.2-0 and Table 7.3B.

2.3.5.2-1 the reference sensitivity is defined only for the specific uplink and downlink test points specified in Table 7.3B.2.3.5.1-1, Table 7.3B.2.3.5.2-0 and Table 7.3B.

2.3.5.2-1. For these test points the reference sensitivity levels specified in clause 7.3.1 in TS 36.101 [4] and 7.3.2 of TS 38.101-1 [2] for the corresponding channel bandwidths or in clause 7.3.1 of TS 36.101 [4] are relaxed by the amount of the parameter MSD given in Table 7.3B.2.3.5.1-1, Table 7.3B.2.3.5.2-0 and Table 7.3B.

2.3.5.2-1.

The throughput on each of the CGs shall be $\geq 95\%$ of the maximum throughput of the respective reference measurement channels as specified in Annex A of TS 38.101-1 [2] and Annex A of TS 36.101 [4], with parameters specified in Table 7.3B.2.3.5.1-1, Table 7.3B.2.3.5.2-0 and Table 7.3B.2.3.5.2-1 with dual UL transmissions overlapping in time unless otherwise stated.

7.3B.2.3.5.1 MSD test points for intermodulation interference due to dual uplink operation for EN-DC in NR FR1 involving two bands

Table 7.3B.2.3.5.1-1: MSD test points for PCell due to dual uplink operation for EN-DC in NR FR1 (two bands)

NR or E-UTRA Band / Channel bandwidth / N_{RB} / MSD							
EN-DC Configuration	EUTRA or NR band	UL F_c (MHz)	UL/DL BW (MHz)	UL L_{CRB}	DL F_c (MHz)	MSD (dB)	IMD order
DC_1A_n77A	1	1950	5	25	2140	29.8	IMD2 ³
	n77	4090	10	50	4090	N/A	N/A
DC_1A_n77A, DC_1A_n78A, DC_1A_SUL_n78A- n84A	1	1950	5	25	2140	8.0	IMD4 ³
	n77, n78	3710	10	50	3710	N/A	N/A
DC_2A_n66A	2	1855	5	25	1935	20	IMD3
	n66	1775	5	25	2175	N/A	N/A
DC_2A_n66A	2	1883.3	5	25	1963.3	N/A	N/A
	n66	1750	5	25	2150	4	IMD5
DC_2A_n78A	2	1855	5	25	1935	26	IMD2 ³
	n78	3790	10	50	3790	N/A	N/A
DC_2A_n78A	2	1885	5	25	1965	8.0	IMD4 ³
	n78	3690	10	50	3690	N/A	N/A

DC_3A_n7A	3	1730	5	25	1825	N/A	N/A
	n7	2535	10	50	2655	10.2	IMD4
DC_3A_n77A, DC_3A_n78A, DC_3A-SUL_n78A- n80A, DC_3C_n78A	3	1740	5	25	1835	26	IMD2 ³
	n77, n78	3575	10	50	3575	N/A	N/A
DC_3A_n77A, DC_3A_n78A, DC_3A-SUL_n78A- n80A, DC_3C_n78A	3	1765	5	25	1860	8.0	IMD4 ³
	n77, n78	3435	10	50	3435	N/A	N/A
DC_5A_n66A	5	838	5	25	883	30	IMD2 ³
	n66	1721	5	25	2121	N/A	N/A
DC_5A_n78A	5	844	5	25	889	8.3	IMD4
	n78	3421	10	50	3421	N/A	N/A
DC_8A_n77A, DC_8A_n78A, DC_8A-SUL_n78A- n81A	8	897.5	5	25	942.5	8.3	IMD4
	n77, n78	3635	10	50	3635	N/A	N/A
DC_8A_n79A, DC_8A-SUL_n79A- n81A	8	897.5	5	25	942.5	4.8	IMD5
	n79	4532.5	40	216	4532.5	N/A	N/A
DC_18A_n77A DC_18A_n78A	18	N/A	N/A	N/A	N/A	N/A	IMD4
	n77, n78	N/A	N/A	N/A	N/A	N/A	N/A
DC_19A_n77A	19	836.5	5	25	881.5	13.6	IMD4 ³
	n77	3391	10	50	3391	N/A	N/A
DC_19A_n78A	19	836.5	5	25	881.5	13.6	IMD4
	n78	3391	10	50	3391	N/A	N/A
DC_20A_n8A	20	849.5	5	25	808.5	25	IMD3
	n8	892.5	5	25	937.5	25	IMD3
DC_20A_n77A, DC_20A_n78A, DC_20A-SUL_n78A- n82A	20	850	5	25	809	11	IMD4
	n77, n78	3359	10	50	3359	N/A	N/A
DC_20A_n77A	20	840	5	25	799	6.5	IMD5
	n77	4159	10	50	4159	N/A	N/A
DC_21A_n79A	21	1457.5	5	25	1505.5	18.4	IMD3
	n79	4420.5	40	216	4420.5	N/A	N/A
DC_26A_n41A	26	839	5	25	884	15.6	IMD3 ³
	n41	2562	10	50	2562	N/A	N/A
DC_28A_n51A	28	742.3	5	25	797.3	5	IMD4
	n51	1429.5	5	25	1429.5	N/A	N/A
DC_26A_n77A, DC_26A_n78A	26	836.5	5	25	881.5	11.1	IMD4
	n77, n78	3391	10	50	3391	N/A	N/A
DC_28A_n77A, DC_28A_n78A, DC_28A-SUL_n78A- n83A	28	705.5	5	25	760.5	5.5	IMD5
	n77, n78	3582.5	10	50	3582.5	N/A	N/A
DC_66A_n5A	n5	838	5	25	883	30	IMD2 ³
	66	1721	5	25	2121	N/A	N/A
DC_66A_n71A	66	1750	5	25	2150	5	IMD4
	n71	675	5	25	629	N/A	N/A
DC_66A_n78A	66	1730	5	25	2130	5.0	IMD5
	n78	3660	5	50	3660	N/A	N/A
NOTE 1: Both of the transmitters shall be set min(+20 dBm, P _{C_{MAX},L,c}) as defined in clause 6.2.5A.							
NOTE 2: RB _{start} = 0							
NOTE 3: This band is subject to IMD5 also which MSD is not specified.							
NOTE 4: Void.							

NOTE 5: Void
NOTE 6: For NR band, UL/DL BW and UL L_{CRB} can be adjusted according to the supported BW and lowest SCS supported by the UE.

7.3B.2.3.5.2 MSD test points for intermodulation interference due to dual uplink operation for EN-DC in NR FR1 involving three bands

Table 7.3B.2.3.5.2-0: MSD test points for Pcell due to dual uplink operation for EN-DC in NR FR1 (three bands)

NR or E-UTRA Band / Channel bandwidth / N_{RB} / MSD							
EN-DC Configuration	EUTRA/NR band	UL F_c (MHz)	UL/DL BW (MHz)	UL L_{CRB}	DL F_c (MHz)	MSD (dB)	IMD order
DC_66A_(n)71AA	66	1750	5	25	2150	5	IMD4
	n71	678	10	10 ($RB_{start}=0$)	632	N/A	N/A

NOTE 1: For NR band, UL/DL BW and UL L_{CRB} can be adjusted according to the supported BW and lowest SCS supported by the UE.

Table 7.3B.2.3.5.2-1: MSD test points for Scell due to dual uplink operation for EN-DC in NR FR1 (three bands)

NR or E-UTRA Band / Channel bandwidth / N_{RB} / MSD							
EN-DC Configuration	EUTRA / NR band	UL F_c (MHz)	UL/DL BW (MHz)	UL L_{CRB}	DL F_c (MHz)	MSD (dB)	IMD order
DC_1A-3A_n28A	1	1975	5	25	2165	N/A	N/A
	n28	710.5	5	25	765.5	N/A	N/A
	3	1723.5	5	25	1818.5	4.0	IMD5
DC_1A-3A_n28A	3	1780	5	25	1875	N/A	N/A
	n28	710.5	5	25	765.5	N/A	N/A
	1	1949	5	25	2139	11.0	IMD4
DC_1A-7A_n28A	1	1935	5	25	2125	N/A	N/A
	n28	718	5	25	773	N/A	N/A
	7	2533	10	50	2653	30.0	IMD2
DC_1A-3A_n77A DC_1A-3A_n77C	1	1950	5	25	2140	N/A	N/A
	3	1712.5	5	25	1807.5	31.5	IMD2
	n77	3757.5	10	50	3757.5	N/A	N/A
	1	1950	5	25	2140	N/A	N/A
	3	1775	5	25	1870	8.5	IMD4
	n77	3980	10	50	3980	N/A	N/A
	1	1950	5	25	2140	31.0	IMD2
	3	1775	5	25	1870	N/A	N/A
	n77	3915	10	50	3915	N/A	N/A
DC_1A-3A_n78A DC_1A-3A_n78C DC_1A-3C_n78A	1	1950	5	25	2140	N/A	N/A
	3	1712.5	5	25	1807.5	31.2	IMD2
	n78	3757.5	10	50	3757.5	N/A	N/A
	1	1935	5	25	2125	2.8	IMD5
	3	1775	5	25	1870	N/A	N/A

	n78	3725	10	50	3725	N/A	N/A
DC_1A-5A_n78A	1	1932	5	25	2122	18.1	IMD3
	5	829	5	25	874	N/A	N/A
	n78	3780	10	50	3780	N/A	N/A
	1	1975	5	25	2165	N/A	N/A
	5	840	5	25	885	3.1	IMD5
	n78	3405	10	50	3405	N/A	N/A
DC_1A-7A_n78A	1	1977.5	5	25	2167.5	N/A	N/A
	7	2507.5	5	25	2627.5	9.1	IMD4
	n78	3305	10	50	3305	N/A	N/A
	1	1950	5	25	2140	8.7	IMD4
	7	2510	10	50	2630	N/A	N/A
	n78	3580	10	50	3580	N/A	N/A
DC_1A-8A_n78A	1	1945	5	25	2135	N/A	N/A
	8	900	5	25	945	N/A	N/A
	n78	3745	10	52	3745	14.9	IMD3
	1	1940	5	25	2130	N/A	N/A
	8	895	5	25	940	3.3	IMD5
	n78	3380	10	52	3330	N/A	N/A
DC_1A-3A_n79A	1	1950	5	25	2140	3.6	IMD5
	3	1750	5	25	1845	N/A	N/A
	n79	4860	40	216	4860	N/A	N/A
DC_1A-18A_n77A	1	N/A	N/A	N/A	N/A	N/A	N/A
	18	N/A	N/A	N/A	N/A	N/A	IMD5
	n77	N/A	N/A	N/A	N/A	N/A	N/A
	1	1930	5	25	2120	16.4	IMD3
	18	825	5	25	870	N/A	N/A
	n77	3770	10	50	3770	N/A	N/A
DC_1A-18A_n78A	1	N/A	N/A	N/A	N/A	N/A	N/A
	18	N/A	N/A	N/A	N/A	N/A	IMD5
	n78	N/A	N/A	N/A	N/A	N/A	N/A
	1	1930	5	25	2120	16.4	IMD3
	18	819	5	25	864	N/A	N/A
	n78	3758	10	50	3758	N/A	N/A
DC_1A-18A_n79A	1	1935	5	25	2125	N/A	N/A
	18	822.5	5	25	867.5	18.3	IMD3
	n79	4737.5	40	216	4737.5	N/A	N/A
	1	1930	5	25	2120	N/A	N/A
	18	820	5	25	865	8.9	IMD4
	n79	4925	40	216	4925	N/A	N/A
	1	1935	5	25	2125	8.1	IMD4
	18	822.5	5	25	867.5	N/A	N/A
DC_1A-19A_n77A DC_1A-19A_n78A	n79	4592.5	40	216	4592.5	N/A	N/A
	1	1940	5	25	2130	17.8	IMD3
	19	832.5	5	25	877.5	N/A	N/A
	n77, n78	3795	10	50	3795	N/A	N/A
	1	1940	5	25	2130	N/A	N/A
	19	835	5	25	880	5.1	IMD5

	1	1977.5	5	25	2167.5	1.2	IMD4
	28	745.5	5	25	800.5	N/A	N/A
	n79	4420	40	216	4420	N/A	N/A
	1	1935	5	25	2125	4.5	IMD5
	28	718	5	25	773	N/A	N/A
	n79	4807	40	216	4807	N/A	N/A
DC_1A-41A_n77A	1	1970	5	25	2160	N/A	N/A
	n77	3400	10	50	3400	N/A	
	41	2510	5	25	2510	11.0	IMD4
	1	1950	5	25	2140	9.3	IMD4
	n77	3710	10	50	3710	N/A	N/A
	41	2640	5	25	2640	N/A	N/A
	1	1930	5	25	2120	N/A	N/A
	n77	4150	10	50	4150	N/A	
DC_1A-41A_n78A	41	2510	5	25	2510	3.6	IMD5
	1	1950	5	25	2140	9.3	IMD4
	41	2640	5	25	2640	N/A	N/A
	n78	3710	10	50	3710	N/A	N/A
	1	1975	5	25	2165	N/A	N/A
	41	2515	5	25	2515	12	IMD4
DC_1A-41A_n78A	n78	3410	10	50	3410	N/A	N/A
	1	1955	5	25	2145	8.7	IMD4
	41	2507.5	10	50	2507.5	N/A	N/A
DC_1A-41A_n79A	n78	3580	10	50	3580	N/A	N/A
	1	1970	5	25	2160	N/A	N/A
	n79	4500	40	216	4500	N/A	
DC_1A-42A_n79A	41	2530	5	25	2530	29.4	IMD2
	1	1977.5	5	25	2167.5	N/A	N/A
	n79	4420	40	216	4420	N/A	N/A
	42	3490	5	25	3490	4.8	IMD5
	42	3402.5	5	25	3402.5	N/A	N/A
	n79	4640	40	216	4640	N/A	N/A
	1	1975	5	25	2165	15.5	IMD3
	42	3450	5	25	3450	N/A	N/A
	n79	4520	40	216	4520	N/A	N/A
DC_1A_n78A-n79A	1	1950	5	25	2140	9.3	IMD4
	1	1950	5	25	2140	N/A	N/A
	n78	3410	10	50	3410	N/A	N/A
	n79	4870	40	216	4870	15.9	IMD3
	1	1950	5	25	2140	N/A	N/A
	n79	4670	40	216	4670	N/A	N/A
DC_2A-12A_n66A	n78	3490	10	50	3490	4.6	IMD5
	2	N/A	N/A	N/A	N/A	N/A	IMD4
	12	N/A	N/A	N/A	N/A	N/A	N/A
DC_3A-5A_n78A	n66	N/A	N/A	N/A	N/A	N/A	N/A
	3	N/A	N/A	N/A	N/A	N/A	IMD3
	5	N/A	N/A	N/A	N/A	N/A	N/A
DC_3A-7A_n28A	n78	N/A	N/A	N/A	N/A	N/A	N/A
	3	1712.5	5	25	1807.5	N/A	N/A

	n28	743	5	25	798	N/A	N/A
	7	2562	10	50	2682	16.9	IMD3
	7	2543	10	50	2663	N/A	N/A
	n28	710.5	5	25	765.5	N/A	N/A
	3	1737.5	5	25	1832.5	26.0	IMD2
DC_3A-7A_n78A DC_3C-7A_n78A DC_3C-7C_n78A	3	1725	5	25	1820	17.6	IMD3
	7	2565	5	25	2685	N/A	N/A
	n78	3310	10	50	3310	N/A	N/A
	3	1725	5	25	1820	8.6	IMD4
	7	2565	5	25	2685	N/A	N/A
	n78	3475	10	50	3475	N/A	N/A
DC_3A-8A_n78A	8	910	5	25	955	N/A	N/A
	n78	3640	10	50	3640	N/A	N/A
	3	1725	5	25	1820	16.5	IMD3
DC_3A-19A_n77A DC_3A-19A_n78A	3	1775	5	25	1850	17.3	IMD3
	19	835	5	25	880	N/A	N/A
	N77, n78	3520	10	50	3520	N/A	N/A
DC_3A-19A_n79A	3	1775	5	25	1870	N/A	N/A
	19	840	5	25	885	18.5	IMD3
	n79	4435	40	216	4435	N/A	N/A
	3	1782.5	5	25	1877.5	5.5	IMD4
	19	842.5	5	25	887.5	N/A	N/A
	n79	4420	40	216	4420	N/A	N/A
DC_3A-20A_n28A	20	852	5	25	811	N/A	N/A
	n28	728	5	25	783	N/A	N/A
	3	1733	5	25	1828	9.4	IMD4
DC_3A-20A_n78A DC_3C-20A_n78A	3	1725	5	25	1820	17.3	IMD3
	20	845	5	25	804	N/A	N/A
	n78	3510	10	50	3510	N/A	N/A
DC_3A-21A_n77A DC_3A-21A_n78A	3	1767.5	5	25	1862.5	N/A	N/A
	21	1459.5	5	25	1507.5	8.8	IMD4 ²
	n77, n78	3795	10	50	3795	N/A	N/A
	3	1767.5	5	25	1862.5	30.8	IMD2
	21	1459.5	5	25	1507.5	N/A	N/A
	N77, n78	3322	10	50	3322	N/A	N/A
DC_3A-21A_n77A	3	1771.6	5	25	1866.6	3.4	IMD5
	21	1450.4	5	25	1498.4	N/A	N/A
	n77	3935	10	50	3935	N/A	N/A
DC_3A-21A_n79A	3	N/A	N/A	N/A	N/A	N/A	N/A
	21	N/A	N/A	N/A	N/A	N/A	IMD3
	n79	N/A	N/A	N/A	N/A	N/A	N/A
	3	1774.2	5	25	1869.2	17.8	IMD3
	21	1450.4	5	25	1498.4	N/A	N/A
	n79	4770	40	216	4770	N/A	N/A
DC_3A-28A_n77A	3	1712.5	5	25	1807.5	N/A	N/A
	28	715	5	25	770	15.3	IMD3
	n77	4195	10	50	4195	N/A	N/A
	3	1755	5	25	1850	17.0	IMD3
	28	735	5	25	790	N/A	N/A
	n77	3320	10	50	3320	N/A	N/A

DC_3A-28A_n78A	3	1775	5	25	1870	17.3	IMD3
	28	740	5	25	760	N/A	N/A
	n78	3350	10	25	3350	N/A	N/A
DC_3A-28A_n79A	3	1770	5	25	1865	N/A	N/A
	28	725	5	25	780	10.3	IMD4
	n79	4530	40	216	4530	N/A	N/A
	3	1775	5	25	1870	5.7	IMD5
	28	725	5	25	780	N/A	N/A
	n79	4770	40	216	4770	N/A	N/A
DC_3A_n28A-n78A	3	1750	5	25	1845	N/A	N/A
	n28	743	5	25	798	N/A	N/A
	n78	3764	10	50	3764	4.5	IMD5
DC_3A-41A_n78A	41	2620	5	25	2620	N/A	N/A
	n78	3400	10	50	3400	N/A	N/A
	3	1745	5	25	1840	16.4	IMD3
DC_3A_n78A-n79A	3	1770	5	25	1865	N/A	N/A
	n78	3340	10	50	3340	N/A	N/A
	n79	4910	40	216	4910	16.3	IMD3
	3	1770	5	25	1865	N/A	N/A
	n79	4510	40	216	4510	N/A	N/A
	n78	3710	10	50	3710	4.2	IMD5
DC_3A-SUL_n78A-n82A	3	1775	5	25	1870	4	IMD4
	n82	840	5	25		N/A	N/A
DC_5A-7A_n78A	5	844	5	25	889	N/A	N/A
	7	2525	10	50	2645	30.1	IMD2
	n78	3489	10	50	3489	N/A	N/A
	5	834	5	25	879	30.2	IMD2
	7	2550	10	50	2670	N/A	N/A
	n78	3429	10	50	3429	N/A	N/A
	5	830	5	25	875	3.3	IMD5
	7	2525	10	50	2645	N/A	N/A
	n78	3350	10	50	3350	N/A	N/A
DC_5A-41A_n78A	5	860	5	25	885	30.2	IMD2
	41	2615	20	100	2615	N/A	N/A
	n78	3500	10	50	3500	N/A	N/A
	5	856.5	5	25	881.5	3.1	IMD5
	41	2620.5	20	100	2620.5	N/A	N/A
	n78	3490	10	50	3490	N/A	N/A
DC_7A-20A_n28A	20	842	5	25	801	N/A	N/A
	n28	728	5	25	783	N/A	N/A
	7	2520	10	50	2640	5.9	IMD5
DC_7A-20A_n78A	7	2560	5	25	2680	N/A	N/A
	20	851	5	25	810	30.5	IMD2
	n78	3370	10	50	3370	N/A	N/A
DC_7A-20A_n78A	7	2560	5	25	2680	N/A	N/A
	20	851	5	25	810	3.0	IMD5

	n78	3435	10	50	3435	N/A	N/A
DC_7A-20A_n78A	7	2555	5	25	2675	30.8	IMD2
	20	845	5	25	804	N/A	N/A
	n78	3520	10	50	3520	N/A	N/A
DC_7A-28A_n78A	7	2567.5	5	25	2687.5	N/A	N/A
	28	727.5	5	25	782.5	28.8	IMD2
	n78	3350	10	50	3350	N/A	N/A
	7	2567.5	5	25	2687.5	N/A	N/A
	28	727.5	5	25	782.5	3.0	IMD5
	n78	3460	10	50	3460	N/A	N/A
	7	2530	5	25	2650	30.5	IMD2
	28	740	5	25	795	N/A	N/A
	n78	3390	10	50	3390	N/A	N/A
DC_7A_n28A-n78A	7	2565	5	25	2685	N/A	N/A
	n28	745	5	25	800	N/A	N/A
	n78	3310	10	50	3310	29.7	IMD2
	7	2565	5	25	2685	N/A	N/A
	n78	3365	10	50	3365	N/A	N/A
	n28	745	5	25	800	28.8	IMD2
DC_7A-46A_n78A ⁶	7	N/A	N/A	N/A	N/A	N/A	N/A
	46	N/A	N/A	N/A	N/A	N/A	IMD2, IMD5
	n78	N/A	N/A	N/A	N/A	N/A	N/A
DC_18A-28A_n77A	18	820	5	25	865	N/A	N/A
	28	723	5	25	778	4.4	IMD5
	n77	4058	10	50	4058	N/A	N/A
DC_18A-28A_n77A	18	820	5	25	865	3.9	IMD5
	28	723	5	25	778	N/A	N/A
	n77	3757	10	50	3757	N/A	N/A
DC_18A-28A_n78A	18	819	5	25	864	3.8	IMD5
	28	723	5	25	778	N/A	N/A
	n78	3756	10	50	3756	N/A	N/A
DC_19A-21A_n77A DC_19A-21A_n78A	19	837.5	5	25	882.5	18.7	IMD3
	21	1450.4	5	25	1498.4	N/A	N/A
	n77, n78	3783.3	10	50	3783.3	N/A	N/A
	19	837.5	5	25	882.5	13.2	IMD4
	21	1450.4	5	25	1498.4	N/A	N/A
	n77, n78	3468.7	10	50	3468.7	N/A	N/A
DC_19A-21A_n77A	19	837.5	5	25	882.5	N/A	N/A
	21	1454.5	5	25	1502.5	9.0	IMD4
	n77	4015	10	50	4015	N/A	N/A
DC_19A-21A_n79A	19	N/A	N/A	N/A	N/A	N/A	IMD5
	21	N/A	N/A	N/A	N/A	N/A	N/A
	n79	N/A	N/A	N/A	N/A	N/A	N/A
	19	837.5	5	25	882.2	N/A	N/A
	21	1452	5	25	1500	3.8	IMD5
	n79	4850	40	216	4850	N/A	N/A
DC_21A-28A_n77A DC_21A-28A_n78A	21	1452	5	25	1500	N/A	N/A
	28	730.5	5	25	785.5	16.9	IMD3
	n77/n78	3689.5	10	50	3689.5	N/A	N/A

	21	1450.5	5	25	1498.5	9.9	IMD4
	28	730.5	5	25	785.5	N/A	N/A
	n77/n78	3690	10	50	3690	N/A	N/A
DC_21A-28A_n79A	21	1450	5	25	1498	5.2	IMD5
	28	730.5	5	25	785.5	N/A	N/A
	n79	4420	40	216	4420	N/A	N/A
DC_28A-42A_79A	28	730	5	25	785	N/A	N/A
	42	3420	5	25	3420	15.3	IMD3
	n79	4880	40	216	4880	N/A	N/A
	28	745	5	25	800	16.2	IMD2
	42	3597.5	5	25	3597.5	N/A	N/A
	n79	4420	40	216	4420	N/A	N/A
DC_19A_n78A-n79A	19	835	5	25	880	N/A	N/A
	n78	3680	10	50	3680	N/A	N/A
	n79	4515	40	216	4515	29.3	IMD2
	19	835	5	25	880	N/A	N/A
	n79	4550	40	216	4550	N/A	N/A
	n78	3715	10	50	3715	28.8	IMD2
DC_20A_n28A-n78A, DC_20A_SUL_n78A- n83A	20	857	5	25	816	N/A	N/A
	n28, n83	743	5	25	798	N/A	N/A
	n78	3314	10	50	3314	8.7	IMD4
	20	837	5	25	796	N/A	N/A
	n78	3310	10	50	3310	N/A	N/A
	n28	744	5	25	799	9.4	IMD4
DC_21A_n78A-n79A	21	1453	5	25	1501	N/A	N/A
	n78	3420	10	50	3420	N/A	N/A
	n79	4873	40	216	4873	30.1	IMD2
	21	1453	5	25	1501	N/A	N/A
	n79	4940	40	216	4940	N/A	N/A
	n78	3487	10	50	3487	29.8	IMD2
NOTE 1: For NR band, UL/DL BW and UL L_{CRB} can be adjusted according to the supported BW and lowest SCS supported by the UE.							
NOTE 2: This band is subject to IMD5 also which MSD is not specified.							

7.3B.2.3.5.3 Void

7.3B.2.3.6 Reference sensitivity exceptions due to Tx non-linearity interference in 1st or 2nd adjacent channel of UL band for EN-DC in NR FR1

Sensitivity degradation is allowed for a band if it is impacted by Tx non-linearity interference in 1st or 2nd adjacent channel from another UL band part of the same EN-DC configuration. Reference sensitivity exceptions for the victim band (DL band) are specified in Table 7.3B.2.3.6-1 with uplink configuration of the aggressor band (UL band) specified in Table 7.3B.2.3.6-1.

Table 7.3B.2.3.6-1: Reference sensitivity exceptions (MSD) due to Tx non-linearity interference in 1st or 2nd adjacent channel of UL band for EN-DC in NR FR1

UL band	DL band	SCS of UL band (kHz)	L_{CRB} of UL band	Applicable UL BW(MHz)	MSD value of DL band (dB)	The adjacent channel of UL band
n40	1	30	25	≥ 70	21.5	2 nd adjacent channel

NOTE 1: For interference in 2nd adjacent channel, the MSD exceptions are applicable to the case that interference located in 2nd adjacent channel of UL band falls into the DL channels. (The victim frequency of DL band can be expressed as f_{v} , where f_c is the centre frequency of UL channel and f_a is the allocated transmission frequency of UL band).

7.3B.2.3a Inter-band NE-DC within FR1

Reference sensitivity exceptions are specified for the condition when there is uplink transmission only in the aggressor band. This clause addresses directly only NE-DC configurations that don't have a corresponding specified EN-DC configuration or specific NE-DC exceptions.

7.3B.2.3a.1 Reference sensitivity exceptions due to UL harmonic interference for NE-DC in NR FR1

Sensitivity degradation is allowed for a band if it is impacted by UL harmonic interference from another band part of the same NE-DC configuration. For the NE-DC configurations that have an EN-DC defined configuration, the reference sensitivity exceptions for the victim band (high) are specified in Table 7.3B.2.3.1-1 with uplink configuration of the aggressor band (low) specified in Table 7.3B.2.3.1-2 are applicable.

7.3B.2.4 Inter-band EN-DC including FR2

7.3B.2.4.1 Void

7.3B.2.5 Inter-band EN-DC including both FR1 and FR2

7.3B.2.5.1 Reference sensitivity exceptions due to UL harmonic interference for EN-DC including both FR1 and FR2

For inter-band EN-DC of E-UTRA and NR in both FR1 and FR2, the UE is allowed to apply each sensitivity degradation for EN-DC in FR1 specified in clause 7.3B.2.3 TS 38.101-3 and for EN-DC including FR2 specified in clause 7.3B.2.3 of TS 38.101-3 independently.

7.3B.3 $\Delta R_{IB,c}$, ΔR_{IBNC} for DC

7.3B.3.0 General

For the UE which supports inter-band EN-DC or NE-DC configuration, the minimum requirement for reference sensitivity in Table 7.3.1-1 and Table 7.3.1-1a in TS 36.101 [4], clause 7.3.2, 7.3A.2, 7.3C.2 in TS 38.101-1 [2] and clause 7.3.2, 7.3A.2 in TS 38.101-2 [3] shall be increased by the amount given in $\Delta R_{IB,c}$, ΔR_{IBNC} in Tables below where unless otherwise stated, the same $\Delta R_{IB,c}$, ΔR_{IBNC} are applicable to NR band(s) part for DC configurations which have the same NR operating band combination. Unless otherwise stated, $\Delta R_{IB,c}$ or ΔR_{IBNC} is set to zero.

In case the UE supports more than one of band combinations for CA, SUL or DC, and an operating band belongs to more than one band combinations then

- When the operating band frequency range is ≤ 1 GHz, the applicable additional $\Delta R_{IB,c}$ shall be the average value for all band combinations defined in clause 7.3A, 7.3B, 7.3C in TS 38.101-1 [2] and 7.3A, 7.3B in this specification, truncated to one decimal place that apply for that operating band among the supported band combinations. In case there is a harmonic relation between low band UL and high band DL, then the maximum $\Delta R_{IB,c}$ among the different supported band combinations involving such band shall be applied
- When the operating band frequency range is > 1 GHz, the applicable additional $\Delta R_{IB,c}$ shall be the maximum value for all band combinations defined in clause 7.3A, 7.3B, 7.3C in TS 38.101-1 [2] and 7.3A, 7.3B in this specification for the applicable operating bands.

Unless $\Delta R_{IB,c}$ is specified for the NE-DC configuration, the specified $\Delta R_{IB,c}$ for the EN-DC configuration including same bands as the corresponding NE-DC configuration is applicable for the NE-DC configuration.

7.3B.3.1 Intra-band contiguous EN-DC

7.3B.3.2 Intra-band non-contiguous EN-DC

Table 7.3B.3.2-1: Intra-band non-contiguous EN-DC with one uplink configuration on E-UTRA for reference sensitivity

DC configuration	Aggregated channel bandwidth		W_{gap} / (MHz)	UL E-UTRA allocation	ΔR_{IBNC} (dB)	Duplex mode
	E-UTRA	NR				
DC_3A_n3A	5 MHz	5 MHz	$45.0 < W_{gap} \leq 65.0$	12^1	4.7	FDD
			$0.0 < W_{gap} \leq 45.0$	25^1	0	
	5 MHz	10 MHz	$40.0 < W_{gap} \leq 60.0$	12^1	3.8	
			$0.0 < W_{gap} \leq 40.0$	25^1	0	
	5 MHz	15 MHz	$35.0 < W_{gap} \leq 55.0$	12^1	3.6	
			$0.0 < W_{gap} \leq 35.0$	25^1	0	
	5 MHz	20 MHz	$30.0 < W_{gap} \leq 50.0$	12^1	3.4	
			$0.0 < W_{gap} \leq 30.0$	25^1	0	
	5 MHz	25 MHz	$25.0 < W_{gap} \leq 45.0$	12^1	3.2	
			$0.0 < W_{gap} \leq 25.0$	25^1	0	
	5 MHz	30 MHz	$20.0 < W_{gap} \leq 40.0$	12^1	3.0	
			$0.0 < W_{gap} \leq 20.0$	25^1	0	
	10 MHz	5 MHz	$30.0 < W_{gap} \leq 60.0$	12^5	5.1	
			$0.0 < W_{gap} \leq 30.0$	32^1	0	
	10 MHz	10MHz	$25.0 < W_{gap} \leq 55.0$	12^5	4.3	
			$0.0 < W_{gap} \leq 25.0$	32^1	0	
	10 MHz	15 MHz	$20.0 < W_{gap} \leq 50.0$	12^5	3.8	
			$0.0 < W_{gap} \leq 20.0$	32^1	0	
	10 MHz	20 MHz	$15.0 < W_{gap} \leq 45.0$	12^5	3.5	
			$0.0 < W_{gap} \leq 15.0$	32^1	0	
	10 MHz	25 MHz	$10.0 < W_{gap} \leq 40.0$	12^5	3.2	
			$0.0 < W_{gap} \leq 10.0$	32^1	0	
	10 MHz	30 MHz	$5.0 < W_{gap} \leq 35.0$	12^5	2.8	
			$0.0 < W_{gap} \leq 5.0$	32^1	0	
	15 MHz	5 MHz	$25.0 < W_{gap} \leq 55.0$	12^6	6.0	

	15 MHz	10 MHz	$0.0 < W_{\text{gap}} \leq 25.0$	32^1	0
			$20.0 < W_{\text{gap}} \leq 50.0$	12^6	4.7
			$0.0 < W_{\text{gap}} \leq 20.0$	32^1	0
	15 MHz	15 MHz	$15.0 < W_{\text{gap}} \leq 45.0$	12^6	4.2
			$0.0 < W_{\text{gap}} \leq 15.0$	32^1	0
	15 MHz	20 MHz	$10.0 < W_{\text{gap}} \leq 40.0$	12^6	3.8
			$0.0 < W_{\text{gap}} \leq 10.0$	32^1	0
	15 MHz	25 MHz	$5.0 < W_{\text{gap}} \leq 35.0$	12^6	3.5
			$0.0 < W_{\text{gap}} \leq 5.0$	32^1	0
	15 MHz	30 MHz	$0.0 < W_{\text{gap}} \leq 30.0$	12^6	3.3
	20 MHz	5 MHz	$15.0 < W_{\text{gap}} \leq 50.0$	16^7	6.5
			$0.0 < W_{\text{gap}} \leq 15.0$	32^1	0
	20 MHz	10 MHz	$10.0 < W_{\text{gap}} \leq 45.0$	16^7	5.1
			$0.0 < W_{\text{gap}} \leq 10.0$	32^1	0
	20 MHz	15 MHz	$5.0 < W_{\text{gap}} \leq 40.0$	16^7	4.5
			$0.0 < W_{\text{gap}} \leq 5.0$	32^1	0
	20 MHz	20 MHz	$0.0 < W_{\text{gap}} \leq 35.0$	16^7	4.1
	20 MHz	25 MHz	$0.0 < W_{\text{gap}} \leq 30.0$	16^7	3.8
	20 MHz	30 MHz	$0.0 < W_{\text{gap}} \leq 25.0$	16^7	3.6

NOTE 1: UL resource blocks shall be located as close as possible to the downlink operating band but confined within the transmission.

NOTE 2: W_{gap} is the sub-block gap between the two sub-blocks.

NOTE 3: The table only applies when the center frequency of E-UTRA carrier is higher than the NR carrier, and the ΔR_{IBNC} applies to the NR DL carrier only

NOTE 4: All combinations of channel bandwidths defined in Table 5.3B.1.3-1.

NOTE 5: UL resource blocks shall be located at $\text{RB}_{\text{start}} = 25$.

NOTE 6: UL resource blocks shall be located at $\text{RB}_{\text{start}} = 35$.

NOTE 7: UL resource blocks shall be located at $\text{RB}_{\text{start}} = 50$.

7.3B.3.3 Inter-band EN-DC within FR1

7.3B.3.3.1 $\Delta R_{\text{IB,c}}$ for EN-DC in two bands

Table 7.3B.3.3.1-1: $\Delta R_{\text{IB,c}}$ due to EN-DC(two bands)

Inter-band EN-DC configuration	E-UTRA or NR Band	$\Delta R_{\text{IB,c}}$ (dB)
DC_1_n28	n28	0.2
DC_1_n51	n51	0.1
DC_1_n77	1	0.2
	n77	0.5
DC_1_n78	n78	0.5
DC_2_n66	2	0.3
	n66	0.3
DC_2_n78	2	0.2
	n78	0.5
DC_3_n51	3	0.2
	n51	0.2
DC_3_n77	3	0.2
	n77	0.5
DC_3_n78	3	0.2

	n78	0.5
DC_5_n78	5	0.2
	n78	0.5
DC_7_n51	n51	0.2
DC_7_n77	n77	0.5
DC_7_n78 DC_7-7_n78	n78	0.5
DC_8_n77	8	0.2
	n77	0.5
DC_8_n78	8	0.2
	n78	0.5
DC_11_n77	n77	0.5
DC_11_n78	n78	0.5
DC_12_n5	12	0.3
	n5	0.5
DC_12_n66	12	0.5
DC_18_n77	n77	0.5
DC_19_n77	n77	0.5
DC_19_n78	n78	0.5
DC_20_n51	n51	0.2
DC_20_n77	n77	0.5
DC_20_n78	n78	0.5
DC_21_n77	n77	0.5
DC_21_n78	n78	0.5
DC_25_n41	n41	0 ¹
		0.5 ²
DC_26_n77	n77	0.5
DC_26_n78	n78	0.5
DC_28_n51	n51	0.2
DC_28_n77	28	0.2
	n77	0.5
DC_28_n78	28	0.2
	n78	0.5
DC_30_n66	30	0.5
	n66	0.4
DC_38_n78	38	0.4
	n78	0.5
DC_39_n78	n78	0.5
DC_39_n79	n79	0.5
DC_40_n77	40	0.4
	n77	0.5
DC_41_n77	n77	0.5
DC_41_n78	n78	0.5
DC_41_n79	n79	0.5
DC_42_n51	n51	0.2
DC_66_n78	66	0.2
	n78	0.5

NOTE 1: The requirement is applied for UE transmitting on the frequency range of 2545 – 2690 MHz.

NOTE 2: The requirement is applied for UE transmitting on the frequency range of 2496 – 2545 MHz.

7.3B.3.3.2 $\Delta R_{IB,c}$ for EN-DC three bands

Table 7.3B.3.3.2-1: $\Delta R_{IB,c}$ due to EN-DC (three bands)

Inter-band EN-DC configuration	E-UTRA or NR Band	$\Delta R_{IB,c}$ (dB)
DC_1-3_n28	n28	0.2
DC_1-3_n77	1	0.2
	3	0.2
	n77	0.5
DC_1-3_n78	1	0.2
	3	0.2
	n78	0.5
DC_1-5_n78	1	0.2
	5	0.2
	n78	0.5
DC_1-7_n28	n28	0.2
DC_1-7_n78 DC_1-7-7_n78	1	0.2
	7	0.2
	n78	0.5
DC_1-8_n78	8	0.2
	n78	0.5
DC_1-18_n77	n77	0.5
DC_1-18_n78	n78	0.5
DC_1-19_n77	n77	0.5
DC_1-19_n78	n78	0.5
DC_1-19_n79	1	0.3
	19	0.3
DC_1-20_n28		
	20	0.2
	n28	0.2
DC_1-20_n78	n78	0.5
DC_1-21_n77	n77	0.5
DC_1-21_n78	1	0.2
	n78	0.5
DC_1-28_n77	28	0.2
	n77	0.5
DC_1-28_n78 DC_1_n28-n78	28 or n28	0.2
	n78	0.5
DC_1_n28-n79	1	0.3
	28	0.3
DC_1-42_n77	1	0.2
	42	0.5
	n77	0.5
DC_1-41_n77	n77	0.5
DC_1-41_n78	n78	0.5
DC_1-42_n78	1	0.2
	42	0.5
	n78	0.5
DC_1-42_n79	42	0.5
DC_1_n77-n79	1	0.2
	n77	0.5
DC_1_n78-n79	n78	0.5
DC_1-SUL_n78-n84	n78	0.5
DC_2_5_n66	2	0.3
	n66	0.3

DC_2_30_n66	2	0.4
	30	0.5
	n66	0.4
DC_2-66_n71B	2	0.3
	66	0.3
DC_3_n3-n77	3	0.2
	n3	0.2
	n77	0.5
DC_3_n3-n78	3	0.2
	n3	0.2
	n78	0.5
DC_3-5_n78	3	0.2
	5	0.2
	n78	0.5
DC_3-7_n78, DC_3-7-7_n78	3	0.2
	7	0.2
	n78	0.5
DC_3-8_n78	3	0.2
	8	0.2
	n78	0.5
DC_3-19_n77	3	0.2
	n77	0.5
DC_3-19_n78	3	0.2
	n78	0.5
DC_3-20_n28	20	0.1
	n28	0.1
DC_3-20_n78	3	0.2
	n78	0.5
DC_3-21_n77	3	0.3
	21	0.5
	n77	0.5
DC_3-21_n78	3	0.3
	21	0.5
	n78	0.5
DC_3-21_n79	3	0.3
	21	0.5
DC_3-28_n78 DC_3_n28-n78	3	0.2
	n78	0.5
DC_3-38_n78	3	0.2
	38	0.4
	n78	0.5
DC_3-41_n78	3	0.2
	41	0 ¹
		0.5 ²
	n78	0.5
DC_3-42_n77	3	0.2
	42	0.5
	n77	0.5
DC_3-42_n78	3	0.2
	42	0.5
	n78	0.5
DC_3-42_n79	3	0.2
	42	0.5

DC_3_n77-n79	3	0.2
	n77	0.5
DC_3_n78-n79	3	0.2
	n78	0.5
DC_3-SUL_n78-n80	3	0.2
	n78	0.5
DC_3-SUL_n78-n82	3	0.2
	n78	0.5
DC_5-7_n78	5	0.2
	7	0.2
	n78	0.5
DC_5_30_n66	30	0.5
	n66	0.4
DC_7-20_n28	20	0.2
	n28	0.2
DC_7-20_n78	n78	0.5
DC_7-28_n78	n78	0.5
DC_7_n28-n78	n78	0.5
DC_7-46_n78	n78	0.5
DC_8A-SUL_n78-n81	8	0.2
	n78	0.2
DC_18-28_n77	n77	0.5
DC_18-28_n78	n78	0.5
DC_19-21_n77	n77	0.5
DC_19-21_n78	n78	0.5
DC_19-42_n77	42	0.5
	n77	0.5
DC_19-42_n78	42	0.5
	n78	0.5
DC_19-42_n79	42	0.5
DC_19_n77-n79	n77	0.5
DC_19_n78-n79	n78	0.5
DC_20_n28-n75	n28	0.2
DC_20_n28-n78	20	0.2
	n28	0.2
	n78	0.5
DC_20_n75-n78	n78	0.5
DC_20_n76-n78	n78	0.5
DC_20-SUL_n78-n82	n78	0.5
DC_20-SUL_n78-n83	20	0.2
	n78	0.5
DC_21-42_n77	42	0.5
	n77	0.5
DC_21-42_n78	42	0.5
	n78	0.5
DC_21-42_n79	42	0.5
DC_21_n77-n79	n77	0.5
DC_21_n78-n79	n78	0.5
DC_28-SUL_n78-n83	28	0.2
	n78	0.5
DC_28-42_n77	28	0.2
	42	0.5
	n77	0.5

DC_28-42_n78	28	0.2
	42	0.5
	n78	0.5
DC_28-42_n79	28	0.2
	42	0.5
DC_41-42_n77	42	0.5
	n77	0.5
DC_41-42_n78	42	0.5
	n78	0.5
DC_41-42_n79	42	0.5
DC_66-SUL_n78-n86	66	0.2
	n78	0.5
NOTE 1: The requirement is applied for UE transmitting on the frequency range of 2545 – 2690 MHz.		
NOTE 2: The requirement is applied for UE transmitting on the frequency range of 2496 – 2545 MHz.		

7.3B.3.3.3 $\Delta R_{IB,c}$ for EN-DC four bands

Table 7.3B.3.3.3-1: $\Delta R_{IB,c}$ due to EN-DC (four bands)

Inter-band EN-DC configuration	E-UTRA or NR Band	$\Delta R_{IB,c}$ (dB)
DC_1-3-5_n78	1	0.2
	3	0.2
	n78	0.5
DC_1-3-7_n28	n28	0.2
DC_1-3-7_n78 DC_1-3-7-7_n78	1	0.3
	3	0.3
	7	0.3
	n78	0.5
DC_1-3-8_n78	1	0.2
	3	0.2
	8	0.2
	n78	0.5
DC_1-3-19_n78	1	0.2
	3	0.2
	n78	0.5
DC_1-3-20_n28	20	0.2
	n28	0.2
DC_1-3-20_n78	1	0.2
	3	0.2
	n78	0.5
DC_1-3-21_n77	1	0.2
	3	0.3
	21	0.5
	n77	0.5
DC_1-3-21_n78	1	0.2
	3	0.3
	21	0.5
	n78	0.5
DC_1-3-21_n79	3	0.3
	21	0.5
DC_1-3-28_n77	1	0.2
	3	0.2
	28	0.2

	n77	0.5
DC_1-3-28_n78 DC_1-3_n28-n78	1	0.2
	3	0.2
	28 or n28	0.2
	n78	0.5
DC_1-3-28_n79	1	0.2
	3	0.2
	28	0.2
DC_1-3-42_n77	1	0.2
	3	0.2
	42	0.5
	n77	0.5
DC_1-3-42_n78	1	0.2
	3	0.2
	42	0.5
	n78	0.5
DC_1-3-42_n79	1	0.2
	3	0.2
	42	0.5
DC_1-5-7_n78 DC_1-5-7-7_n78	1	0.2
	5	0.2
	7	0.2
	n78	0.5
DC_1-7-20_n28	20	0.2
	n28	0.2
DC_1-7-20_n78	1	0.2
	7	0.2
	20	0.2
	n78	0.5
DC_1-7_n28-n78	1	0.2
	7	0.2
	n28	0.2
	n78	0.5
DC_1-18-28_n77	n77	0.5
DC_1-18-28_n78	n78	0.5
DC_1-19-42_n77	1	0.2
	42	0.5
	n77	0.5
DC_1-19-42_n78	42	0.5
	n78	0.5
DC_1-19-42_n79	42	0.5
DC_1-20_n28-n78	20	0.2
	n28	0.2
	n78	0.5
DC_1-21-42_n77	1	0.2
	42	0.5
	n77	0.5
DC_1-21-42_n78	42	0.5
	n78	0.5
DC_1-21-42_n79	42	0.5
DC_1-28-42_n77	1	0.2
	28	0.2
	42	0.5

	n77	0.5
DC_1-28-42_n78	28	0.2
	42	0.5
	n78	0.5
DC_1-28-42_n79	28	0.2
	42	0.5
DC_1-41-42_n77	42	0.5
	n77	0.5
DC_1-41-42_n78	42	0.5
	n78	0.5
DC_1-41-42_n79	42	0.5
DC_1-41-42_n79	42	0.5
DC_2-66-(n)71	2	0.3
	66	0.3
DC_3-5-7_n78, DC_3-5-7-7_n78	3	0.2
	5	0.2
	7	0.2
	n78	0.5
DC_3-7-7_n78	3	0.2
	7	0.2
	n78	0.5
DC_3-7-20_n28	20	0.2
	n28	0.1
DC_3-7-20_n78	3	0.2
	7	0.2
	n78	0.5
DC_3-7-28_n78 DC_3-7_n28-n78	3	0.2
	7	0.2
	28 or n28	0.2
	n78	0.5
DC_3-19-21_n77	3	0.3
	21	0.5
	n77	0.5
DC_3-19-21_n78	3	0.3
	21	0.5
	n78	0.5
DC_3-19-21_n79	3	0.3
	21	0.5
DC_3-19-42_n77	3	0.2
	42	0.5
	n77	0.5
DC_3-19-42_n78	3	0.2
	42	0.5
	n78	0.5
DC_3-19-42_n79	3	0.2
	42	0.5
DC_3-20_n28-n78	3	0.2
	20	0.2
	n28	0.2
	n78	0.5
DC_3-21-42_n77	3	0.3
	21	0.5
	42	0.5

	n77	0.5
DC_3-21-42_n78	3	0.3
	21	0.5
	42	0.5
	n78	0.5
DC_3-21-42_n79	3	0.3
	21	0.5
	42	0.5
DC_3-28-42_n77	3	0.2
	28	0.2
	42	0.5
	n77	0.5
DC_3-28-42_n78	3	0.2
	28	0.2
	42	0.5
	n78	0.5
DC_3-28-42_n79	3	0.2
	28	0.2
	42	0.5
DC_5-7-7_n78	5	0.2
	7	0.2
	n78	0.5
DC_7-20_n28-n78	20	0.2
	n28	0.2
	n78	0.5
DC_19-21-42_n77	42	0.5
	n77	0.5
DC_19-21-42_n78	42	0.5
	n78	0.5
DC_19-21-42_n79	42	0.5
DC_21-28-42_n77	28	0.2
	42	0.5
	n77	0.5
DC_21-28-42_n78	28	0.2
	42	0.5
	n78	0.5
DC_21-28-42_n79	28	0.2
	42	0.5

7.3B.3.3.4 $\Delta R_{IB,c}$ for EN-DC five bands

Table 7.3B.3.3.4-1: $\Delta R_{IB,c}$ due to EN-DC (five bands)

Inter-band EN-DC configuration	E-UTRA or NR Band	$\Delta R_{IB,c}$ (dB)
DC_1-3-5-7_n78, DC_1-3-5-7-7_n78	1	0.2
	3	0.2
	5	0.2
	7	0.2
	n78	0.5
DC_1-3-7-20_n28	20	0.2
	n28	0.2
DC_1-3-7-20_n78	1	0.2
	3	0.2

	7	0.2
	n78	0.5
DC_1-3-7_n28-n78	1	0.2
	3	0.2
	7	0.2
	n28	0.2
	n78	0.5
DC_1-3-19-21_n77	1	0.2
	3	0.3
	21	0.5
	n77	0.5
DC_1-3-19-21_n78	1	0.2
	3	0.3
	21	0.5
	n78	0.5
DC_1-3-19-21_n79	3	0.3
	21	0.5
DC_1-3-19-42_n77	1	0.2
	3	0.2
	42	0.5
	n77	0.5
DC_1-3-19-42_n78	1	0.2
	3	0.2
	42	0.5
	n78	0.5
DC_1-3-19-42_n79	1	0.2
	3	0.2
	42	0.5
DC_1-3-28-42_n77	1	0.2
	3	0.2
	28	0.2
	42	0.5
	n77	0.5
DC_1-3-28-42_n78	1	0.2
	3	0.2
	28	0.2
	42	0.5
	n78	0.5
DC_1-3-28-42_n79	1	0.2
	3	0.2
	28	0.2
	42	0.5
DC_1-3-20_n28-n78	1	0.2
	3	0.2
	20	0.2
	n28	0.2
	n78	0.5
DC_1-3-21-42_n77	1	0.2
	3	0.3
	21	0.5
	42	0.5
	n77	0.2
DC_1-3-21-42_n78	1	0.2
	3	0.3
	21	0.5

	42	0.5
	n78	0.2
DC_1-3-21-42_n79	1	0.2
	3	0.3
	21	0.5
	42	0.5
DC_1-7-20_n28-n78	1	0.2
	7	0.2
	20	0.2
	n28	0.2
	n78	0.5
DC_1-19-21-42_n77	1	0.2
	42	0.5
	n77	0.5
DC_1-19-21-42_n78	42	0.5
	n78	0.5
DC_1-19-21-42_n79	42	0.5
DC_1-21-28-42_n77	1	0.2
	28	0.2
	42	0.5
	n77	0.5
DC_1-21-28-42_n78	28	0.2
	42	0.5
	n78	0.5
DC_1-21-28-42_n79	28	0.2
	42	0.5
DC_3-7-20_n28-n78	3	0.2
	7	0.2
	20	0.2
	n28	0.2

7.3B.3.3.5 $\Delta R_{IB,c}$ for EN-DC six bands

Table 7.3B.3.3.5-1: $\Delta R_{IB,c}$ due to EN-DC (six bands)

Inter-band EN-DC configuration	E-UTRA or NR Band	$\Delta R_{IB,c}$ (dB)
DC_1-3-7-20_n28-n78	1	0.2
	3	0.2
	7	0.2
	20	0.2
	n28	0.2
	n78	0.5

7.3B.3.3a Inter-band NE-DC within FR1

Unless $\Delta R_{IB,c}$ is specified in this clause, the value of $\Delta R_{IB,c}$ for the correspondingly specified EN-DC configuration in clause 7.3B.3.3 is applicable.

7.3B.3.4 Inter-band EN-DC including FR2

7.3B.3.4.1 $\Delta R_{IB,c}$ for EN-DC in two bands

Unless otherwise stated, $\Delta R_{IB,c}$ for E-UTRA and FR2 NR bands of inter-band EN-DC combinations defined in table 5.5B.5.1-1 is set to zero.

Table 7.3B.3.4.1-1: Void

7.3B.3.4.2 $\Delta R_{IB,c}$ for EN-DC three bands

Unless otherwise stated, $\Delta R_{IB,c}$ for FR2 NR bands is set to zero, and $\Delta R_{IB,c}$ for constituent E-UTRA bands for inter-band EN-DC defined in table 5.5B.5.3-1 is the same as those for the corresponding E-UTRA CA configuration specified in TS 36.101 [4], without the FR2 NR bands.

Table 7.3B.3.4.2-1: Void

7.3B.3.4.3 $\Delta R_{IB,c}$ for EN-DC four bands

Unless otherwise stated, $\Delta R_{IB,c}$ for FR2 NR bands is set to zero, and $\Delta R_{IB,c}$ for constituent E-UTRA bands for inter-band EN-DC defined in table 5.2B.5.3-1 is the same as those for the corresponding E-UTRA CA configuration specified in TS 36.101 [4], without the FR2 NR bands.

Table 7.3B.3.4.3-1: Void

7.3B.3.4.4 $\Delta R_{IB,c}$ for EN-DC five bands

Unless otherwise stated, $\Delta R_{IB,c}$ for FR2 NR bands is set to zero, and $\Delta R_{IB,c}$ for constituent E-UTRA bands for inter-band EN-DC defined in table 5.5B.5.4-1 is the same as those for the corresponding E-UTRA CA configuration specified in TS 36.101 [4], without the FR2 NR bands.

Table 7.3B.3.4.4-1: Void

7.3B.3.4.5 Void

7.3B.3.5 Inter-band EN-DC including both FR1 and FR2

7.3B.3.5.2 $\Delta R_{IB,c}$ for EN-DC three bands

Unless otherwise stated, for inter-band EN-DC configurations defined in table 5.5B.6.2-1, $\Delta R_{IB,c}$ for constituent FR2 NR bands is set to zero, and $\Delta R_{IB,c}$ for constituent E-UTRA and FR1 NR bands is the same as those for the corresponding inter band EN-DC configuration without the FR2 bands specified in 7.3B.3.3.

Table 7.3B.3.5.2-1: Void

7.3B.3.5.3 $\Delta R_{IB,c}$ for EN-DC four bands

Unless otherwise stated, for inter-band EN-DC configurations defined in table 5.5B.6.3-1, $\Delta R_{IB,c}$ for constituent FR2 NR bands is set to zero, and $\Delta R_{IB,c}$ for constituent E-UTRA and FR1 NR bands is the same as those for the corresponding inter band EN-DC configuration without the FR2 bands specified in 7.3B.3.3.

7.3B.3.5.4 $\Delta R_{IB,c}$ for EN-DC five bands

Unless otherwise stated, for a certain inter-band EN-DC configurations defined in table 5.5B.6.4-1, $\Delta R_{IB,c}$ for constituent FR2 NR bands is set to zero, and $\Delta R_{IB,c}$ for constituent E-UTRA and FR1 NR bands is the same as those for the corresponding inter band EN-DC configuration without the FR2 bands specified in 7.3B.3.3.

7.3B.3.5.5 $\Delta R_{IB,c}$ for EN-DC six bands

Unless otherwise stated, for inter-band EN-DC configurations defined in table 5.5B.6.5-1, $\Delta R_{IB,c}$ for constituent FR2 NR bands is set to zero, and $\Delta R_{IB,c}$ for constituent E-UTRA and FR1 NR bands is the same as those for the corresponding inter band EN-DC configuration without the FR2 bands specified in 7.3B.3.3.

7.4 Void

7.4A Maximum input level for CA

For inter-band NR CA between FR1 and FR2, the maximum input level specified in TS 38.101-1 [2] and TS 38.101-2 [3] apply for FR1 and FR2 respectively.

7.4B Maximum input level for DC in FR1

7.4B.1 Intra-band contiguous EN-DC in FR1

Intra-band contiguous EN-DC maximum input level requirement and parameters are defined in Table 7.4B.1-1.

Table 7.4B.1-1: Maximum Input

Power in Largest CC, E-UTRA or NR, dBm	X^1
Power in each other CC, dBm	$X^1 - 10 \cdot \log_{10}(N_x SCS_x / N_y SCS_y)$
NOTE 1: Power in Largest E-UTRA or NR bandwidth CC, listed in Table 7.4-1 [2]	
NOTE 2: N_x , SCS_x is the number of RB's and Sub carrier spacing in the largest carrier bandwidth and could be E-UTRA or NR carrier	
NOTE 3: N_y , SCS_y is the number of RB's in any other carrier.	
NOTE 4: Void.	
NOTE 5: Void.	

7.4B.2 Intra-band non-contiguous EN-DC in FR1

For the E-UTRA sub-block containing one or multiple CC's, the requirement is defined in clause 7.4.1 for single carrier operation and in clause 7.4.1A for CA in TS 36.101 [4].

For the NR sub-block, the requirement is defined in clause 7.4 in TS 38.101-1 [2].

7.4B.3 Inter-band EN-DC within FR1

Maximum input level requirement for E-UTRA single carrier and CA operation specified in clauses 7.4.1 and 7.4.1A of TS 36.101 [4] and for NR single carrier and CA operation specified in clauses 7.4 and 7.4A of TS 38.101-1 [2] apply.

7.4B.3a Inter-band NE-DC within FR1

Maximum input level requirement for E-UTRA single carrier and CA operation specified in clauses 7.4.1 and 7.4.1A of TS 36.101 [4] and for NR single carrier and CA operation specified in clauses 7.4 and 7.4A of TS 38.101-1 [2] apply.

7.4B.4 Inter-band EN-DC including FR2

Maximum input level requirement for E-UTRA single carrier and CA operation specified in clauses 7.4.1 and 7.4.1A of TS 36.101 [4] and for NR single carrier and CA operation specified in clauses 7.4, 7. and 7.4B of TS 38.101-2 [3] apply.

7.4B.5 Inter-band EN-DC including both FR1 and FR2

Maximum input level requirement for E-UTRA single carrier and CA operation specified in clauses 7.4.1 and 7.4.1A of TS 36.101 [4] and for NR single carrier and CA operation specified in clauses 7.4, 7.4A and 7.4B of TS 38.101-1 [2] and TS 38.101-2 [3] apply.

7.5 Void

7.5A Adjacent channel selectivity for CA

For inter-band NR CA between FR1 and FR2, the adjacent channel selectivity specified in TS 38.101-1 [2] and TS 38.101-2 [3] apply for FR1 and FR2 respectively.

7.5B Adjacent channel selectivity for DC in FR1

7.5B.1 Intra-band contiguous EN-DC in FR1

Intra-band contiguous EN-DC ACS requirement and parameters are defined for test case 1 in Table 7.5B.1-1 and for test case 2 in Table 7.5B.1-2.

Table 7.5B.1-1: ACS test case 1

EN-DC Aggregated Bandwidth, MHz	<=100	>100, <=120	>120, <=140	>140, <=160
ACS, dB	X ¹	19.2	18.5	17.9
P _{interferer} , dBm	P _i ²	Aggregated power + 17.7 dB	Aggregated power + 17 dB	Aggregated power + 16.4dB
Pw in Transmission BW configuration, per CC, dBm	REFSENS +14dB			
NOTE 1: X is ACS level at the specified EN-DC aggregated bandwidth from Table 7.5.1A-1 in TS 36.101 [4]				
NOTE 2: P _i is from Table 7.5.1A-2 in TS 36.101 [4]				
NOTE 3: Jammer BW and offset is from Table 7.5.1A-2 [4] and is applied from the lowest edge of the lowest carrier and the highest edge of the highest carrier				
NOTE 4: Void.				
NOTE 5: Void.				

Table 7.5B.1-2: ACS test case 2

EN-DC Aggregated Bandwidth, ENBW, MHz	≤ 100	$>100, \leq 120$	$>120, \leq 140$	$>140, \leq 160$
P_w in Transmission Bandwidth Configuration, perCC, dBm	P_w^1	$-42.7 + 10\log_{10}(N_{RB,c}/N_{RB_agg})$	$-42 + 10\log_{10}(N_{RB,c}/N_{RB_agg})$	$-41.4 + 10\log_{10}(N_{RB,c}/N_{RB_agg})$

$P_{\text{interferer}}$, dBm	-25
NOTE 1: P_W is wanted signal power level at the specified EN-DC aggregated Bandwidth from Table 7.5.1A-3 in TS 36.101 [4]	
NOTE 2: Jammer BW and offset is from Table 7.5.1A-3 [4] and is applied from the lowest edge of the lowest carrier and the highest edge of the highest carrier	
NOTE 3: Void.	
NOTE 4: Void.	

7.5B.2 Intra-band non-contiguous EN-DC in FR1

For the E-UTRA sub-block containing one or multiple CC's, the requirement is defined in clause 7.5.1 for single carrier operation and in clause 7.5.1A for CA in TS 36.101 [4].

For the NR sub-block, the requirement is defined in clause 7.5 in TS 38.101-1 [2].

The blocker configuration is defined in the general clause 7.1.

7.5B.3 Inter-band EN-DC within FR1

Adjacent channel selectivity requirement for E-UTRA single carrier and CA operation specified in clauses 7.5.1 and 7.5.1A of TS 36.101 [4] and for NR single carrier and CA operation specified in clauses 7.5 and 7.5A of TS 38.101-1 [2] apply.

7.5B.3a Inter-band NE-DC within FR1

Adjacent channel selectivity requirement for E-UTRA single carrier and CA operation specified in clauses 7.5.1 and 7.5.1A of TS 36.101 [4] and for NR single carrier and CA operation specified in clauses 7.5 and 7.5A of TS 38.101-1 [2] apply.

7.5B.4 Inter-band EN-DC including FR2

Adjacent channel selectivity requirement for E-UTRA single carrier and CA operation specified in clauses 7.5.1 and 7.5.1A of TS 36.101 [4] and for NR single carrier and CA operation specified in clauses 7.5, 7.5A and 7.5B of TS 38.101-2 [3] apply.

7.5B.5 Inter-band EN-DC including both FR1 and FR2

Adjacent channel selectivity requirement for E-UTRA single carrier and CA operation specified in clauses 7.5.1 and 7.5.1A of TS 36.101 [4] and for NR single carrier and CA operation specified in clauses 7.5, 7.5A and 7.5B of TS 38.101-1 [2] and TS 38.101-2 [3] apply.

7.6 Void

7.6A Blocking characteristics for CA

For inter-band NR CA between FR1 and FR2, the in-band blocking characteristics specified in TS 38.101-1 [2] and TS 38.101-2 [3] apply for FR1 and FR2 respectively. The narrow band blocking and out-of-band blocking specified in TS 38.101-1 [2] apply for FR1.

7.6B Blocking characteristics for DC in FR1

7.6B.1 General

7.6B.2 Inband blocking for DC in FR1

7.6B.2.1 Intra-band contiguous EN-DC in FR1

Intra-band contiguous EN-DC in-band blocking requirement and parameters are defined in Table 7.6B.2.1-1.

Table 7.6B.2.1-1: In-band blocking

EN-DC Aggregated Bandwidth, MHz	≤100	>100, ≤120	>120, ≤140	>140, ≤160
P _w in Transmission Bandwidth Configuration, perCC, dBm	P _w ¹	REFSENS + Aggregated BW specific value below		
		16.8	17.5	18
NOTE 1: P _w is wanted signal power level at the specified EN-DC aggregated Bandwidth from Table 7.6.1.1A-1 in TS 36.101 [4]				
NOTE 2: Interferer values are specified from Table 7.6.1.1A-2 in TS 36.101 [4]				
NOTE 3: Jammer BW and offset is from Table 7.6.1.1A-1 [4] and is applied from the lowest edge of the lowest carrier and the highest edge of the highest carrier				
NOTE 4: Void.				
NOTE 5: Void.				

7.6B.2.2 Intra-band non-contiguous EN-DC in FR1

For the E-TRA sub-block containing one or multiple CC's, the requirement is defined in clause 7.6.1.1 for single carrier operation and in clause 7.6.1.1A for CA in TS 36.101 [4].

For the NR sub-block, the requirement is defined in clause 7.6.2 in TS 38.101-1 [2].

The blocker configuration is defined in the general clause 7.1.

7.6B.2.3 Inter-band EN-DC within FR1

Inband blocking requirement for E-UTRA single carrier and CA operation specified in clauses 7.6.1.1 and 7.6.1.1A of TS 36.101 [4] and for NR single carrier and CA operation specified in clauses 7.6.2 and 7.6A.2 of TS 38.101-1 [2] apply.

7.6B.2.3a Inter-band NE-DC within FR1

Inband blocking requirement for E-UTRA single carrier and CA operation specified in clauses 7.6.1.1 and 7.6.1.1A of TS 36.101 [4] and for NR single carrier and CA operation specified in clauses 7.6.2 and 7.6A.2 of TS 38.101-1 [2] apply.

7.6B.2.4 Inter-band EN-DC including FR2

Inband blocking requirement for E-UTRA single carrier and CA operation specified in clauses 7.6.1.1 and 7.6.1.1A of TS 36.101 [4] and for NR single carrier and CA operation specified in clauses 7.6.2, 7.6A.2 and 7.6B.2 of TS 38.101-2 [3] apply.

7.6B.2.5 Inter-band EN-DC including both FR1 and FR2

Inband blocking requirement for E-UTRA single carrier and CA operation specified in clauses 7.6.1.1 and 7.6.1.1A of TS 36.101 [4] and for NR single carrier and CA operation specified in clauses 7.6.2, 7.6A.2 and 7.6B.2 of TS 38.101-1 [2] and TS 38.101-2 [3] apply.

7.6B.3 Out-of-band blocking for DC in FR1

7.6B.3.1 Intra-band contiguous EN-DC in FR1

Intra-band contiguous EN-DC out-of-band requirement and parameters are defined in Table 7.6B.3.1-1.

Table 7.6B.3.1-1: Out-of-band blocking

EN-DC Aggregated Bandwidth, MHz	≤100	>100, ≤120	>120, ≤140	>140, ≤160
Pw in Transmission Bandwidth Configuration, perCC, dBm	REFSENS + Aggregated BW specific value below			
	9			
NOTE 1: Interferer values and offsets are specified from Table 7.6.2.1A-2 in TS 36.101 [4]				
NOTE 2: Void.				
NOTE 3: Void.				

For Table 7.6.2.1A-2 from TS 36.101 [4] in frequency range 1, 2 and 3, up to exceptions are allowed for spurious response frequencies in each assigned frequency channel when measured using a 1MHz step size. For these exceptions the requirements of subclause 7.7B.1 Spurious response are applicable.

7.6B.3.2 Intra-band non-contiguous EN-DC in FR1

For the E-UTRA sub-block containing one or multiple CC's, the requirement is defined in clause 7.6.2.1 for single carrier operation and in clause 7.6.2.1A for CA in TS 36.101 [4].
For the NR sub-block, the requirement is defined in clause 7.6.3 is [2].

7.6B.3.3 Inter-band EN-DC within FR1

Out-of band blocking requirements for E-UTRA single carrier and CA operation specified in clauses 7.6.2.1 and 7.6.2.1A of TS 36.101 [4] and for NR single carrier and CA operation specified in clauses 7.6.3 and 7.6A.3 of TS 38.101-1 [2] apply for lowest level EN-DC fallbacks (two bands) in clause 5.5B.4.1 with following conditions

- one E-UTRA uplink carrier with the output power set to 4 dB below $P_{\text{CMAX_L,C}}$ and the NR band whose downlink is being tested has its uplink carrier output power set to 29 dB below $P_{\text{CMAX_L,f,C}}$.

one NR uplink carrier with the output power set to 4 dB below $P_{\text{CMAX}_L, f, c}$ on the NR band with both E-UTRA and NR downlinks being tested with E-UTRA output power set to 29 dB below $P_{\text{CMAX}_L, c}$.

If CW interferer falls in a gap between $F_{\text{DL_high}}$ of the E-UTRA or NR band and $F_{\text{DL_low}}$ of the NR or EUTRA band, where the corresponding OOB ranges 1 and 2 overlap, then the lower level interferer limit of the overlapping OOB ranges applies.

If $F_{\text{DL_high}}$ of the lower E-UTRA or NR band is greater than or equal to the $F_{\text{DL_low}}$ of the upper NR or E-UTRA band as in overlapping RX frequency ranges, then the OOB range shall start from the $F_{\text{DL_low}}$ of the lower E-UTRA or NR band, and from the $F_{\text{DL_high}}$ of the upper NR or E-UTRA band.

For ENDC combination listed in Table 7.6B.3.3-1 under the first test condition above, exceptions to the requirement specified in Table 7.6B.3.3-2 are allowed when the second order intermodulation product of the lower frequency band UL carrier and the CW interfering signal fully or partially overlaps with the higher frequency band DL carrier.

Table 7.6B.3.3-1: ENDC combination with exceptions allowed

EN-DC combination
DC_5_n78
DC_8_n77
DC_8_n78
DC_8_n79
DC_11_n77
DC_18_n77
DC_18_n78
DC_18_n79
DC_19_n77
DC_19_n78
DC_19_n79
DC_20_n77
DC_20_n78
DC_21_n77
DC_26_n77
DC_26_n78
DC_26_n79
DC_28_n77
DC_28_n78
DC_28_n79

Parameter	Unit	Level
$P_{\text{Interferer}} \text{ (CW)}$	dBm	-44 ¹
NOTE 1: The requirement applies when , where and are the carrier frequencies for lower frequency band UL and higher frequency band DL, respectively. and are the channel bandwidths configured for lower frequency band UL carrier and higher frequency band DL carrier in MHz, respectively.		

For each of the two test cases in clauses 7.6.2.1 and 7.6.2.1A of TS 36.101 [4] and for NR single carrier and CA operation specified in clauses 7.6.3 and 7.6A.3 of TS 38.101-1 [2] for all interferer frequency ranges a maximum of

exceptions are allowed for spurious response frequencies in each assigned frequency channel when measured using a step size of MHz with N_{RB} the number of resource blocks in the downlink transmission bandwidth configuration, CBW the bandwidth of the frequency channel in MHz and $n = 1, 2, 3$ for SCS = 15, 30, 60 kHz, respectively. For these exceptions, the requirements in clause 7.7 apply.

7.6B.3.3a Inter-band NE-DC within FR1

Out-of band blocking requirements for E-UTRA single carrier and CA operation specified in clauses 7.6.2.1 and 7.6.2.1A of TS 36.101 [4] and for NR single carrier and CA operation specified in clauses 7.6.3 and 7.6A.3 of TS 38.101-1 [2] apply for lowest level NE-DC fallbacks (two bands) in clause 5.5B.4a.1 with following conditions

one E-UTRA uplink carrier with the output power set to 4 dB below $P_{CMAX_L,c}$ and the NR band whose downlink is being tested has its uplink carrier output power set to 29 dB below $P_{CMAX_L,f,c}$.

one NR uplink carrier with the output power set to 4 dB below $P_{CMAX_L,f,c}$ on the NR band with both E-UTRA and NR downlinks being tested with E-UTRA output power set to 29 dB below $P_{CMAX_L,c}$.

7.6B.3.4 Inter-band EN-DC including FR2

Out-of band blocking requirements specified for E-UTRA single carrier and CA operation specified in clauses 7.6.2.1 and 7.6.2.1A of TS 36.101 [4] apply for lowest level EN-DC fallbacks (two bands) in clause 5.5B.5.1 with only E-UTRA UL with output power as in TS 36.101 [4] (4 dB below P_{CMAX_L}).

7.6B.3.5 Inter-band EN-DC including both FR1 and FR2

Out-of band blocking requirements specified for E-UTRA single carrier and CA operation specified in clauses 7.6.2.1 and 7.6.2.1A of TS 36.101 [4] and for NR single carrier and CA operation specified in clauses 7.6.3 and 7.6A.3 of TS 38.101-1 [2] apply for lowest level EN-DC fallbacks (three bands) in clause 5.5B.6.2 with only E-UTRA UL with output power as in TS 36.101 [4] (4 dB below P_{CMAX_L}).

7.6B.4 Narrow band blocking for DC in FR1

7.6B.4.1 Intra-band contiguous EN-DC in FR1

Intra-band contiguous EN-DC narrow band blocking requirement and parameters are defined in Table 7.6B.4.1-1.

Table 7.6B.4.1-1: Narrow band blocking parameters

EN-DC Aggregated Bandwidth, MHz	≤100	>100, ≤120	>120, ≤140	>140, ≤160
Pw in Transmission Bandwidth Configuration, perCC, dBm	REFSENS + Aggregated BW specific value below			
	16			

P_{UW} , dBm (CW)	-55
NOTE 1: Jammer offset is from Table 7.6.3.1A-1 [4] and is applied from the lowest edge of the lowest carrier and the highest edge of the highest carrier	
NOTE 2: Void.	
NOTE 3: Void.	
NOTE 4: If NR carrier BW > 40 MHz, no narrow band blocking requirements apply when blocker is applied at the edge of the NR carrier.	

7.6B.4.2 Intra-band non-contiguous EN-DC in FR1

For the E-TRA sub-block containing one or multiple CC's, the requirement is defined in clause 7.6.3.1 for single carrier operation and in clause 7.6.3.1A for CA in TS 36.101 [4].

For the NR sub-block, the requirement is defined in clause 7.6.4 in TS 38.101-1 [2].

The blocker configuration is defined in the general clause 7.1.

7.6B.4.3 Inter-band EN-DC within FR1

Narrow band blocking requirement for E-UTRA single carrier and CA operation specified in clauses 7.6.3.1 and 7.6.3.1A of TS 36.101 [4] and for NR single carrier and CA operation specified in clauses 7.6.4 and 7.6A.4 of TS 38.101-1 [2] apply.

7.6B.4.3a Inter-band NE-DC within FR1

Narrow band blocking requirement for E-UTRA single carrier and CA operation specified in clauses 7.6.3.1 and 7.6.3.1A of TS 36.101 [4] and for NR single carrier and CA operation specified in clauses 7.6.4 and 7.6A.4 of TS 38.101-1 [2] apply.

7.6B.4.4 Inter-band EN-DC including FR2

Narrow band blocking requirement for E-UTRA single carrier and CA operation specified in clauses 7.6.3.1 and 7.6.3.1A of TS 36.101 [4] apply.

7.6B.4.5 Inter-band EN-DC including both FR1 and FR2

Narrow band blocking requirement for E-UTRA single carrier and CA operation specified in clauses 7.6.3.1 and 7.6.3.1A of TS 36.101 [4] and for NR single carrier and CA operation specified in clauses 7.6.4 and 7.6A.4 of TS 38.101-1 [2] apply.

7.7 Void

7.7A Spurious response for CA

For inter-band NR CA between FR1 and FR2, the spurious response specified in TS 38.101-1 [2] apply for FR1.

7.7B Spurious response for DC in FR1

7.7B.1 Intra-band contiguous EN-DC in FR1

Intra-band contiguous EN-DC spurious response requirement and parameters are defined in Table 7.7B.1-1.

Table 7.7B.1-1: Spurious Response Parameters

EN-DC Aggregated Bandwidth, MHz	≤ 100	$> 100, \leq 120$	$> 120, \leq 140$	$> 140, \leq 160$
P_{w} in Transmission Bandwidth Configuration, perCC, dBm	REFSENS + Aggregated BW specific value below			

	9
P_{interferer}, dBm (CW)	-44
NOTE 1: Void.	
NOTE 2: Void.	

7.7B.2 Intra-band non-contiguous EN-DC in FR1

For the E-UTRA sub-block containing one or multiple CC's, the requirement is defined in clause 7.7.1 for single carrier operation and in clause 7.7.1A for CA in TS 36.101 [4].

For the NR sub-block, the requirement is defined in clause 7.7 is [2].

7.7B.3 Inter-band EN-DC within FR1

Spurious response requirement for E-UTRA single carrier and CA operation specified in clauses 7.7.1 and 7.7.1A of TS 36.101 [4] and for NR single carrier and CA operation specified in clauses 7.7 and 7.7A of TS 38.101-1 [2] apply for lowest level EN-DC fallbacks (two bands) in clause 5.5B.4.1 with following conditions

- one E-UTRA uplink carrier with the output power set to 4 dB below $P_{\text{CMAX_L,c}}$ and the NR band whose downlink is being tested has its uplink carrier output power set to 29 dB below $P_{\text{CMAX_L,f,c}}$.
- one NR uplink carrier with the output power set to 4 dB below $P_{\text{CMAX_L,f,c}}$ on the NR band with both E-UTRA and NR downlinks being tested with E-UTRA output power set to 29 dB below $P_{\text{CMAX_L,c}}$.

7.7B.3a Inter-band NE-DC within FR1

Spurious response requirement for E-UTRA single carrier and CA operation specified in clauses 7.7.1 and 7.7.1A of TS 36.101 [4] and for NR single carrier and CA operation specified in clauses 7.7 and 7.7A of TS 38.101-1 [2] apply for lowest level NE-DC fallbacks (two bands) in clause 5.5B.4a.1 with following conditions

- one E-UTRA uplink carrier with the output power set to 4 dB below $P_{\text{CMAX_L,c}}$ and the NR band whose downlink is being tested has its uplink carrier output power set to 29 dB below $P_{\text{CMAX_L,f,c}}$.
- one NR uplink carrier with the output power set to 4 dB below $P_{\text{CMAX_L,f,c}}$ on the NR band with both E-UTRA and NR downlinks being tested with E-UTRA output power set to 29 dB below $P_{\text{CMAX_L,c}}$.

7.7B.4 Inter-band EN-DC including FR2

Spurious response requirement for E-UTRA single carrier and CA operation specified in clauses 7.7.1 and 7.7.1A of TS 36.101 [4] apply for lowest level EN-DC fallbacks (two bands) in clause 5.5B.5.1 with only E-UTRA UL with output power as in TS 36.101 [4] (4 dB below $P_{\text{CMAX_L}}$).

7.7B.5 Inter-band EN-DC including both FR1 and FR2

Spurious response requirement for E-UTRA single carrier and CA operation specified in clauses 7.7.1 and 7.7.1A of TS 36.101 [4] and for NR single carrier and CA operation specified in clauses 7.7 and 7.7A of TS 38.101-1 [2] apply for lowest

level EN-DC fallbacks (three bands) in clause 5.5B.6.2 with only E-UTRA UL with output power as in TS 36.101 [4] (4 dB below P_{CMAX_L}).

7.8 Void

7.8A Intermodulation characteristics for CA

For inter-band NR CA between FR1 and FR2, the intermodulation characteristics specified in TS 38.101-1 [2] apply for FR1.

7.8B Intermodulation characteristics for DC in FR1

7.8B.1 General

7.8B.2 Wide band Intermodulation

7.8B.2.1 Intra-band contiguous EN-DC in FR1

Intra-band contiguous EN-DC wide band intermodulation requirement and parameters are defined in Table 7.8B.2.1-1.

Table 7.8B.2.1-1: Wide band intermodulation

EN-DC Aggregated Bandwidth, MHz	<=100	>100, <=120	>120, <=140	>140, <=160
P _w in Transmission Bandwidth Configuration, perCC, dBm	P _w ¹	REFSENS + Aggregated BW specific value below		
		16.8	17.5	18.0
P _{interferer 1} , dBm (CW) ²	-46			
P _{interferer 2} , dBm (Modulated) ²	-46			
NOTE 1: P _w is wanted signal power level from Table 7.8.1A-1 in TS 36.101 [4]				
NOTE 2: Jammer BW and offsets is from Table 7.8.1A-1 [4] and is applied from the lowest edge of the lowest carrier and the highest edge of the highest carrier				
NOTE 3: Void.				
NOTE 4: Void.				

7.8B.2.2 Intra-band non-contiguous EN-DC in FR1

For the E-UTRA sub-block containing one or multiple CC's, the requirement is defined in clause 7.8.1 for single carrier operation and in clause 7.8.1A for CA in TS 36.101 [4].

For the NR sub-block, the requirement is defined in clause 7.8.2 in TS 38.101-1 [2].

The blocker configuration is defined in the general clause 7.1 and the requirement only apply for out of gap interferers.

7.8B.2.3 Inter-band EN-DC within FR1

Wide band Intermodulation requirement for E-UTRA single carrier and CA operation specified in clauses 7.8.1 and 7.8.1A of TS 36.101 [4] and for NR single carrier and CA operation specified in clauses 7.8.2 and 7.8A.2 of TS 38.101-1 [2] apply.

7.8B.2.3a Inter-band NE-DC within FR1

Wide band Intermodulation requirement for E-UTRA single carrier and CA operation specified in clauses 7.8.1 and 7.8.1A of TS 36.101 [4] and for NR single carrier and CA operation specified in clauses 7.8.2 and 7.8A.2 of TS 38.101-1 [2] apply.

7.8B.2.4 Inter-band EN-DC including FR2

Wide band Intermodulation requirement for E-UTRA single carrier and CA operation specified in clauses 7.8.1 and 7.8.1A of TS 36.101 [4] apply.

7.8B.2.5 Inter-band EN-DC including both FR1 and FR2

Wide band Intermodulation requirement for E-UTRA single carrier and CA operation specified in clauses 7.8.1 and 7.8.1A of TS 36.101 [4] and for NR single carrier and CA operation specified in clauses 7.8.2 and 7.8A.2 of TS 38.101-1 [2] apply.

7.9 Void

7.9A Spurious emissions for CA

For inter-band NR CA between FR1 and FR2, the spurious emission specified in TS 38.101-1 [2] and TS 38.101-2 [3] apply for FR1 and FR2 respectively.

7.9B Spurious emissions for DC in FR1

7.9B.1 Intra-band contiguous EN-DC in FR1

The requirement is defined in clause 7.9A.1 in TS 38.101-1 [2].

7.9B.2 Intra-band non-contiguous EN-DC in FR1

Spurious emissions requirement for E-UTRA single carrier and CA operation specified in clauses 7.9.1 and 7.9.1A of TS 36.101 [4] and for NR single carrier and CA operation specified in clauses 7.9 and 7.9A of TS 38.101-1 [2] apply.

7.9B.3 Inter-band EN-DC within FR1

E-UTRA requirements from TS 36.101 [4] and NR requirements from TS 38.101-1 [2] apply.

7.9B.3a Inter-band NE-DC within FR1

E-UTRA requirements from TS 36.101 [4] and NR requirements from TS 38.101-1 [2] apply.

7.9B.4 Inter-band EN-DC including FR2

Spurious emissions requirement for E-UTRA single carrier and CA operation specified in clauses 7.9.1 and 7.9.1A of TS 36.101 [4] and for NR single carrier and CA operation specified in clause 7.9 of TS 38.101-2 [3] apply.

7.9B.5 Inter-band EN-DC including both FR1 and FR2

Spurious emissions requirement for E-UTRA single carrier and CA operation specified in clauses 7.9.1 and 7.9.1A of TS 36.101 [4] and for NR single carrier and CA operation specified in clauses 7.9 and 7.9A of TS 38.101-1 [2] and TS 38.101-2 [3] apply.